

The Portsmouth Papers: A Hands-On Tour of PostgreSQL Beyond the Relational Model

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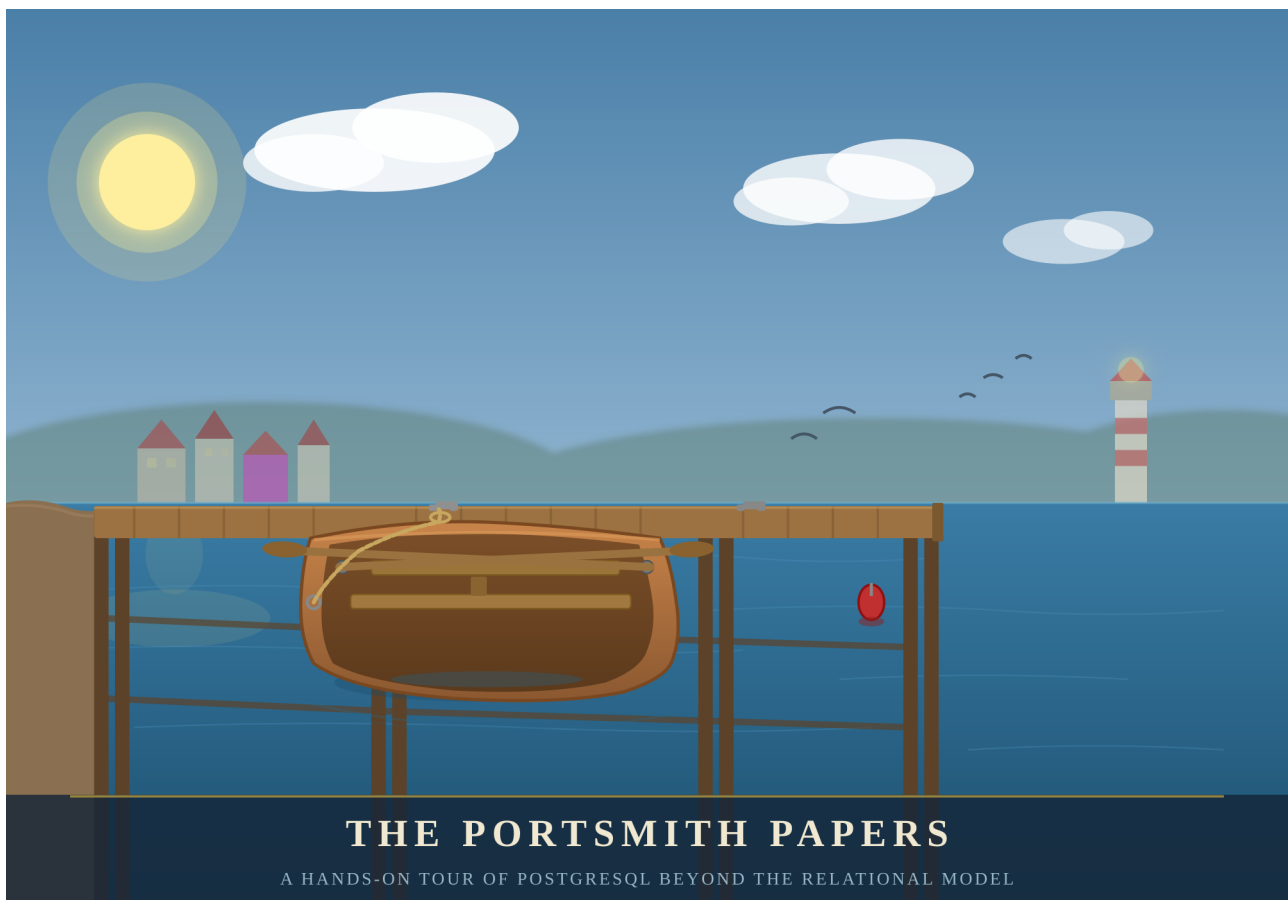
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The Portsmouth Papers

A Hands-On Tour of PostgreSQL Beyond the Relational Model



*"PostgreSQL is not a database with extensions.
It is an extensible data platform that happens to speak SQL."*

About This Book

Most PostgreSQL tutorials end where the interesting work begins.

Once you know how to `SELECT`, `JOIN`, and `GROUP BY`, you have mastered perhaps twenty percent of what PostgreSQL can do. The remaining eighty percent — geospatial queries, semantic search, real-time notifications, fuzzy matching, vector embeddings, distributed coordination, and more — lives in a ecosystem of extensions, index types, and language features that most practitioners never discover.

This book is a guided tour of that other eighty percent.

Each chapter is built around a concrete engineering problem faced by the fictional city of **Portsmith** and its data platform team. We store business directories as semi-structured JSON documents. We route emergency services using geospatial proximity queries. We build a job queue with no message broker. We search municipal records with fuzzy matching that survives typos and OCR errors. We expose the entire platform as a REST API with zero application code.

Every chapter follows the same structure: synthetic data is generated first, giving you a realistic dataset to work against, and then a series of exercises walks you from first principles to production-ready technique. The exercises are written to be done — not just read.

By the end, you will see PostgreSQL not as a place to store rows, but as a programmable data infrastructure layer capable of doing work that most teams reach for separate specialized systems to handle.

What you will need:

- A working PostgreSQL 16 installation (setup covered in Appendix A)
 - Python 3.12+
 - A Debian-based Linux environment
 - Familiarity with basic SQL (`SELECT`, `INSERT`, `JOIN`, `GROUP BY`)
 - Curiosity about what else is in there
-

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Chapter 1 — JSONB: Semi-Structured Data Without a Schema Tax

“A schema is a prediction about the future. JSONB lets you hedge.”

Background

Relational databases enforce a contract: every row in a table has exactly the same columns. That contract is usually a feature — it catches mistakes and makes queries predictable. But sometimes you genuinely don’t know what columns you’ll need until the data arrives. A business directory is a good example. A restaurant has hours, cuisine, and a reservations policy. A hardware shop has trade accounts and parking. A hotel has star ratings and room types. Forcing them all into the same set of columns means drowning in NULLs or building a rats’ nest of one-to-many extension tables.

PostgreSQL’s `JSONB` type lets you store arbitrary JSON documents in a column while keeping the rest of your row fully relational. It is not a NoSQL escape hatch — it is an indexed, queryable, patchable document field that sits inside a SQL table. You can `WHERE` on it, join against it, aggregate over it, and update individual keys without rewriting the whole document.

This chapter works through all the operators and patterns you’ll reach for most often. The exercises are short. Run every one, then try a variation of your own.

The Scenario

Portsmouth’s city council maintains a public business directory. Every business gets a short relational record — an ID, a name, an address, and a neighbourhood — but the supplementary detail varies so widely by business type that a single fixed schema would be unworkable.

The `businesses` table therefore stores all of that variable metadata in a `details` `JSONB` column. Each business category contributes its own keys:

Category	JSONB keys present
restaurant	cuisine, price_range, rating, hours, tags, accepts_reservations, outdoor_seating
retail	subcategory, rating, hours, tags, payment_methods, has_parking
service	subcategory, rating, hours, tags, appointment_required, specialties
accommodation	subcategory, star_rating, rating, hours, amenities, room_types, price_per_night
entertainment	subcategory, rating, hours, tags, live_music, age_restriction

The `hours` value is itself a nested object keyed by day of week (`mon–sun`). Each day is either `null` (closed) or `{"open": "HH:MM", "close": "HH:MM"}`. Several businesses also carry optional keys that appear only for their category, like `social` (social media handles), `happy_hour`, or `approved_brands`.

This heterogeneity is intentional — it is what makes the data a good vehicle for learning JSONB operators.

Exercise Goals

By the end of this chapter you will be able to:

- Extract values from a JSONB document using `->`, `->>`, and `#>>`, and explain why the operator choice changes the result type.
 - Filter rows using the containment operator `@>` and the key-existence operators `?`, `?|`, and `?&`.
 - Create a GIN index on a JSONB column and confirm that PostgreSQL uses it.
 - Update a document in place with `jsonb_set` and the `||` merge operator without rewriting the whole column.
 - Expand a JSONB array into a set of rows with `jsonb_array_elements` and aggregate the results.
 - Write `jsonb_path_query` expressions to filter on nested values.
-

Installation

1 — PostgreSQL

If PostgreSQL 16 is not already installed:

```
sudo apt update
sudo apt install -y postgresql-16 postgresql-client-16
```

Confirm the server is running:

```
pg_lsclusters
```

You should see a cluster at version 16 in the `online` state.

2 — Python 3.12 and psycopg

```
sudo apt install -y python3.12 python3.12-venv
```

Create a virtual environment in the book's working directory and install the PostgreSQL driver:

```
python3.12 -m venv .venv
source .venv/bin/activate
pip install "psycopg[binary]"
```

Note: `psycopg` (version 3) is the modern Python driver for PostgreSQL. It is not available as an `apt` package for Python 3.12, so we install it via `pip` into the virtual environment. The `[binary]` extra bundles a pre-compiled C extension so you don't need `libpq-dev`.

Loading the Data

Create the database

```
# Create a role matching your OS user (skip if it already exists)
sudo -u postgres createuser --createdb "$(whoami)"

# Create the portsmith database
createdb portsmith
```

Run the seed script

From the `book/` directory, with the virtual environment active:

```
python data/ch01_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 48 businesses ...
Done - 48 rows in businesses.
```

Verify the load

Open a `psql` session:

```
psql portsmith
```

Run these checks. If all three pass, the data is correct.

Check 1 — Row count and column types:

```
\d businesses
```

Table "public.businesses"				
Column	Type	Collation	Nullable	Default
id	integer		not null	nextval('businesses_id_seq'::regclass)
name	text		not null	
address	text		not null	
neighbourhood	text		not null	
details	jsonb		not null	

Check 2 — Counts by neighbourhood:

```
SELECT neighbourhood, COUNT(*) AS businesses
FROM   businesses
GROUP BY neighbourhood
ORDER BY neighbourhood;
```

neighbourhood	businesses
Harbour District	9
Industrial Port	7
Northgate	9
Old Town	9

Riverside		9
University Quarter		5

(6 rows)

Check 3 — Counts by category:

```
SELECT details->>'category' AS category, COUNT(*) AS businesses
FROM businesses
GROUP BY category
ORDER BY category;
```

category		businesses
-----+-----		
accommodation		5
entertainment		6
restaurant		15
retail		12
service		10

(5 rows)

If all three match, proceed to the exercises.

Exercises

Exercise 1 — The Three Extraction Operators

JSONB has two families of accessor. The arrow operators (-> and ->>) navigate one key at a time. The path operator (#>)>) jumps straight to a deeply nested value. The difference between them is the **type they return**.

1.1 — Arrow operators

Run these two queries and compare the result types:

```
-- -> returns JSONB
SELECT name,
       details -> 'rating' AS rating_jsonb
FROM businesses
LIMIT 5;
```

```
-- ->> returns TEXT
SELECT name,
       details ->> 'rating' AS rating_text
```

```
FROM businesses
LIMIT 5;
```

Look carefully at the column headings in `psql` — one will show `jsonb`, the other `text`. This distinction matters the moment you try to do arithmetic:

```
-- This works: cast text to numeric
SELECT name,
       (details ->> 'rating')::numeric AS rating
FROM   businesses
ORDER BY rating DESC
LIMIT 5;
```

name	rating
Lighthouse Bookshop	4.9
Finch & Sons Barbers	4.9
Portsmouth Vet. Clinic	4.8
River Bend Bakery	4.8
Quarter Note Jazz Club	4.8

(5 rows)

Key point: Use `->` when you want to keep working with JSONB (e.g., to navigate deeper). Use `->>` when you need the final value as text for comparisons, casting, or display.

1.2 — The path operator `#>>`

Navigating nested keys with chained `->` quickly becomes unreadable. The path operator takes an array of keys and returns the leaf as text in one step.

```
-- Chained arrows (verbose)
SELECT name,
       details -> 'hours' -> 'fri' ->> 'close' AS friday_close
FROM   businesses
WHERE  details ->> 'category' = 'restaurant'
LIMIT 8;
```

```
-- Path operator (equivalent, cleaner)
SELECT name,
       details #>> '{hours,fri,close}' AS friday_close
FROM   businesses
WHERE  details ->> 'category' = 'restaurant'
LIMIT 8;
```

Both produce the same result. Notice that some rows return `NULL` — those restaurants are closed on Fridays (the `fri` key is JSON `null`).

name	friday_close
-----+	
The Gilded Clam	22:30
Anchor & Oar Tavern	01:00
Bella Napoli	23:00
Le Petit Bistro	14:30
Dragon Palace	23:00
Spice Garden	23:00
Sol y Mar	23:00
Mango Bay Caribbean	22:30
(8 rows)	

Exercise 2 — Filtering with Containment and Key Existence

2.1 — Containment: @>

The containment operator checks whether the left JSONB document contains all the key-value pairs in the right document. It is the most common way to filter on JSONB values.

Find all waterfront businesses:

```
SELECT name, neighbourhood
FROM businesses
WHERE details @> '{"tags": ["waterfront"]}'
ORDER BY neighbourhood, name;
```

name	neighbourhood
-----+	
Anchor & Oar Tavern	Harbour District
Harbour Inn	Harbour District
Harbour View Theater	Harbour District
Mariners Rest B&B	Harbour District
Portsmith Fish Market	Harbour District
Saltbox Gallery	Harbour District
The Gilded Clam	Harbour District
Tidal Wave Surf Shop	Harbour District
Old Brewery Tap	Industrial Port
The Rusty Anchor	Industrial Port
(10 rows)	

Why this works: The @> operator checks whether the tags *array* on the left contains the tags array ["waterfront"] on the right — array containment is supported natively.

2.2 — Key existence: ?

The `?` operator tests whether a key exists in a JSONB object (or a value exists in a JSONB array). Unlike `@>`, it does not check the value — only presence.

Find all businesses that publish social media handles:

```
SELECT name,
       details -> 'social' AS social_links
FROM   businesses
WHERE  details ? 'social'
ORDER BY name;
```

name	social_links
Bella Napoli	{"instagram": "@bellanapoli_portsmith"}
Le Petit Bistro	{"instagram": "@lepetitbistro"}
Quarter Note Jazz Club	{"instagram": "@quarternote_portsmith"}
Spice Garden	{"instagram": "@spicegarden_portsmith"}
The Gilded Clam	{"facebook": "thegildedclam", "instagram": "@gildedclam"}
The Riverside Vegan	{"instagram": "@riverside_vegan"}

(6 rows)

2.3 — Any-key and all-key existence: `?|` and `?&`

`?|` returns true if *at least one* of the listed keys exists. `?&` returns true only if *all* listed keys exist.

```
-- Businesses that offer either delivery OR takeaway
SELECT name
FROM   businesses
WHERE  details ?| ARRAY['delivery', 'takeaway']
ORDER BY name;
```

```
-- Businesses that offer BOTH delivery AND takeaway
SELECT name
FROM   businesses
WHERE  details ?& ARRAY['delivery', 'takeaway']
ORDER BY name;
```

Run both and note the difference in result count. The `?&` set is a strict subset of the `?|` set.

Exercise 3 — Missing Keys vs. NULL Values

JSONB has two distinct ways to represent “no value”: a JSON `null` and a **missing key**. They behave differently in queries.

In our data, a business closed on Monday can be represented two ways:

- `"mon": null` — the key exists, the value is JSON null
- the `mon` key is absent entirely

The seed data uses `"mon": null` for closures, so we can observe this.

3.1 — The difference matters for ?

```
-- ? checks key EXISTENCE, not value
SELECT name
FROM   businesses
WHERE  (details -> 'hours') ? 'sun'    -- key exists (even if null)
ORDER BY name;
```

Run it. You should get **43 rows** — every business that uses a day-keyed hours object. The five accommodation businesses are excluded because their hours look like `{"reception": "07:00-23:00"}` and have no `sun` key.

Now try what seems like a natural follow-up:

```
-- ⚠ This does NOT do what you might expect
SELECT name
FROM   businesses
WHERE  (details -> 'hours' -> 'sun') IS NOT NULL
ORDER BY name;
```

Run it. You still get **43 rows** — identical to the first query.

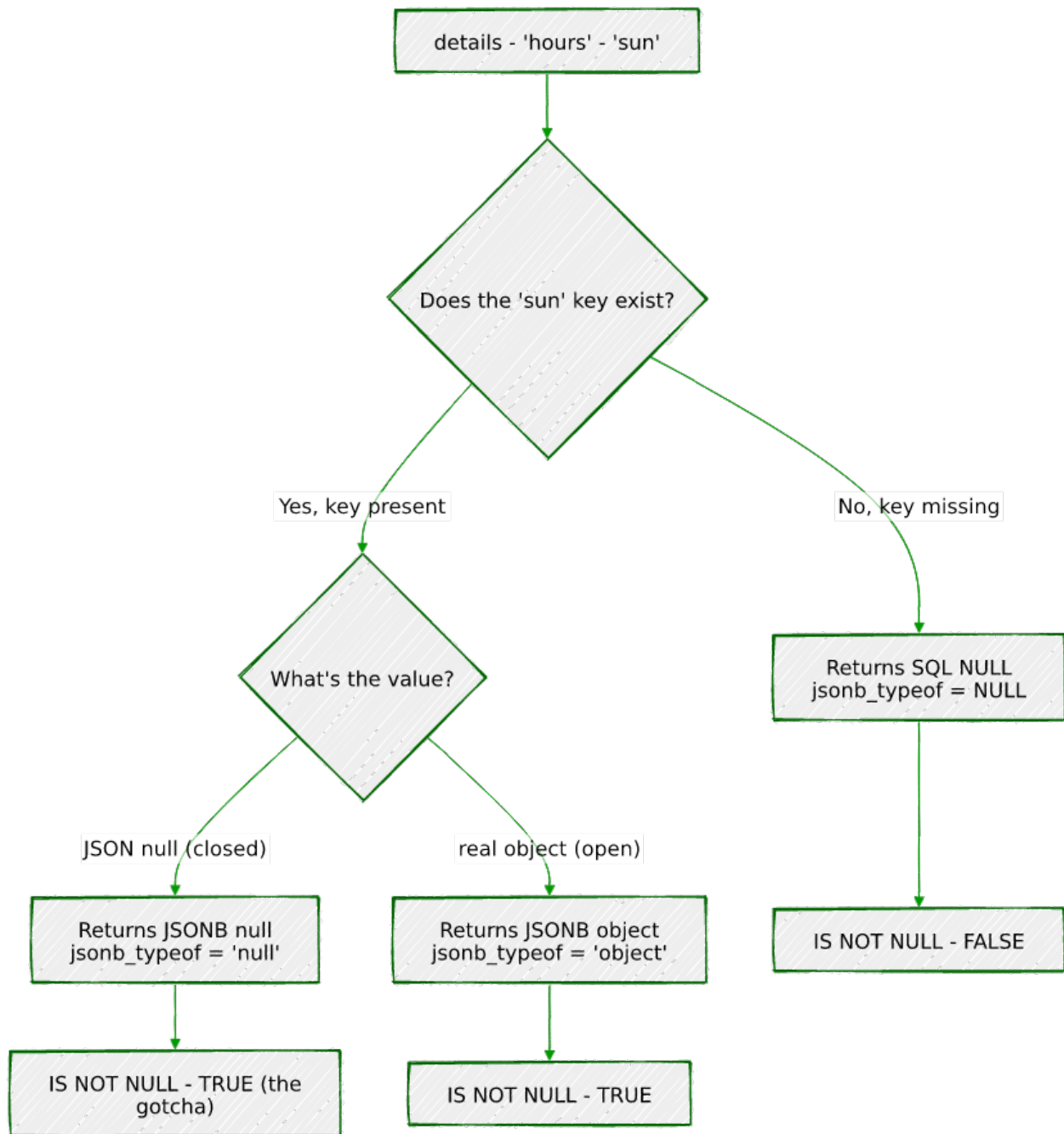
Why? This is one of JSONB's most important gotchas:

JSONB `null` is not SQL `NULL`.

When the seed script stores `"sun": None` in Python, `json.dumps` produces `"sun": null` — a JSON null *value* under an existing key. When PostgreSQL evaluates `details -> 'hours' -> 'sun'` on such a row, it returns the JSONB value `null`. That is a real value — not the absence of a value — so `IS NOT NULL` evaluates to `TRUE`.

SQL `NULL` only appears when the key itself is **missing** from the object. That is what happens for the five accommodation businesses: `details -> 'hours'` returns `{"reception": "..."}` , and then `-> 'sun'` on that object finds no such key and returns SQL `NULL`, so `IS NOT NULL` is `FALSE`.

The same logic, traced as a flowchart instead of read as prose:



To correctly distinguish the three states, use `jsonb_typeof()`:

State	details -> 'hours' -> 'sun'	jsonb_typeof(...)	IS NOT NULL
Key missing (accommodation)	SQL NULL	SQL NULL	FALSE
Key present, closed (null)	JSONB null	'null'	TRUE ← gotcha
Key present, open (object)	JSONB object	'object'	TRUE

The correct query for “businesses actually open on Sunday”:

```
SELECT name
FROM businesses
WHERE jsonb_typeof(details -> 'hours' -> 'sun') = 'object'
ORDER BY name;
```

This returns **25 rows** — only the businesses with a real hours object for Sunday.

`jsonb_typeof` returns `'object'` only for JSON objects, `'null'` for JSON null, and SQL `NULL` for a missing key (which then fails the `=` test).

Rule of thumb: Never use `IS NOT NULL` to test whether a JSONB navigation result is a “real” value. Use `jsonb_typeof(...)` to check the actual JSON type, or navigate one level deeper (e.g., check `details #>> '{hours,sun,open}'`) — the `#>>` operator converts JSON null to SQL `NULL`, so a deeper path naturally filters it out.

3.2 — Handling accommodation differently

Hotels and hostels store hours as a single `"reception": "HH:MM-HH:MM"` string rather than a day-keyed object. This means the `? 'sun'` test returns false for them even though they may be open 24/7. Write a query that finds all businesses that are either:

- Open on Sunday (day-keyed hours with a non-null `sun`), **or**
- An accommodation with 24/7 reception

```
SELECT name, neighbourhood,
CASE
    WHEN jsonb_typeof(details -> 'hours' -> 'sun') = 'object'
    THEN (details #>> '{hours,sun,open}') || '-' ||
         (details #>> '{hours,sun,close}')
    WHEN details #>> '{hours,reception}' = '24/7'
    THEN '24/7'
    ELSE details #>> '{hours,reception}'
END AS sunday_availability
FROM businesses
WHERE jsonb_typeof(details -> 'hours' -> 'sun') = 'object'
    OR (details -> 'hours') ? 'reception'
ORDER BY neighbourhood, name;
```

Exercise 4 — GIN Indexes

Without an index, every JSONB query is a sequential scan — PostgreSQL reads every row and evaluates the expression. For a 48-row toy dataset that is unnoticeable, but with millions of rows it becomes a bottleneck.

A **GIN** (Generalised Inverted Index) index over a JSONB column indexes every key and value inside every document. It accelerates `@>`, `?`, `?|`, and `?&` queries.

4.1 — Observe the plan before indexing

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  details @> '{"tags": ["waterfront"]}';
```

With 48 rows you will see `seq scan` (sequential scan, i.e. reading every row) — that is expected. On a larger table this would say `seq scan` with a high cost estimate.

4.2 — Create the GIN index

```
CREATE INDEX idx_businesses_details_gin
ON   businesses
USING GIN (details);
```

4.3 — Observe the plan after indexing

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  details @> '{"tags": ["waterfront"]}';
```

On a small table PostgreSQL may still choose a sequential scan (the planner knows the table fits in memory). Force it to use the index to see the mechanism:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  details @> '{"tags": ["waterfront"]}';

SET enable_seqscan = on;  -- always restore this
```

You should now see `Bitmap Index Scan on idx_businesses_details_gin` in the plan. The GIN index will pay for itself in production as the table grows.

What GIN indexes: The default `jsonb_ops` operator class indexes every key and value in the document, supporting `@>`, `?`, `?|`, `?&`. A `jsonb_path_ops` class indexes only values (not keys), producing a smaller index that is faster for `@>` but cannot accelerate `?` queries. Use `jsonb_path_ops` when you only ever query by value containment and index size matters.

```
-- Alternative: value-only index (smaller, faster @>, no ? support)
CREATE INDEX idx_businesses_details_pathops
  ON businesses
  USING GIN (details jsonb_path_ops);
```

Exercise 5 — Updating Documents in Place

JSONB columns are immutable at the document level — you cannot change a single key without rewriting the whole value. PostgreSQL gives you two tools to do that rewrite concisely.

5.1 — `jsonb_set`: replace or add one value

The Lighthouse Bookshop just received a flood of five-star reviews. Update its rating to 5.0:

```
UPDATE businesses
SET   details = jsonb_set(details, '{rating}', '5.0')
WHERE name = 'Lighthouse Bookshop';
```

Verify:

```
SELECT name, details ->> 'rating' AS rating
FROM   businesses
WHERE  name = 'Lighthouse Bookshop';
```

`jsonb_set(target, path, new_value)` — the path is a text array of keys. If the key already exists it is replaced; if it does not exist it is added (by default — there is a fourth parameter to suppress creation).

5.2 — `jsonb_set` on a nested key

The Gilded Clam is now closing an hour later on Sundays:

```
UPDATE businesses
SET   details = jsonb_set(
      details,
      '{hours,sun,close}',
```

```

        "21:00" -- note: a JSON string must be double-quoted
    )
WHERE name = 'The Gilded Clam';

```

```

SELECT name,
       details #>> '{hours,sun,close}' AS new_sunday_close
FROM   businesses
WHERE  name = 'The Gilded Clam';

```

5.3 — || merge operator: add or overwrite multiple keys at once

The `||` operator merges two JSONB objects, with the right side winning on conflicts. Use it to add a `verified` flag and update `review_count` in a single statement:

```

UPDATE businesses
SET   details = details || '{"verified": true, "review_count": 999}'
WHERE name = 'Anchor & Oar Tavern';

```

```

SELECT name,
       details ->> 'verified' AS verified,
       details ->> 'review_count' AS reviews
FROM   businesses
WHERE  name = 'Anchor & Oar Tavern';

```

Caution: `||` does a *shallow* merge. If `details` contains a nested object like `hours` and you merge another object that also has `hours`, the entire `hours` value is replaced — not deep-merged. Use `jsonb_set` for targeted nested updates.

Exercise 6 — Expanding Arrays with `jsonb_array_elements`

The `tags` key in most business documents is a JSON array. To query across tags — for example, to find the most common ones across the whole directory — you need to expand the array into individual rows.

6.1 — Expand one business's tags

```

SELECT name,
       jsonb_array_elements_text(details -> 'tags') AS tag
FROM   businesses
WHERE  name = 'The Gilded Clam';

```

name	tag
The Gilded Clam	waterfront
The Gilded Clam	seafood
The Gilded Clam	romantic


```
The Gilded Clam | outdoor_seating
(4 rows)
```

`jsonb_array_elements_text` is a set-returning function: each element of the array becomes its own row. The result has more rows than the input.

6.2 — Count tags across all businesses

```
SELECT tag, COUNT(*) AS occurrences
FROM   businesses,
       jsonb_array_elements_text(details -> 'tags') AS tag
GROUP BY tag
ORDER BY occurrences DESC
LIMIT 15;
```

tag	occurrences
waterfront	10
takeaway	5
family_friendly	4
family_owned	4
organic	3
...	

(15 rows)

Your exact numbers will differ — this shows the query pattern. The comma between `businesses` and the `jsonb_array_elements_text(...)` call is a **lateral join** (PostgreSQL expands the function for each input row automatically when used in the `FROM` clause this way).

6.3 — Only businesses with multiple specific tags

Find restaurants that are both `vegan_options` and `takeaway`:

```
SELECT name, details ->> 'cuisine' AS cuisine
FROM   businesses
WHERE  details @> '{"tags": ["vegan_options"]}'
AND    details @> '{"tags": ["takeaway"]}';
```

Because each `@>` call is independent, both conditions must hold. This is more efficient than expanding the array when the GIN index is in place.

Exercise 7 — Path Queries with `jsonb_path_query`

The `@?` operator and `jsonb_path_query()` function implement **SQL/JSON path language** — a mini-query language for navigating and filtering within a JSONB document. It is more expressive than chained arrow operators for conditional navigation.

7.1 — Simple path existence

Find businesses where the social links include Instagram:

```
SELECT name
FROM   businesses
WHERE  details @? '$.social.instagram';
```

`$.social.instagram` means: start at the root (`$`), descend to `social`, then to `instagram`. `@?` returns true if the path resolves to at least one value.

7.2 — Filter on a nested value

Find restaurants with a rating above 4.5:

```
SELECT name,
       details ->> 'rating' AS rating
FROM   businesses
WHERE  details @? '$ ? (@.category == "restaurant" && @.rating > 4.5)';
```

name	rating
Bella Napoli	4.6
Le Petit Bistro	4.7
River Bend Bakery	4.8
Spice Garden	4.6
The Gilded Clam	4.5

(5 rows)

The `?` (filter) syntax inside a path expression is equivalent to a `WHERE` clause inside the document.

7.3 — Find businesses open late on Friday

Businesses that close at or after 21:00 on Friday (using string comparison, which works correctly for 24-hour `HH:MM` strings):

```
SELECT name,
       details #>> '{hours,fri,close}' AS friday_close
FROM   businesses
```

```
WHERE details @? '$.hours.fri.close ? (@ >= "21:00")'
ORDER BY friday_close DESC, name;
```

name	friday_close
Bella Napoli	23:00
Dragon Palace	23:00
Harbour View Theater	23:00
The Hungry Scholar	23:00
Spice Garden	23:00
Sol y Mar	23:00
Mango Bay Caribbean	22:30
The Gilded Clam	22:30
Thai Orchid	22:00
The Riverside Vegan	22:00
Northgate Grocers	21:00
Lotus Spa & Wellness	20:00

(12 rows)

Limitation to notice: Businesses that close *after midnight* — Anchor & Oar, The Clocktower Pub, The Rusty Anchor, and others — show a `close` of `"01:00"` or `"02:00"`. Lexicographically, `"02:00" < "21:00"`, so these businesses are *excluded* from the above results even though they are open very late. This is a design trade-off of storing times as plain strings. One solution is to store closing times past midnight as `"25:00"`, `"26:00"`, etc. — an unusual but practical convention for 24-hour time arithmetic.

7.4 — Extract values with `jsonb_path_query`

`@?` is a boolean test. `jsonb_path_query()` returns the actual matched values:

```
SELECT name,
       jsonb_path_query(details, '$.hours.fri.close') AS friday_close_json
FROM   businesses
WHERE  details @? '$.hours.fri'
ORDER BY name
LIMIT 8;
```

The returned values are JSONB (note the quotes around the time strings). Use `jsonb_path_query_first(...)` `#>> '{}'` to extract a single scalar as text without the surrounding quotes.

Summary — What You Should Now Know

You have worked through the core JSONB toolkit. Here is what each operator and function you used actually does:

Tool	What it does
<code>-> 'key'</code>	Navigate one level; returns JSONB
<code>->> 'key'</code>	Navigate one level; returns text
<code>#>> '{a,b,c}'</code>	Navigate a path; returns text ; JSON null becomes SQL NULL
<code>@> '{"k":"v"}'</code>	Containment test; accelerated by GIN
<code>? 'key'</code>	Key exists test (regardless of value); accelerated by GIN
<code>? / ?&</code>	Any-key / all-keys exist; accelerated by GIN
<code>jsonb_typeof(val)</code>	Returns 'object', 'array', 'string', 'number', 'boolean', or 'null'; SQL NULL for a missing key
<code>jsonb_set(col, path, val)</code>	Replace or insert a nested value
<code>col \ '{"k":"v"}'</code>	Shallow-merge a document
<code>jsonb_array_elements_text(col)</code>	Expand a JSON array into rows
<code>@? 'path'</code>	SQL/JSON path existence test
<code>jsonb_path_query(col, 'path')</code>	SQL/JSON path — return matched values

Remember: JSONB `null` $\neq$ SQL NULL. Use `jsonb_typeof()` — not `IS NOT NULL` — when you need to distinguish a missing key from a key that exists but holds a JSON null value.

The key design insight from this chapter is that JSONB lets you keep a relational spine — the columns your queries always need (`id`, `name`, `neighbourhood`) — while storing everything else in a document whose shape can vary row by row. The GIN index means you do not pay a query performance penalty for that flexibility.

In the next chapter you will add point geometry to the `businesses` table and use PostGIS to answer spatial questions: which businesses are within 500 metres of the harbour, and which neighbourhood does each one belong to.

Going further: PostgreSQL 14+ supports the `jsonpath` type natively. The `jsonb_path_query_array` and `jsonb_path_query_first` variants are useful for pagination and

single-value extraction. For write-heavy workloads, profile whether JSONB or a computed stored column (Chapter 16) gives better INSERT/UPDATE throughput on your hardware.

Chapter 2 — PostGIS: Geospatial Queries on Real Geometry

“A city is not a list of rows. It is a shape on the ground.”

Background

Every interesting question about a city is ultimately a spatial question. Which businesses are near the harbour? Which neighbourhood is this address in? How large is the industrial waterfront? Relational databases answer these badly when location is stored as a text field or a pair of float columns — there is no native concept of “within”, “contains”, or “distance”.

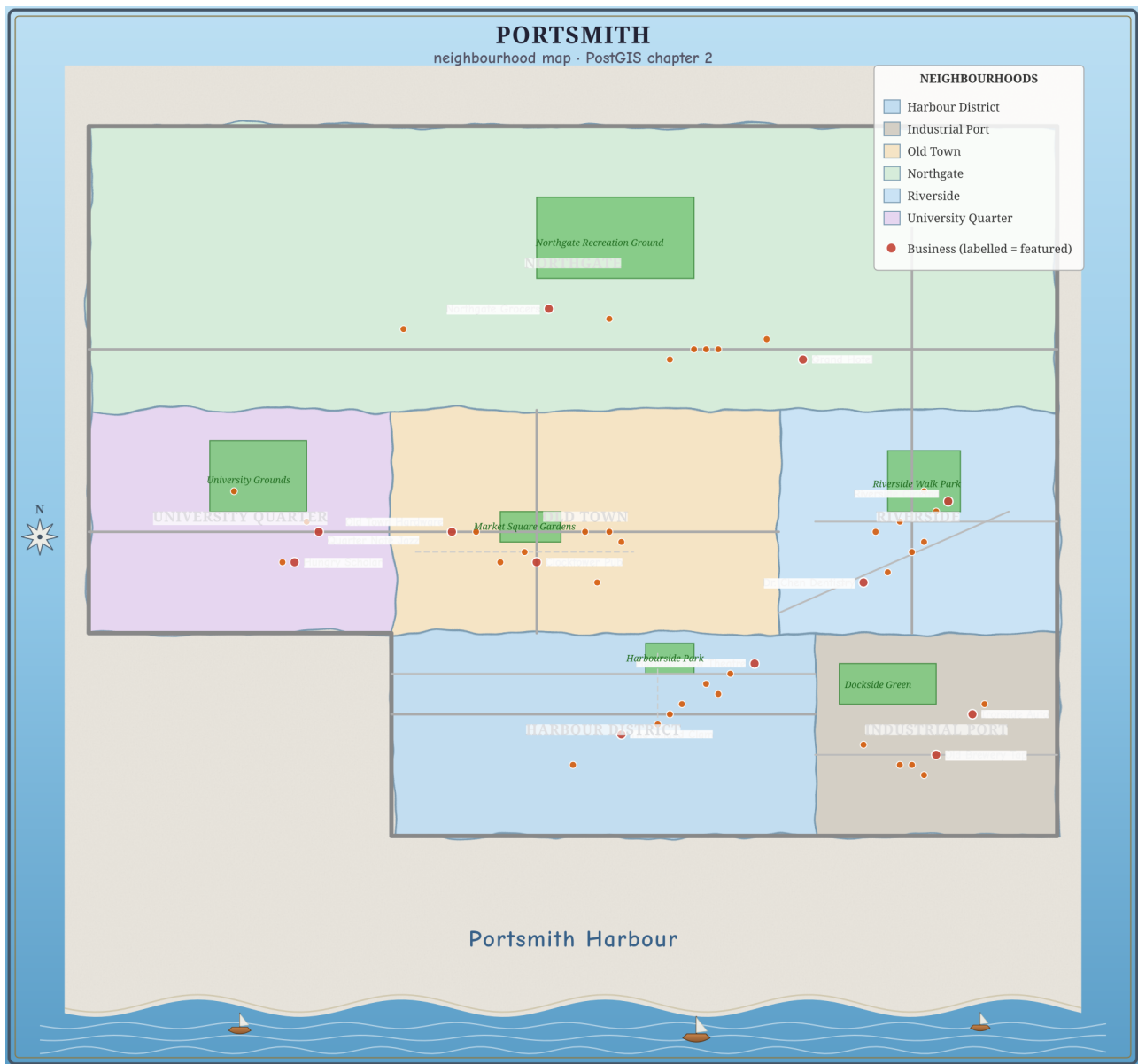
PostGIS is a PostgreSQL extension that adds first-class geometry and geography types, plus several hundred spatial functions and operators. It turns PostgreSQL into a full spatial database: you can store points, lines, and polygons; index them with GIST; and answer proximity, containment, and area queries in SQL without an external GIS system.

This matters in practice far beyond mapping applications. Address geocoding, logistics routing, fraud detection (is this login coming from the expected region?), real estate valuation, and urban planning all reach for spatial queries. PostGIS is the standard tool for all of them in the PostgreSQL ecosystem.

The Scenario

The Portsmouth business directory from Chapter 1 stores each business’s neighbourhood as a plain text column. That works for simple filtering, but it cannot answer *where* questions: it cannot find businesses near a given coordinate, cannot verify that a business address actually falls inside its declared neighbourhood, and cannot measure distances.

This chapter adds a point geometry to every business record, then introduces three new spatial tables:



Portsmouth's six neighbourhoods, parks, and featured businesses — rendered directly from this chapter's own neighborhoods, parks, and businesses tables via `utils/render_map.py`. Every shape and dot on this map is a row you're about to query.

Table	Geometry type	What it holds
neighborhoods	POLYGON	Boundary polygons for Portsmouth's six neighbourhoods
parks	POLYGON	Six public parks and green spaces
city_infrastructure	LINESTRING	Twelve named road segments

All coordinates are in WGS-84 (SRID 4326), the same coordinate system used by GPS and most web mapping APIs.

Exercise Goals

By the end of this chapter you will be able to:

- Store and inspect `POLYGON` geometry using `WKT` (Well-Known Text) and `ST_GeomFromText`.
 - Run proximity searches with `ST_DWithin`, and understand why the `::geography` cast matters for distance accuracy.
 - Perform spatial joins using `ST_Within` and `ST_Contains`, and explain the difference between them.
 - Compute polygon areas in square kilometres using `ST_Area` with the `geography` type.
 - Find the nearest feature for every row using `ST_Distance` and a `CROSS JOIN LATERAL`.
 - Create a GIST spatial index and confirm that PostgreSQL uses it.
-

Installation

1 — PostGIS server package

PostGIS is a separate package from PostgreSQL. On Debian/Ubuntu:

```
sudo apt install -y postgresql-16-postgis-3
```

2 — Enable PostGIS in the database

Connect to the `portsmith` database and enable the extension:

```
psql portsmith
```

```
CREATE EXTENSION IF NOT EXISTS postgis;
```

Verify it loaded:

```
SELECT postgis_full_version();
```

You should see a long string beginning with `POSTGIS="3.x.x"`. If you see an error about the extension not existing, the server package is not installed.

Loading the Data

Prerequisites

Chapter 1's seed script must have been run first — the `businesses` table must exist. If it does not:

```
python data/ch01_seed.py
```

Run the Chapter 2 seed

```
python data/ch02_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Enabling PostGIS extension ...
Applying DDL ...
Inserting 6 neighbourhoods ...
Inserting 6 parks ...
Inserting 12 road segments ...
Updating 48 business locations ...

Done:
  businesses with geometry : 48
  neighbourhoods           : 6
  parks                    : 6
  road segments            : 12
```

Verify the load

Open `psql portsmith` and run these four checks.

Check 1 — businesses now has a geometry column:

```
\d businesses
```

You should see a new `geom` column of type `geometry(Point,4326)`.

Check 2 — neighbourhood table structure and row count:

```
SELECT name, population,
       ST_AsText(geom) AS wkt_preview
```

```
FROM neighborhoods
ORDER BY name;
```

name	population	wkt_preview
Harbour District	4200	POLYGON((-1.805 50.69,...))
Industrial Port	2100	POLYGON((-1.77 50.69,...))
Northgate	18500	POLYGON((-1.83 50.732,...))
Old Town	6800	POLYGON((-1.805 50.71,...))
Riverside	9300	POLYGON((-1.773 50.71,...))
University Quarter	11200	POLYGON((-1.83 50.71,...))

(6 rows)

Check 3 — parks and roads:

```
SELECT COUNT(*) FROM parks;
SELECT COUNT(*) FROM city_infrastructure;
```

Both should return 6 and 12 respectively.

Check 4 — SRID on all tables:

```
SELECT f_table_name, f_geometry_column, srid, type
FROM geometry_columns
WHERE f_table_schema = 'public'
ORDER BY f_table_name;
```

f_table_name	f_geometry_column	srid	type
businesses	geom	4326	POINT
city_infrastructure	geom	4326	LINESTRING
neighborhoods	geom	4326	POLYGON
parks	geom	4326	POLYGON

(4 rows)

If all four pass, proceed to the exercises.

Exercises

Exercise 1 — Neighbourhood Polygons and WKT

Geometry in PostGIS is often loaded from **Well-Known Text** (WKT), a human-readable representation of shapes. Understanding WKT lets you read geometry from SQL output, write it in queries, and reason about what you stored.

1.1 — Read a polygon in WKT

```
SELECT name, ST_AsText(geom) AS wkt
FROM neighborhoods
WHERE name = 'Harbour District';
```

name	wkt
Harbour District	POLYGON((-1.805 50.69,-1.77 50.69,-1.77 50.71,...))

WKT for a polygon is `POLYGON((x1 y1, x2 y2, ...))`. In geographic coordinates the convention is *longitude latitude* (x then y), matching the X/Y convention in maths. The ring must **close** — the last coordinate must equal the first.

1.2 — Inspect the bounding box

`ST_Envelope` returns the minimum bounding rectangle for any geometry:

```
SELECT name,
       ST_XMin(ST_Envelope(geom)) AS west,
       ST_XMax(ST_Envelope(geom)) AS east,
       ST_YMin(ST_Envelope(geom)) AS south,
       ST_YMax(ST_Envelope(geom)) AS north
FROM neighborhoods
ORDER BY name;
```

Use this output to confirm each polygon sits in the right part of the coordinate space (longitudes around -1.75 to -1.83 , latitudes around 50.69 to 50.76).

1.3 — Count vertices

```
SELECT name,
       ST_NPoints(geom) AS vertex_count
FROM neighborhoods
ORDER BY name;
```

Each neighbourhood polygon has five vertices (four corners plus the closing duplicate). A real city boundary imported from an OS/census shapefile might have thousands.

1.4 — Understand the SRID

`ST_SRID` returns the spatial reference identifier stored with the geometry:

```
SELECT name, ST_SRID(geom) AS srid
FROM neighborhoods
LIMIT 3;
```

SRID 4326 is the WGS-84 system used by GPS. Every geometry in these tables carries the same SRID, which means they can be compared and joined directly. If SRIDs differ, PostGIS will return an error — a deliberate safety check.

The geometry/geography distinction: In PostGIS, `geometry` stores coordinates in whatever units the SRS defines. For SRID 4326 that means degrees. When you ask for a distance between two `geometry` points in 4326, you get a value in *degrees* — nearly useless for human-scale distances. The `geography` type, by contrast, always works on a spheroid and returns distances in *metres*. You can cast any SRID-4326 geometry to geography with `::geography` to get metre-based calculations. The exercises use this cast throughout.

Exercise 2 — Proximity Search with `ST_DWithin`

`ST_DWithin(a, b, distance)` returns true when the distance between `a` and `b` is at most `distance`. With geography inputs the distance is in metres.

2.1 — Find businesses within 500 m of Portsmouth Pier

Portsmouth's main pier entrance sits at approximately `(-1.785, 50.700)`. Find every business within 500 metres of it:

```
SELECT b.name,
       b.neighbourhood,
       ROUND(
         ST_Distance(b.geom::geography,
                     ST_Point(-1.785, 50.700)::geography)::numeric
       ) AS distance_m
FROM   businesses b
WHERE  ST_DWithin(b.geom::geography,
                  ST_Point(-1.785, 50.700)::geography,
                  500)
ORDER BY distance_m;
```

name	neighbourhood	distance_m
The Gilded Clam	Harbour District	70
Anchor & Oar Tavern	Harbour District	179
Tidal Wave Surf Shop	Harbour District	307
Harbour Inn	Harbour District	437
Portsmouth Fish Market	Harbour District	484

(5 rows)

All five results are in the Harbour District — exactly what you would expect for a search centred on the pier.

2.2 — Why `::geography` matters

Try the same query without the cast. Note that `ST_Point(...)` without an explicit SRID gets SRID 0, which PostGIS refuses to compare against SRID-4326 geometry — so you must use `ST_SetSRID`:

```
SELECT b.name,
       ROUND(ST_Distance(b.geom,
                        ST_SetSRID(ST_Point(-1.785, 50.700), 4326))::numeric,
            6) AS distance_degrees
FROM   businesses b
WHERE  ST_DWithin(b.geom,
                ST_SetSRID(ST_Point(-1.785, 50.700), 4326),
                500)
ORDER BY distance_degrees;
```

The `WHERE` clause now has a threshold of 500 — but in *degrees*. One degree of latitude is about 111 km, so this matches businesses within roughly 55,000 km: all 48 of them, from the whole city. The `distance_degrees` values in the output are tiny fractions like 0.001000, not useful numbers.

rows returned: 48 (the entire city, not 5)

Rule: For any distance query on SRID-4326 data, always cast to `::geography` in `ST_DWithin` and `ST_Distance`. The cast has negligible performance cost at the scales used in city-level data.

2.3 — Try different radii

How many businesses fall within 1 km of the pier? 2 km?

```
SELECT radius_m,
       COUNT(*) AS business_count
FROM   (VALUES (500), (1000), (2000)) AS radii(radius_m)
CROSS JOIN LATERAL (
    SELECT 1
    FROM    businesses b
    WHERE   ST_DWithin(b.geom::geography,
                    ST_Point(-1.785, 50.700)::geography,
                    radius_m)
) AS matches
GROUP BY radius_m
ORDER BY radius_m;
```

radius_m	business_count
500	5
1000	8
2000	17

(3 rows)

The pier sits in the southern Harbour District. Doubling the radius to 2 km pulls in more of the central neighbourhoods, but the northern Northgate businesses are nearly 5 km away — a reminder that Portsmouth stretches a significant distance north from the waterfront.

Exercise 3 — Spatial Joins with `ST_Within` and `ST_Contains`

A **spatial join** links rows from two tables based on a geometric relationship rather than a key match. The most common are containment tests: does point A fall inside polygon B?

3.1 — Which neighbourhood is each business in?

```
SELECT b.name,
       b.neighbourhood AS declared_neighbourhood,
       n.name          AS postgis_neighbourhood
FROM   businesses b
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom)
ORDER BY n.name, b.name;
```

`ST_Within(a, b)` returns true when geometry a lies completely inside geometry b. The query should return all 48 businesses, each matched to its neighbourhood.

Confirm the declared `neighbourhood` column always matches `postgis_neighbourhood`:

```
SELECT COUNT(*) AS mismatches
FROM   businesses b
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom)
WHERE  b.neighbourhood <> n.name;
```

mismatches
0

(1 row)

Zero mismatches: the text column from Chapter 1 and the geometry are consistent.

3.2 — `ST_Contains` is the inverse

`ST_Contains(a, b)` returns true when polygon a contains geometry b — it is the mirror of `ST_Within`. These two queries produce identical results:

```
-- Point inside polygon
SELECT b.name FROM businesses b
JOIN  neighborhoods n ON ST_Within(b.geom, n.geom)
WHERE n.name = 'Old Town'
ORDER BY b.name;

-- Polygon contains point
SELECT b.name FROM businesses b
JOIN  neighborhoods n ON ST_Contains(n.geom, b.geom)
WHERE n.name = 'Old Town'
ORDER BY b.name;
```

```
      name
-----
Bella Napoli
Finch & Sons Barbers
Le Petit Bistro
Old Town Hardware
Portsmouth Accountancy Ltd.
Portsmouth Arms Hotel
Portsmouth Legal Group
Portsmouth Tailors
The Clocktower Pub
(9 rows)
```

Both return the same 9 Old Town businesses.

The subtle difference: `ST_Contains(A, B)` requires that no point of B lies on the boundary of A. A point sitting exactly *on* a polygon edge would fail `ST_Contains` but pass `ST_Covers`. In practice, coordinates rarely land precisely on a boundary, so the two functions behave identically for most point-in-polygon work. Reach for `ST_Covers` if you need to include boundary-touching cases explicitly.

3.3 — Check for gaps: businesses in no neighbourhood

This query finds any businesses whose geometry falls outside every neighbourhood polygon — useful for catching data quality problems:

```
SELECT b.name, b.neighbourhood
FROM  businesses b
WHERE NOT EXISTS (
    SELECT 1
    FROM    neighborhoods n
```



```
WHERE ST_Within(b.geom, n.geom)
);
```

```
name | neighbourhood
-----+-----
(0 rows)
```

All 48 businesses are inside a neighbourhood polygon.

Exercise 4 — Computing Area in Square Kilometres

`ST_Area` returns the area of a polygon. Called on `geometry` (degrees), it returns a value in square degrees — meaningless to most people. Called on `geography`, it returns square metres.

4.1 — Incorrect: area in square degrees

```
SELECT name,
       ROUND(ST_Area(geom)::numeric, 6) AS area_sq_degrees
FROM   neighborhoods
ORDER BY area_sq_degrees DESC;
```

The numbers look tiny (around 0.0007). They are geometrically correct but impossible to interpret as real-world area because degrees are not uniform units.

4.2 — Correct: area in square metres, then kilometres

```
SELECT name,
       ROUND((ST_Area(geom::geography) / 1e6)::numeric, 2) AS area_km2
FROM   neighborhoods
ORDER BY area_km2 DESC;
```

```
name      | area_km2
-----+-----
Northgate |    17.59
Old Town  |     5.53
Harbour District |    5.50
University Quarter |    4.32
Riverside |     3.98
Industrial Port |    3.14
(6 rows)
```

Northgate is by far the largest neighbourhood — it spans the entire northern width of the city. Industrial Port is the smallest, a tight strip of dockside land. Old Town and Harbour District are almost identical in area despite having very different characters.

Why the numbers differ slightly from manual estimates: `ST_Area` on geography uses the WGS-84 spheroid, which accounts for the fact that the Earth is not a perfect sphere. At latitude 50.7° the correction is small (under 0.3%) but present.

4.3 — Population density

Combine the computed area with the `population` column:

```
SELECT name,
       population,
       ROUND((ST_Area(geom::geography) / 1e6)::numeric, 2) AS area_km2,
       ROUND(
         (population / (ST_Area(geom::geography) / 1e6))::numeric
       ) AS pop_per_km2
FROM   neighborhoods
ORDER BY pop_per_km2 DESC;
```

name	population	area_km2	pop_per_km2
University Quarter	11200	4.32	2592
Riverside	9300	3.98	2340
Old Town	6800	5.53	1230
Northgate	18500	17.59	1052
Harbour District	4200	5.50	763
Industrial Port	2100	3.14	668

(6 rows)

The University Quarter and Riverside are the most densely populated neighbourhoods — student housing and riverside apartments pack a lot of people into compact areas. The Industrial Port is the least dense despite its small size; much of it is warehouse and dock rather than residential.

Exercise 5 — Nearest Park Using `ST_Distance` and a Lateral Join

Finding the nearest feature from another table requires a **lateral join** — a subquery that can reference columns from the outer query row by row.

5.1 — Distance from one business to one park

`ST_Distance` between two geography values returns metres:

```
SELECT b.name          AS business,
       p.name          AS park,
```

```

        ROUND(ST_Distance(b.geom::geography,
                           p.geom::geography)::numeric) AS distance_m
FROM    businesses b,
        parks p
WHERE   b.name = 'The Gilded Clam'
ORDER BY distance_m;

```

business	park	distance_m
The Gilded Clam	Harbourside Park	682
The Gilded Clam	Dockside Green	1315
The Gilded Clam	Market Square Gardens	2143
The Gilded Clam	Riverside Walk Park	2899
The Gilded Clam	University Grounds	3060
The Gilded Clam	Northgate Recreation Ground	5006

(6 rows)

The nearest park to The Gilded Clam is Harbourside Park, 682 m away. The `ST_Distance` to a polygon returns the distance from the point to the nearest point on the polygon's boundary — zero when the point is inside the polygon.

5.2 — Nearest park for a single business (lateral pattern)

The canonical pattern for “nearest one thing” is `ORDER BY ... LIMIT 1` inside a lateral subquery:

```

SELECT b.name,
       nearest.park_name,
       nearest.distance_m
FROM    businesses b
CROSS JOIN LATERAL (
    SELECT p.name                                AS park_name,
           ROUND(ST_Distance(b.geom::geography,
                              p.geom::geography)::numeric) AS distance_m
    FROM    parks p
    ORDER BY b.geom::geography <-> p.geom::geography
    LIMIT 1
) AS nearest
WHERE   b.name = 'Quarter Note Jazz Club';

```

name	park_name	distance_m
Quarter Note Jazz Club	University Grounds	233

(1 row)

The `<->` operator is the KNN (k-nearest-neighbour) distance operator. When used in an `ORDER BY` clause inside a lateral join, PostgreSQL can accelerate it with the GIST index (which

you will create in Exercise 6). Using `<->` in `ORDER BY` is preferred over `ORDER BY ST_Distance(...)` for this reason.

5.3 — Nearest park for every business

Remove the `WHERE` filter to run across all 48 businesses:

```
SELECT b.name                                AS business,
       b.neighbourhood,
       nearest.park_name,
       nearest.distance_m
FROM   businesses b
CROSS JOIN LATERAL (
  SELECT p.name                                AS park_name,
         ROUND(ST_Distance(b.geom::geography,
                           p.geom::geography)::numeric) AS distance_m
  FROM   parks p
  ORDER BY b.geom::geography <-> p.geom::geography
  LIMIT 1
) AS nearest
ORDER BY b.neighbourhood, b.name;
```

Scan the results. You should see a clean pattern: each neighbourhood’s businesses point to the park in that same neighbourhood. Notice that three Riverside businesses (Portsmouth Pharmacy, Portsmouth Veterinary Clinic, Riverside Cinema) and University Bookshop all show `distance_m = 0` — their coordinates fall *inside* the park polygon. `ST_Distance` to a polygon returns zero when the point is contained within it.

5.4 — Aggregate: average walking distance to a park, by neighbourhood

```
SELECT b.neighbourhood,
       ROUND(AVG(nearest.distance_m)::numeric) AS avg_distance_m
FROM   businesses b
CROSS JOIN LATERAL (
  SELECT ROUND(ST_Distance(b.geom::geography,
                           p.geom::geography)::numeric) AS distance_m
  FROM   parks p
  ORDER BY b.geom::geography <-> p.geom::geography
  LIMIT 1
) AS nearest
GROUP BY b.neighbourhood
ORDER BY avg_distance_m;
```

This gives a rough “walkability to green space” metric per neighbourhood. Old Town ranks first — Market Square Gardens sits centrally within the neighbourhood. Northgate

comes last despite having the largest park, because the park is in the far north of the district while businesses cluster near the southern edge.

Exercise 6 — GIST Indexes and Spatial Query Plans

Without an index, every spatial query is a sequential scan: PostgreSQL reads every row, applies the geometry function, and discards non-matching rows. For a 48-row dataset that is instant, but at 48 million rows it is not.

PostGIS spatial queries are accelerated by **GIST** (Generalised Search Tree) indexes. A GIST index on a geometry column builds a tree of bounding boxes, allowing the planner to quickly eliminate large portions of the table.

6.1 — Observe the plan before indexing

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom:geography,
                ST_Point(-1.785, 50.700)::geography,
                500);
```

On the 48-row `businesses` table you will see something like:

```
Seq Scan on businesses  (cost=0.00..4.60 rows=1 width=...) ...
  Filter: (st_dwithin(...))
  Rows Removed by Filter: 43
```

A sequential scan is expected on a tiny table — the planner knows the overhead of an index lookup exceeds the cost of reading all 48 rows.

6.2 — Create GIST indexes on the geometry columns

```
CREATE INDEX idx_businesses_geom
  ON businesses USING GIST (geom);

CREATE INDEX idx_neighborhoods_geom
  ON neighborhoods USING GIST (geom);

CREATE INDEX idx_parks_geom
  ON parks USING GIST (geom);

CREATE INDEX idx_city_infrastructure_geom
  ON city_infrastructure USING GIST (geom);
```

6.3 — The geometry/geography index split

With `enable_seqscan` off, verify which plan the proximity query uses:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom::geography,
                 ST_Point(-1.785, 50.700)::geography,
                 500);

SET enable_seqscan = on;
```

You will see — perhaps surprisingly — that the planner *still* uses a sequential scan, even though `idx_businesses_geom` exists:

```
Seq Scan on businesses  (cost=10000000000.00...) ...
  Filter: st_dwithin((geom)::geography, ...)
```

Why? The GIST index was built on `geom` (type `geometry`). The query filters on `geom::geography` (type `geography`). These are different types — the index is not usable for geography operations.

To accelerate geography-based distance queries you need a **functional index** on the cast:

```
CREATE INDEX idx_businesses_geom_geography
ON businesses USING GIST (CAST(geom AS geography));
```

Now with `enable_seqscan` off:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom::geography,
                 ST_Point(-1.785, 50.700)::geography,
                 500);

SET enable_seqscan = on;
```

```
Index Scan using idx_businesses_geom_geography on businesses
  Index Cond: ((geom)::geography &&
               _st_expand('...', '500'))
  Filter: st_dwithin((geom)::geography, ...)
  Rows Removed by Filter: 2
```

The index is now used. The index condition uses `&&` (bounding-box overlap on the spheroid) to quickly discard most rows, then `st_dwithin` rechecks the exact distance for the survivors.

6.4 — Verify the spatial join plan

The containment join uses geometry-to-geometry operations, so `idx_businesses_geom` (the plain geometry index) *is* used there:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT b.name, n.name AS neighbourhood
FROM   businesses b
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom);

SET enable_seqscan = on;
```

Nested Loop ...

```
-> Seq Scan on neighborhoods n  (6 rows - tiny table)
-> Index Scan using idx_businesses_geom on businesses b
    Index Cond: (geom @ n.geom)
    Filter: st_within(geom, n.geom)
```

For each neighbourhood polygon, `idx_businesses_geom` is used to find businesses whose bounding box falls inside the neighbourhood's bounding box (`geom @ n.geom`), then `st_within` rechecks the exact containment.

The index rule: A GIST index on `geom` geometry accelerates geometry-to-geometry operations (`ST_Within`, `ST_Contains`, `ST_Intersects`, `&&`). A GIST index ON `CAST(geom AS geography)` accelerates geography operations (`ST_DWithin(...::geography, ...)`, `<->` on geography). If you query both ways, create both indexes — they coexist happily on the same table.

Summary — What You Should Now Know

You have worked through the core PostGIS toolkit for points, polygons, and linear features. Here is a reference for everything used:

Function / operator

`ST_GeomFromText(wkt, srid)`

What it does

Parse WKT into a geometry with the given SRID

Function / operator

`ST_AsText(geom)`

`ST_SetSRID(ST_MakePoint(lon, lat), srid)`

`ST_Point(lon, lat)::geography`

`geom::geography`

`ST_SRID(geom)`

`ST_NPoints(geom)`

`ST_Envelope(geom)`

`ST_XMin/XMax/YMin/YMax(geom)`

`ST_DWithin(a, b, d)`

`ST_Distance(a, b)`

`a::geography <-> b::geography`

`ST_Within(a, b)`

`ST_Contains(a, b)`

`ST_Area(geom::geography)`

`CROSS JOIN LATERAL (... LIMIT 1)`

`USING GIST (geom)`

What it does

Format geometry as WKT for display

Construct a point geometry

Construct a point and cast to geography

Cast a 4326 geometry to geography (distances now in metres)

Return the SRID stored with a geometry

Count vertices in a geometry

Return the bounding-box rectangle

Extract bounding-box extents

True when the distance between a and b is $\leq d$

Distance between two geometries (metres for geography)

KNN distance operator; use in ORDER BY for index acceleration

True when a lies completely inside b

True when a contains b (inverse of ST_Within)

Area in square metres on the spheroid

Nearest-neighbour join pattern

Create a GIST spatial index

Geometry vs geography in one sentence: Use `geometry` for storage and when working with projected coordinate systems where units are already metres or feet. Cast to `geography` any time you need distance, area, or proximity results in real-world units from SRID-4326 data.

The coordinates added in this chapter will carry forward. Chapter 12 extends the `city_infrastructure` road network into a graph and uses a recursive CTE to find the shortest path between two intersections.

Going further: PostGIS supports many more geometry types — `MULTIPOLYGON` for areas with holes, `GEOMETRYCOLLECTION` for mixed types, and 3D geometries with a Z coordinate for elevation data. For routing specifically, `pgRouting` builds on PostGIS to provide Dijkstra, A, and turn-restriction-aware shortest-path algorithms over road network graphs. For importing real boundary data, `shp2pgsql` converts ESRI Shapefiles directly into PostGIS-compatible `INSERT` statements, and `ogr2ogr` handles GeoJSON, KML, GeoPackage, and dozens of other formats.*

Chapter 3 — Job Queues: `FOR UPDATE SKIP LOCKED`

“A queue is just a table that everyone is racing to read.”

Background

Sooner or later almost every application needs a queue: a list of work items that a pool of workers processes one at a time, safely, without two workers ever grabbing the same item. The reflexive answer is to reach for a message broker — Redis, RabbitMQ, SQS, Kafka. Those tools earn their keep at serious scale. But if your data already lives in PostgreSQL, running a second system just to hand out rows to workers is often unnecessary complexity: another service to deploy, monitor, and keep consistent with the database.

PostgreSQL can do this job itself. The `FOR UPDATE SKIP LOCKED` row-locking clause, combined with an ordinary table, gives you an atomic “claim the next item and don’t let anyone else touch it” primitive — the same guarantee a dedicated queue product sells you, built out of two SQL keywords. This chapter builds a job queue from scratch: the schema, the atomic claim query, concurrent worker behaviour, stalled-job recovery, and a dead-letter path for jobs that keep failing.

This is not a toy exercise. This exact pattern — a status column, a claim query, a heartbeat, a dead-letter table — is what libraries like [river](#), [oban](#) (Elixir), and countless in-house job runners implement on top of PostgreSQL in production.

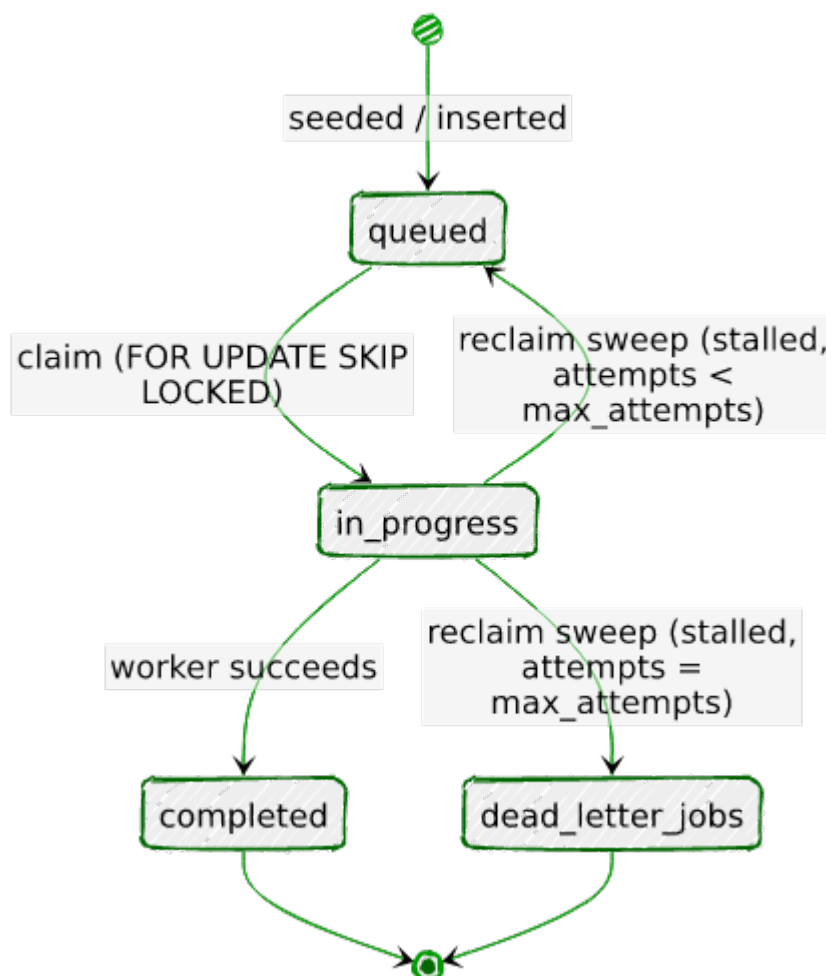
The Scenario

Portsmouth’s permitting office processes a steady stream of permit applications: building work, business licenses, public events, signage, and demolitions. Each application needs to move through a review pipeline, and the office wants that processing to happen asynchronously and reliably — work should never be lost, never double-processed, and a crashed reviewer process shouldn’t leave an application stuck in limbo forever.

The `jobs` table models this as a queue. Every row is one permit application awaiting review. A `status` column tracks its life cycle, a `priority` column lets safety-critical work (demolitions) jump ahead of routine work (sign permits), and a retry counter with a companion `dead_letter_jobs` table handles applications whose processing keeps failing.

Column	Purpose
<code>status</code>	<code>queued</code> → <code>in_progress</code> → <code>completed</code> (or back to <code>queued</code> , or <code>dead-lettered</code>)
<code>priority</code>	1 (most urgent — demolitions) to 5 (least urgent — sign permits)
<code>payload</code>	JSONB — the permit application details
<code>attempts</code> / <code>max_attempts</code>	Retry bookkeeping
<code>claimed_by</code> / <code>claimed_at</code>	Which worker has the job, and since when
<code>heartbeat_at</code>	Updated periodically by the worker while it holds the job

The `status` column's life cycle, drawn out — this is what Exercises 2 through 5 each implement one piece of:



Every arrow here is a specific query you'll write by hand later in this chapter — there's no hidden state machine enforcing this, just the `status` column, the claim query, and the reclaim sweep agreeing on what each value means.

No extensions are required for this chapter — everything here is built on core PostgreSQL locking semantics.

Exercise Goals

By the end of this chapter you will be able to:

- Design a queue table schema, including a partial index tuned for the claim query's access pattern.
 - Write the atomic `UPDATE ... FROM (SELECT ... FOR UPDATE SKIP LOCKED)` claim query, and explain why naive two-step approaches race.
 - Observe, with two concurrent `psql` sessions, that `SKIP LOCKED` lets workers proceed past rows another worker is holding — and contrast that with the blocking behaviour of plain `FOR UPDATE`.
 - Implement a heartbeat/timeout mechanism that reclaims jobs abandoned by a crashed worker.
 - Route jobs that exhaust their retries into a dead-letter table.
 - Benchmark claim throughput at different concurrency levels with `pgbench`.
-

Installation

This chapter needs nothing beyond what Chapter 1 already set up: PostgreSQL 16 and a Python 3.12 virtual environment with `psycopg`. If you skipped Chapter 1, see its Installation section. You will also use `pgbench` for Exercise 6, which ships with the standard PostgreSQL client tools (`postgresql-client-16` on Debian/Ubuntu).

Loading the Data

Run the seed script

From the `book/` directory, with the virtual environment active:

```
python data/ch03_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 45 jobs ...
Done - 45 rows in jobs, all queued.
```

The seed script is self-contained — it does not depend on Chapter 1 or 2’s data.

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d jobs
```

Table "public.jobs"					Default
Column	Type	Collation	Nullable		
id	bigint		not null		
nextval('jobs_id_seq'::regclass)					
job_type	text		not null		
payload	jsonb		not null		
status	text		not null		'queued'::text
priority	smallint		not null		5
attempts	integer		not null		0
max_attempts	integer		not null		3
created_at	timestamp with time zone		not null		clock_timestamp()
claimed_at	timestamp with time zone				
claimed_by	text				
heartbeat_at	timestamp with time zone				
completed_at	timestamp with time zone				
last_error	text				
Indexes:					
"jobs_pkey" PRIMARY KEY, btree (id)					
"idx_jobs_claim_order" btree (priority, created_at, id) WHERE status =					

```
'queued'::text
    "idx_jobs_status" btree (status)
```

Check constraints:

```
"jobs_status_check" CHECK (status = ANY (ARRAY['queued'::text, 'in_progress'::text,
'completed'::text, 'failed'::text]))
```

Check 2 — job counts by type and priority:

```
SELECT job_type, priority, COUNT(*) AS jobs
FROM jobs
GROUP BY job_type, priority
ORDER BY priority;
```

job_type	priority	jobs
demolition_permit	1	4
business_license	2	12
building_permit	3	15
event_permit	4	8
sign_permit	5	6

(5 rows)

Check 3 — everything starts queued:

```
SELECT status, COUNT(*) FROM jobs GROUP BY status;
```

status	count
queued	45

(1 row)

If all three match, proceed to the exercises.

Note: If you re-run `ch03_seed.py` at any point during the exercises to reset to a clean state, it drops and recreates both `jobs` and `dead_letter_jobs`.

Exercises

Exercise 1 — Designing the Queue Schema

1.1 — Why a partial index

The claim query (which you'll write in Exercise 2) only ever looks at rows where `status = 'queued'`, ordered by `priority` then `created_at`. As the queue runs, the vast majority of rows will end up `completed` — a normal B-tree index on `(priority, created_at, id)` would faithfully index every one of those settled rows even though the claim query never looks at them.

`idx_jobs_claim_order` is a **partial index** — it only indexes rows matching `WHERE status = 'queued'`:

```
CREATE INDEX idx_jobs_claim_order
  ON jobs (priority, created_at, id)
  WHERE status = 'queued';
```

This keeps the index small regardless of how many historical jobs pile up in `completed` or `failed` state, because settled rows are never in it.

1.2 — Why `id` is part of the sort key, not just `created_at`

You might expect `ORDER BY priority, created_at` to be enough — oldest job in the highest-priority bucket goes first. But timestamps are not always unique. If two jobs are inserted in the same transaction, both can get an identical `created_at` (more on this below), and `ORDER BY` over tied values has no defined order. Appending the primary key, `id`, as a final tiebreaker guarantees a deterministic order even when timestamps collide:

```
ORDER BY priority ASC, created_at ASC, id ASC
```

A gotcha worth knowing: `now()` returns the **transaction's** start time, not the current wall-clock time — every call to `now()` inside the same transaction returns the same value. The seed script originally used `created_at TIMESTAMPTZ DEFAULT now()`, and because all 45 rows were inserted in one transaction, every single row ended up with an *identical* `created_at`. The fix is `clock_timestamp()`, which returns the actual current time at the moment it's evaluated, differing row to row even within one transaction. `jobs.created_at` uses `clock_timestamp()` for exactly this reason. This is the same family of surprise as the JSONB `null` vs. SQL `NULL` gotcha from Chapter 1: a function name that looks interchangeable with another is not.

1.3 — Confirm the index is used

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id
FROM jobs
WHERE status = 'queued'
```



```
ORDER BY priority ASC, created_at ASC, id ASC
FOR UPDATE SKIP LOCKED
LIMIT 1;
```

```
Limit (cost=0.14..6.17 rows=1 width=24) (actual time=0.022..0.022 rows=1 loops=1)
  Buffers: shared hit=3
  -> LockRows (cost=0.14..12.20 rows=2 width=24) (actual time=0.021..0.021 rows=1
loops=1)
    Buffers: shared hit=3
    -> Index Scan using idx_jobs_claim_order on jobs (cost=0.14..12.18 rows=2
width=24) (actual time=0.010..0.010 rows=1 loops=1)
      Filter: (status = 'queued'::text)
      Buffers: shared hit=2
```

Now drop the index and run the identical query again inside a transaction you roll back (so the drop doesn't stick):

```
BEGIN;
DROP INDEX idx_jobs_claim_order;

EXPLAIN (ANALYZE, BUFFERS)
SELECT id
FROM jobs
WHERE status = 'queued'
ORDER BY priority ASC, created_at ASC, id ASC
FOR UPDATE SKIP LOCKED
LIMIT 1;

ROLLBACK;
```

```
Limit (cost=9.51..9.53 rows=1 width=24) (actual time=0.071..0.072 rows=1 loops=1)
  -> LockRows (cost=9.51..9.54 rows=2 width=24) (actual time=0.070..0.071 rows=1
loops=1)
    -> Sort (cost=9.51..9.52 rows=2 width=24) (actual time=0.067..0.068 rows=1
loops=1)
      Sort Key: priority, created_at, id
      Sort Method: quicksort Memory: 27kB
      -> Bitmap Heap Scan on jobs (cost=4.16..9.50 rows=2 width=24) (actual
time=0.018..0.027 rows=45 loops=1)
        Recheck Cond: (status = 'queued'::text)
        -> Bitmap Index Scan on idx_jobs_status (cost=0.00..4.16 rows=2
width=0) (actual time=0.013..0.013 rows=45 loops=1)
          Index Cond: (status = 'queued'::text)
```

Without `idx_jobs_claim_order`, PostgreSQL still finds the queued rows (via `idx_jobs_status`), but it must then **sort all of them** to find the one with the lowest (`priority, created_at, id`) — an extra Sort node. With the tailored partial index, the rows are already stored in claim order, so PostgreSQL walks the index and stops at the first

match. On a 45-row table the difference is invisible; on a busy production queue with a deep backlog, eliminating the sort on every single claim matters a great deal.

Exercise 2 — The Atomic Claim Query

2.1 — Why a naive two-step claim races

The tempting first approach is to `SELECT` a candidate row, then `UPDATE` it in a second statement:

```
-- ⚠ Do not do this - it has a race condition
SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority, created_at, id LIMIT 1;
-- ... application reads id = 7 ...
UPDATE jobs SET status = 'in_progress' WHERE id = 7;
```

Between the `SELECT` and the `UPDATE`, nothing stops a second worker from running the exact same `SELECT`, reading the same `id = 7`, and also issuing the `UPDATE`. Both workers now believe they own job 7. This is a classic **check-then-act race condition** — the gap between reading and acting is exactly where two workers can interleave.

2.2 — `FOR UPDATE` closes the gap, but blocks

Locking the row as part of the `SELECT` closes the race:

```
SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority, created_at, id
FOR UPDATE LIMIT 1;
```

Now a second worker running the same query, in a separate transaction, **blocks** until the first worker's transaction commits or rolls back — correct, but it means every worker but one sits idle waiting for a lock instead of moving on to a different job. Section 3.1 demonstrates this.

2.3 — `SKIP LOCKED` lets workers move past each other

Adding `SKIP LOCKED` tells PostgreSQL: if the next candidate row is already locked by another transaction, don't wait for it — skip it and consider the row after it.

```
SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority, created_at, id
FOR UPDATE SKIP LOCKED LIMIT 1;
```

This is the piece that makes PostgreSQL usable as a concurrent queue: N workers can run this query at the same instant and each will walk away with a *different* row, with no blocking and no double-claims.

2.4 — The full atomic claim: lock, update, and return in one statement

Locking the row is only half the job — you still need to mark it `in_progress` before releasing the lock, and you want the whole thing to happen as a single round trip. Combine the `SELECT ... FOR UPDATE SKIP LOCKED` with the `UPDATE` using a CTE:

```
WITH next_job AS (  
    SELECT id  
    FROM jobs  
    WHERE status = 'queued'  
    ORDER BY priority ASC, created_at ASC, id ASC  
    FOR UPDATE SKIP LOCKED  
    LIMIT 1  
)  
UPDATE jobs  
SET     status      = 'in_progress',  
        claimed_at  = now(),  
        claimed_by  = 'demo-worker',  
        heartbeat_at = now(),  
        attempts    = attempts + 1  
FROM next_job  
WHERE jobs.id = next_job.id  
RETURNING jobs.id, jobs.job_type, jobs.payload --> 'application_id' AS application_id,  
           jobs.priority, jobs.attempts, jobs.status;
```

id	job_type	application_id	priority	attempts	status
1	demolition_permit	DP-2024-0001	1	1	in_progress

(1 row)

This single statement is the entire claim operation: find the best candidate, skip anything locked, mark it claimed, and hand back its data — atomically, with no window for a race. This is exactly the query `data/ch03_worker.py` runs (see `CLAIM_SQL`). Run it again and the demolition permit at `id = 2` comes back next — `id = 1` is now `in_progress`, so it's no longer a candidate.

Roll this back if you ran it directly in `psql` and want to keep the queue clean for later exercises: wrap it in `BEGIN; ... ROLLBACK;`.

Exercise 3 — Simulating Concurrent Workers

3.1 — SKIP LOCKED vs. plain FOR UPDATE, in two psql sessions

Open two terminals with `psql portsmith` in each. In **Session A**, start a transaction, claim 5 rows, and hold the transaction open (don't commit yet):

```
-- Session A
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE SKIP LOCKED LIMIT 5;
```

```
id |      job_type
----+-----
 1 | demolition_permit
 2 | demolition_permit
 3 | demolition_permit
 4 | demolition_permit
 5 | business_license
(5 rows)
```

Leave that transaction open. In **Session B**, run the identical query:

```
-- Session B (Session A is still open, holding locks on ids 1-5)
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE SKIP LOCKED LIMIT 5;
```

```
id |      job_type
----+-----
 6 | business_license
 7 | business_license
 8 | business_license
 9 | business_license
10 | business_license
(5 rows)
```

Session B returns immediately with a **different** set of rows — it silently skipped 1–5 because they were locked, and moved on to the next five unlocked candidates. Commit or roll back both sessions to release the locks:

```
COMMIT; -- run in both sessions
```

3.2 — Now try it with plain FOR UPDATE (no SKIP LOCKED)

Repeat the same two-session experiment, but drop `SKIP LOCKED`:

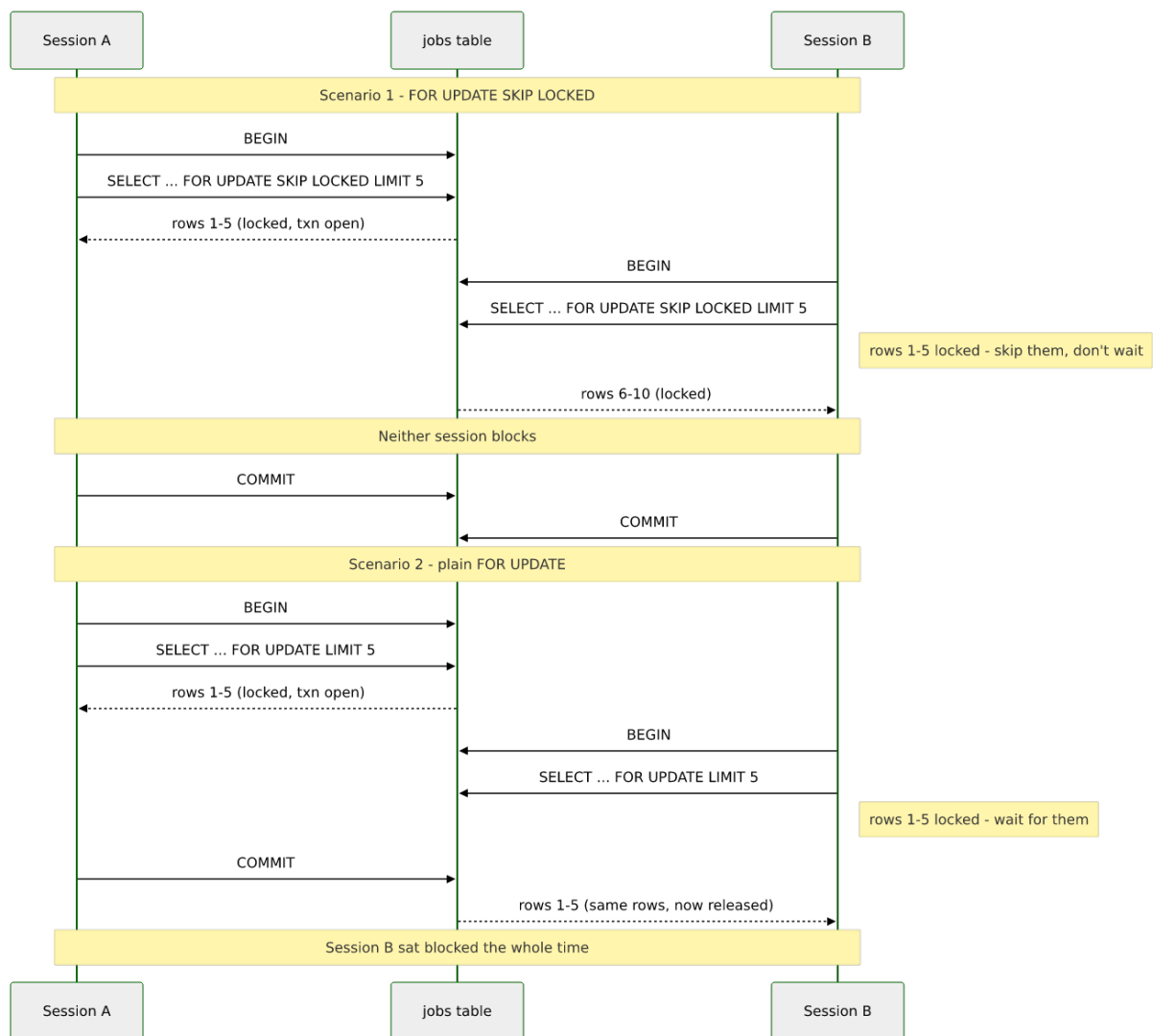
```
-- Session A
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE LIMIT 5;
-- (leave this transaction open)
```

```
-- Session B - this will hang
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE LIMIT 5;
```

Session B does not return. It is blocked, waiting for Session A's row locks to be released. Only once Session A runs `COMMIT` (or `ROLLBACK`) does Session B's query complete — and when it does, it returns the *same* five rows Session A had, now that they're unlocked again:

```
-- (Session B, after Session A commits)
id |      job_type
----+-----
 1 | demolition_permit
 2 | demolition_permit
 3 | demolition_permit
 4 | demolition_permit
 5 | business_license
(5 rows)
```

This is the difference in one sentence: **plain `FOR UPDATE` serializes workers through the lock; `SKIP LOCKED` lets them fan out across the table.** For a job queue, you always want the latter — a blocked worker is a wasted worker. Both runs side by side, as a timeline:



3.3 — Real concurrent workers with `ch03_worker.py`

Reset the data (`python data/ch03_seed.py`) if you ran the manual claim in Exercise 2.4 without rolling it back. Then launch two workers in the background, each capped to 10 jobs so they finish quickly:

```
python data/ch03_worker.py --worker-id w1 --max-jobs 10 --fail-rate 0 &
python data/ch03_worker.py --worker-id w2 --max-jobs 10 --fail-rate 0 &
wait
```

Each prints a running log as it claims and completes jobs, e.g.:

```
[w1] claimed job 2 (demolition_permit, attempt 1/3): DP-2024-0002 - processing for 0.09s
[w1] job 2 completed
[w2] claimed job 1 (demolition_permit, attempt 1/3): DP-2024-0001 - processing for 0.12s
[w2] job 1 completed
```

Once both finish, confirm the split was clean — 20 jobs completed total, no job claimed by both workers:

```
SELECT claimed_by, COUNT(*) FROM jobs WHERE status = 'completed'
GROUP BY claimed_by ORDER BY claimed_by;
```

claimed_by	count
w1	10
w2	10

(2 rows)

```
SELECT status, COUNT(*) FROM jobs GROUP BY status;
```

status	count
completed	20
queued	25

(2 rows)

10 + 10 = 20, matching exactly the `--max-jobs 10` cap on each worker — no overlaps, no double-processing, no lost jobs.

Exercise 4 — Heartbeat and Stalled-Job Recovery

Claiming a job is only half the reliability story. What happens if the worker that claimed a job crashes, gets OOM-killed, or loses its network connection halfway through? The row is stuck at `status = 'in_progress'` forever unless something notices and puts it back.

4.1 — The heartbeat column

`heartbeat_at` exists for exactly this. A well-behaved worker updates it periodically while it holds a job (see the `while` loop in `process_job()` in `ch03_worker.py`, which sends a heartbeat roughly every 2 seconds during simulated processing). If a job has been `in_progress` for a long time *and* its heartbeat has gone stale, that's a strong signal the worker holding it is dead — a live worker would have updated it.

4.2 — Simulate a crashed worker

Pick any queued job and manually walk it through what a real claim would do, then simulate a crash by never sending a heartbeat again:

```
UPDATE jobs
SET    status = 'in_progress', claimed_at = now(),
      claimed_by = 'worker-crashed', heartbeat_at = now(),
      attempts = attempts + 1
WHERE id = 21;
```

```
-- Simulate 5 minutes of silence from the "crashed" worker
UPDATE jobs SET heartbeat_at = now() - interval '5 minutes' WHERE id = 21;
```

4.3 — Run the reclaim sweep

```
python data/ch03_reclaim.py --timeout 30
```

```
Connecting to: dbname=portsmith
  job 21 (building_permit) stalled - requeued (attempt 1)

Done: 1 requeued, 0 dead-lettered.
```

The sweep looks for `in_progress` rows whose `heartbeat_at` is older than `--timeout` seconds (`FIND_STALLED_SQL` in `ch03_reclaim.py`), and since this job's attempts (1) is still below `max_attempts` (3), it goes back to `queued`:

```
SELECT id, status, attempts, claimed_by, heartbeat_at, last_error
FROM   jobs WHERE id = 21;
```

id	status	attempts	claimed_by	heartbeat_at	last_error
21	queued	1			stalled: no heartbeat since ... (last claimed by worker-crashed)

Notice `attempts` stayed at 1 — the reclaim did not reset it. The attempt the crashed worker used is still counted; the job doesn't get a free retry just because the worker that lost it never got to report failure honestly.

4.4 — Why the timeout, not an outright deadline

A fixed timeout on the *last heartbeat* (rather than on total processing time) lets jobs run arbitrarily long as long as they keep proving they're alive. A 10-minute report-generation job with a 30-second heartbeat interval will never be mistaken for stalled, while a worker that dies mid-task is detected within one missed heartbeat window. Run `ch03_reclaim.py` again immediately — it correctly finds nothing to do, since the reclaimed job now has a fresh `queued` state and no stale `in_progress` row exists:

```
No jobs stalled beyond 30s - nothing to do.
```

Exercise 5 — Dead-Lettering Exhausted Jobs

Some jobs will never succeed no matter how many times you retry them — a malformed application, a permanently invalid address, a bug that only triggers on one particular payload. Retrying forever wastes worker time and can mask a real problem. Once a job has used its last attempt, it belongs in `dead_letter_jobs`: out of the active queue, but preserved for a human to inspect.

5.1 — The `dead_letter_jobs` schema

```
\d dead_letter_jobs
```

Table "public.dead_letter_jobs"				
Column	Type	Collation	Nullable	Default
id	bigint		not null	
job_type	text		not null	
payload	jsonb		not null	
priority	smallint		not null	
attempts	integer		not null	
max_attempts	integer		not null	
created_at	timestamp with time zone		not null	
last_error	text			
failed_at	timestamp with time zone		not null	now()

It mirrors `jobs` but adds `failed_at` and drops the in-flight columns (`status`, `claimed_by`, `heartbeat_at`) that no longer mean anything once a job is out of the active queue.

5.2 — Push a job past its retry limit

Continuing with job 21 from Exercise 4 (now back to `queued` with `attempts = 1`), fast-forward it to its last attempt and let it stall again:

```
UPDATE jobs
SET   status = 'in_progress', claimed_at = now(),
      claimed_by = 'worker-crashed-2', heartbeat_at = now() - interval '5 minutes',
      attempts = max_attempts
WHERE id = 21;
```

```
python data/ch03_reclaim.py --timeout 30
```

```
Connecting to: dbname=portsmith
job 21 (building_permit) exhausted retries - dead-lettered
```

```
Done: 0 requeued, 1 dead-lettered.
```

This time `attempts (3)` is no longer less than `max_attempts (3)`, so the sweep's `DEAD_LETTER_SQL` runs instead: it `DELETES` the row from `jobs` and `INSERTS` it into `dead_letter_jobs` in one CTE, so the row is never visible in neither-place or both-places, even under concurrent access.

```
SELECT id FROM jobs WHERE id = 21;
```

```
id
---
(0 rows)
```

```
SELECT id, job_type, payload ->> 'application_id' AS application_id,
       attempts, max_attempts, last_error, failed_at
FROM   dead_letter_jobs WHERE id = 21;
```

id	job_type	application_id	attempts	max_attempts	last_error	failed_at
21	building_permit	BP-2024-0005	3	3	stalled: no heartbeat since ... (last claimed ...)	2026-07-12 22:15:45.613583-04

(1 row)

The exact same dead-lettering logic runs inline inside `ch03_worker.py` when a job fails organically (not via a stall) on its last attempt — see `DEAD_LETTER_SQL` in that script. Whether a job dies from an explicit failure or from going silent, it ends up in the same place, with a record of why.

Exercise 6 — Benchmarking Claim Throughput with `pgbench`

6.1 — A benchmark script that doesn't drain the queue

`pgbench` repeatedly runs a SQL script against the database for a fixed duration, from any number of concurrent client connections — exactly the tool for measuring how the claim query holds up under contention. The catch: if the script just claims a job and leaves it claimed, 45 rows disappear from the pool in well under a second and every worker after that finds nothing to do. `data/ch03_claim_bench.sql` works around this by claiming a job and then **immediately releasing it back to `queued`** in the same transaction:

```
BEGIN;

WITH next_job AS (
  SELECT id
  FROM   jobs
```

```

    WHERE status = 'queued'
    ORDER BY priority ASC, created_at ASC, id ASC
    FOR UPDATE SKIP LOCKED
    LIMIT 1
)
UPDATE jobs
SET   status      = 'in_progress',
      claimed_at   = now(),
      claimed_by   = 'bench-' || :client_id,
      heartbeat_at = now()
FROM   next_job
WHERE  jobs.id = next_job.id
RETURNING jobs.id AS claimed_id \gset

UPDATE jobs
SET   status = 'queued', claimed_at = NULL, claimed_by = NULL, heartbeat_at = NULL
WHERE id = :claimed_id;

COMMIT;

```

`:client_id` is a `pgbench` built-in variable — the number of the simulated client running this script. `\gset` captures the claimed row's `id` into a `pgbench` variable (`:claimed_id`) so the release step can target it. This keeps the pool at a constant 45 claimable rows for the whole benchmark run, so throughput reflects lock contention rather than the queue running dry.

6.2 — Run at increasing concurrency

```

pgbench -n -c 1 -j 1 -T 5 -f data/ch03_claim_bench.sql portsmith
pgbench -n -c 4 -j 4 -T 5 -f data/ch03_claim_bench.sql portsmith
pgbench -n -c 16 -j 16 -T 5 -f data/ch03_claim_bench.sql portsmith

```

`-c` is the number of simulated concurrent workers (clients), `-j` the number of `pgbench` threads driving them, and `-T 5` runs each test for 5 seconds. Results on the machine used to write this chapter:

Clients (`-c`) tps Avg. latency

1	799	1.25 ms
4	1,202	3.33 ms
16	2,570	6.23 ms

Your numbers will differ with your hardware — the shape of the result is what matters. Throughput climbs with concurrency because `SKIP LOCKED` lets additional workers keep finding unlocked rows to claim instead of queueing up behind each other; latency per

transaction also climbs because more workers are contending for the same 45-row table and for CPU. Try dropping `idx_jobs_claim_order` (inside a `BEGIN; ... ROLLBACK;` block, as in Exercise 1.3) and rerunning the `-c 16` case — you should see tps fall, since every claim now pays for a `sort` over all queued rows instead of an index walk.

6.3 — Where this stops being enough

`pgbench` here is exercising claim contention on 45 rows — a stand-in for “the working set the claim query actually scans.” A real permitting-office queue might hold a very different shape of data (thousands of queued rows, heavy skew toward one `priority` bucket), and the honest way to benchmark your own workload is to seed a table that resembles it, not a 45-row toy. Chapter 20 (`pg_stat_statements` and Query Performance) picks this thread back up: rather than a synthetic benchmark, it measures and fixes real slow queries observed in production traffic.

Summary — What You Should Now Know

You built a working, concurrency-safe job queue entirely out of core PostgreSQL features. Here is a reference for the pieces:

Tool	What it does
<code>FOR UPDATE</code>	Locks selected rows; other transactions selecting the same rows block until released
<code>FOR UPDATE SKIP LOCKED</code>	Locks selected rows; other transactions skip already-locked rows instead of blocking
<code>WITH next_job AS (... FOR UPDATE SKIP LOCKED LIMIT 1)</code> <code>UPDATE ... FROM next_job</code>	The atomic claim pattern: find, lock, and mark a row in one round trip
Partial index <code>WHERE status = 'queued'</code>	Keeps the claim-path index small regardless of how many settled rows accumulate
<code>clock_timestamp()</code> VS. <code>now()</code>	<code>now()</code> freezes at transaction start; <code>clock_timestamp()</code> reflects real wall-clock time on every call
<code>heartbeat_at + timeout</code> sweep	Detects and recovers jobs abandoned by a crashed worker
<code>dead_letter_jobs</code>	Removes permanently-failing jobs from the active queue while preserving them for inspection

Tool

```
pgbench -f script.sql
```

What it does

Benchmarks a custom SQL workload at a chosen concurrency level

The key design insight from this chapter is that a reliable queue is not one clever query — it's the combination of an atomic claim, a way to detect workers that go silent, and a place for work that genuinely cannot succeed. PostgreSQL's row-locking primitives give you the first two almost for free; the dead-letter table is just a normal table.

The `jobs` table you built here is reused directly in later chapters: Chapter 13 adds a trigger that fires `NOTIFY` on status changes so listeners can react to queue activity in real time, Chapter 14 uses advisory locks alongside it for leader-election patterns, and Chapter 19 schedules the reclaim sweep you wrote by hand in Exercise 4 to run automatically with `pg_cron`.

Going further: the pattern in this chapter — status column, atomic claim, heartbeat, dead-letter — is exactly what PostgreSQL-backed job-queue libraries like `river` (Go) and `oban` (Elixir) implement, with more polish around scheduling, uniqueness constraints, and observability. If you outgrow a single-database queue — cross-database work distribution, guaranteed ordering at very high throughput, or consumer groups — that's the point at which a dedicated broker starts to earn its operational cost. For most applications below that scale, the table you just built is enough.

Chapter 4 — Full-Text Search: `tsvector`, Stopwords, and Ranking

“Grep finds characters. Full-text search finds meaning — or at least gets a lot closer.”

Background

`LIKE '%harbour%'` finds the literal substring “harbour”. It will not find “Harbour”, “harbours”, or a document about “the harbor” (different spelling) unless you handle every variation yourself, and it cannot tell you whether a match in the title is more relevant than a match buried in paragraph six. As soon as an application needs to search prose — meeting minutes, support tickets, product descriptions, articles — substring matching stops being enough. The usual reflex is to reach for Elasticsearch or a similar dedicated search engine.

PostgreSQL has had a full-text search engine built in since version 8.3. It tokenizes text into words, reduces those words to normalized root forms (*stemming*), discards low-information words like “the” and “of” (*stopwords*), and stores the result in a specialized `tsvector` type that a GIN index can search in milliseconds. A companion `tsquery` type lets you express boolean and phrase searches, and a family of ranking functions score how well each match fits the query. None of this requires an extension — it is core PostgreSQL, the same engine `websearch_to_tsquery`-powered search boxes on production sites are built on.

This chapter builds a search feature over a small document archive: how text becomes a `tsvector`, what stopwords removal actually throws away, how to keep the vector in sync as rows change, how query syntax differs between a developer-facing boolean query language and a plain-text search box, and how to rank and highlight results. The final exercise customizes the stopwords list itself — because in a *city* government’s document archive, the words “city” and “council” appear so often they stop being useful signals, exactly the kind of domain-specific tuning full-text search is designed to support.

The Scenario

Portsmouth’s city clerk publishes an archive of public records: council meeting minutes, zoning ordinances, and public notices. Residents need to search this archive by keyword — “what happened with the Riverside dog park?”, “which ordinances mention Canal Road?” — and get back the most relevant documents first, with a highlighted snippet showing why each one matched.

The `city_documents` table holds all three document types in one place, with a `body` column of plain prose text and a few relational columns for filtering:

Column	Purpose
<code>doc_type</code>	<code>council_minutes</code> , <code>zoning_ordinance</code> , Or <code>public_notice</code>
<code>department</code>	Which city department published it (City Council, Public Works, ...)
<code>title</code>	Short document title
<code>body</code>	Full plain-text document body — the field full-text search runs over
<code>published_date</code>	When the document was published

The documents share recurring topics on purpose — the Riverside dog park, the Harbour District waterfront rezoning, the Canal Road bike lane — so that searches in the exercises return more than one hit, and ranking has something real to sort between. No extensions are required for this chapter: `tsvector`, `tsquery`, and every function used below are core PostgreSQL.

Exercise Goals

By the end of this chapter you will be able to:

- Convert text to a `tsvector` and read the lexeme:position notation it produces.
- Explain exactly what stopwords removal and stemming do to a document, using `ts_debug` to see PostgreSQL’s tokenizer classify each word.
- Maintain a `tsvector` column automatically with a trigger, and index it with GIN for sub-millisecond searches.
- Write boolean queries with `to_tsquery` and understand why `plainto_tsquery` is the safer choice for a plain search box.

- Rank results with `ts_rank` and `ts_rank_cd`, and generate highlighted snippets with `ts_headline`.
 - Build a custom text search configuration that treats domain-specific words as stopwords, and see it change real search results.
-

Installation

This chapter needs nothing beyond what Chapter 1 already set up: PostgreSQL 16 and a Python 3.12 virtual environment with `psycopg`. If you skipped Chapter 1, see its Installation section. Full-text search is core PostgreSQL — there is no extension to enable.

Loading the Data

Run the seed script

From the `book/` directory, with the virtual environment active:

```
python data/ch04_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 30 documents ...
Done - 30 rows in city_documents.
```

The seed script is self-contained — it does not depend on any earlier chapter’s data. Note that it deliberately does **not** create a `tsvector` column or index; you build both by hand in Exercise 3, which is the point of the chapter.

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d city_documents
```

Table "public.city_documents"					Default
Column	Type	Collation	Nullable		
-----+-----+-----+-----					
+-----+-----+-----+-----					
id	integer		not null		
nextval('city_documents_id_seq'::regclass)					
doc_type	text		not null		
department	text		not null		
title	text		not null		
body	text		not null		
published_date	date		not null		
Indexes:					
"city_documents_pkey" PRIMARY KEY, btree (id)					
"idx_city_documents_doc_type" btree (doc_type)					
"idx_city_documents_published_date" btree (published_date)					
Check constraints:					
"city_documents_doc_type_check" CHECK (doc_type = ANY					
(ARRAY['council_minutes'::text, 'zoning_ordinance'::text, 'public_notice'::text]))					

Check 2 — counts by document type:

<pre> SELECT doc_type, COUNT(*) AS documents FROM city_documents GROUP BY doc_type ORDER BY doc_type; </pre>									
<table> <tr> <th>doc_type</th><th>documents</th></tr> <tr> <td>council_minutes</td><td>10</td></tr> <tr> <td>public_notice</td><td>10</td></tr> <tr> <td>zoning_ordinance</td><td>10</td></tr> </table>	doc_type	documents	council_minutes	10	public_notice	10	zoning_ordinance	10	(3 rows)
doc_type	documents								
council_minutes	10								
public_notice	10								
zoning_ordinance	10								

Check 3 — counts by department:

<pre> SELECT department, COUNT(*) AS documents FROM city_documents GROUP BY department ORDER BY department; </pre>													
<table> <tr> <th>department</th><th>documents</th></tr> <tr> <td>City Council</td><td>9</td></tr> <tr> <td>Finance</td><td>1</td></tr> <tr> <td>Parks & Recreation</td><td>3</td></tr> <tr> <td>Planning & Zoning</td><td>12</td></tr> <tr> <td>Public Works</td><td>5</td></tr> </table>	department	documents	City Council	9	Finance	1	Parks & Recreation	3	Planning & Zoning	12	Public Works	5	(5 rows)
department	documents												
City Council	9												
Finance	1												
Parks & Recreation	3												
Planning & Zoning	12												
Public Works	5												

If all three match, proceed to the exercises.

Exercises

Exercise 1 — Converting Text to `tsvector`

1.1 — A basic conversion

`to_tsvector` is the function that turns plain text into PostgreSQL's searchable representation. Try it on one document's body:

```
SELECT title FROM city_documents WHERE id = 1;
```

```
title
```

```
-----  
Council Minutes – Harbour District Waterfront Renovation Budget
```

```
SELECT to_tsvector('english', body) FROM city_documents WHERE id = 1;
```

```
'along':38 'approv':67 'begin':79 'boardwalk':34 'bond':53 'budget':10 'citi':3  
'construct':76 'conven':5 'council':4,41,61 'cover':28 'debat':56 'director':18  
'discuss':43 'district':14 'expect':77 'fall':82 'first':69 'fund':44,74  
'grant':50 'harbour':13,40,72 'includ':46 'infrastructur':49 'light':36  
'measur':54 'member':42 'new':32 'one':65 'pedestrian':33 'phase':26,70  
'plan':27 'portsmith':2 'present':22 'propos':9 'public':20 'renov':16,73  
'repair':30 'review':7 'seawal':29 'six':63 'sourc':45 'state':48  
'three':25 'three-phas':24 'timelin':59 'upgrad':37 'vote':62  
'waterfront':15 'work':21
```

1.2 — Reading the format

Each entry is `'lexeme':position,position,...`. A **lexeme** is a normalized word form, not the literal text — notice `'vote':62` even though the source text says “voted”, `'approv':67` for “approve”, and `'renov':16,73` for both “renovation” (position 16) and “renovation” again later (position 73, from “harbour **renovation** funding”). PostgreSQL stems words to their root so that a search for “vote” also matches documents containing “voted”, “votes”, or “voting” — you don't have to enumerate every inflection yourself.

The **positions** are the word's ordinal location in the original text (1st word, 2nd word, ...). They exist for two reasons: `ts_rank` uses them to compute how spread out or clustered a document's matches are, and phrase search (`<->`, used in Exercise 4) uses them to require words to appear adjacent to each other, not just anywhere in the document.

1.3 — Case and punctuation are already handled

Note that “Council” (capitalized, position 4) and “council” (lowercase, positions 41 and 61) both collapsed into the single lexeme ‘council’ with three positions — `to_tsvector` lowercases everything and strips punctuation as part of tokenization, before stemming ever runs.

Exercise 2 — What Stopword Removal Actually Throws Away

2.1 — Compare `simple` and `english` configurations

PostgreSQL ships several **text search configurations** — named bundles of tokenizer + dictionary rules. `simple` lowercases and tokenizes but does *no* stemming and drops *no* stopwords; `english` does both. Run the same body through each and count the resulting lexemes:

```
SELECT array_length(regexp_split_to_array(body, '\s+'), 1) AS raw_words
FROM   city_documents WHERE id = 1;
```

```
raw_words
```

```
-----
      80
```

```
SELECT array_length(tsvector_to_array(to_tsvector('simple', body)), 1) AS
lexemes_simple,
       array_length(tsvector_to_array(to_tsvector('english', body)), 1) AS
lexemes_english
FROM   city_documents WHERE id = 1;
```

```
lexemes_simple | lexemes_english
```

```
-----+-----
      59 |           49
```

80 raw whitespace-separated words collapse to 59 distinct lexemes under `simple` (repeated words like “the” and “council” count once each, but nothing is removed or stemmed) and to 49 under `english` — ten fewer, because `english` additionally discards stopwords like “the”, “a”, “of”, “to”, “in”, and “and” entirely. They carry no search-relevant meaning on their own, and indexing them would only bloat the index with entries that match nearly every document.

2.2 — Watch the tokenizer classify each word with `ts_debug`

`ts_debug` is the diagnostic function for seeing exactly what a configuration does to a piece of text, word by word — useful whenever a search isn’t matching what you expect and you need to know why. Run it on a short sentence:

```
SELECT alias, token, dictionaries, lexemes
FROM   ts_debug('english', 'The council voted to approve the budget for the harbour
renovation.')
WHERE  alias <> 'blank';
```

alias	token	dictionaries	lexemes
asciiword	The	{english_stem}	{}
asciiword	council	{english_stem}	{council}
asciiword	voted	{english_stem}	{vote}
asciiword	to	{english_stem}	{}
asciiword	approve	{english_stem}	{approv}
asciiword	the	{english_stem}	{}
asciiword	budget	{english_stem}	{budget}
asciiword	for	{english_stem}	{}
asciiword	the	{english_stem}	{}
asciiword	harbour	{english_stem}	{harbour}
asciiword	renovation	{english_stem}	{renov}

Every token gets classified (`asciiword` here — a plain word) and routed to the `english_stem` dictionary. Content words come back with a stemmed lexeme; stopwords (“The”, “to”, “the”, “for”, “the”) come back with an **empty** `lexemes` array — recognized, looked up, and explicitly discarded. That empty array is the entire mechanism: a word is a stopwords precisely when its dictionary entry maps it to nothing.

2.3 — When you’d reach for `simple` instead

`english` is the right default for prose. `simple` is useful for columns where stemming would be actively wrong — product SKUs, tag lists, or anything where “waterfront” and “waterfronts” should *not* be treated as the same token. `city_documents.body` is prose, so the rest of this chapter uses `english`.

Exercise 3 — A Maintained Column and a GIN Index

Computing `to_tsvector(body)` at query time works, but it means recomputing the same tokenization on every single search, over every row, every time. The standard pattern is to store the vector in its own column, keep it current as rows change, and index it.

3.1 — Add the column and backfill it

```
ALTER TABLE city_documents ADD COLUMN search_vector tsvector;

UPDATE city_documents
SET search_vector = to_tsvector('english', title || ' ' || body);
```

Indexing `title` alongside `body` means a search for a word that only appears in the title (not repeated in the body text) still matches.

3.2 — Keep it current with a trigger

A column populated once goes stale the moment anyone edits `title` or `body`. A `BEFORE` trigger recomputes it on every write:

```
CREATE FUNCTION city_documents_search_vector_update() RETURNS trigger AS $$
BEGIN
    NEW.search_vector := to_tsvector('english', NEW.title || ' ' || NEW.body);
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_city_documents_search_vector
    BEFORE INSERT OR UPDATE OF title, body ON city_documents
    FOR EACH ROW
    EXECUTE FUNCTION city_documents_search_vector_update();
```

`OF title, body` scopes the trigger so it only fires when one of the two source columns actually changes — an `UPDATE` that only touches `published_date` doesn't pay for a needless re-tokenization.

Note — the manual way vs. the modern way: This trigger pattern is how every PostgreSQL version has supported derived `tsvector` columns, and it's still exactly right when the derived value depends on more than one column, as it does here (`title` **and** `body`). PostgreSQL 12 added **generated columns** (`GENERATED ALWAYS AS (...) STORED`), which handle the common case — a `tsvector` derived from a single column — with less boilerplate and no trigger function to maintain by hand. Chapter 16 revisits this exact table and replaces this trigger with a generated column once `body` alone is the source. Both approaches produce an identical `tsvector`; the trigger is simply the general-purpose tool that works whenever the derivation is more than one column deep.

3.3 — Index it with GIN

```
CREATE INDEX idx_city_documents_search_vector
ON city_documents USING GIN (search_vector);
```

```
\d city_documents
```

```

Table "public.city_documents"
  Column      | Type          | Collation | Nullable | Default
-----+-----+-----+-----+-----
id            | integer      |           | not null |
nextval('city_documents_id_seq'::regclass)
doc_type      | text         |           | not null |
department    | text         |           | not null |
title         | text         |           | not null |
body          | text         |           | not null |
published_date | date         |           | not null |
search_vector | tsvector     |           |          |
Indexes:
    "city_documents_pkey" PRIMARY KEY, btree (id)
    "idx_city_documents_doc_type" btree (doc_type)
    "idx_city_documents_published_date" btree (published_date)
    "idx_city_documents_search_vector" gin (search_vector)
Check constraints:
    "city_documents_doc_type_check" CHECK (doc_type = ANY
    (ARRAY['council_minutes'::text, 'zoning_ordinance'::text, 'public_notice'::text]))
Triggers:
    trg_city_documents_search_vector BEFORE INSERT OR UPDATE OF title, body ON
    city_documents FOR EACH ROW EXECUTE FUNCTION city_documents_search_vector_update()
```

3.4 — Confirm the index is used

The @@ operator tests whether a tsvector matches a tsquery:

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'harbour & waterfront');
```

QUERY PLAN

```

Seq Scan on city_documents (cost=0.00..8.38 rows=1 width=36) (actual time=0.025..0.047
rows=5 loops=1)
  Filter: (search_vector @@ ''harbour'' & ''waterfront''::tsquery)
  Rows Removed by Filter: 25
  Buffers: shared hit=8
```

On 30 rows the planner reasonably decides a sequential scan is cheaper than an index lookup — same behaviour you saw with the JSONB GIN index in Chapter 1. Force the index to confirm it works, exactly as in that chapter:

```

SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'harbour & waterfront');

SET enable_seqscan = on;  -- always restore this

```

Query Plan

```

-----
Bitmap Heap Scan on city_documents  (cost=12.98..16.99 rows=1 width=36) (actual
time=0.024..0.033 rows=5 loops=1)
  Recheck Cond: (search_vector @@ '''harbour' & 'waterfront'''::tsquery)
  Heap Blocks: exact=5
  Buffers: shared hit=10
  -> Bitmap Index Scan on idx_city_documents_search_vector  (cost=0.00..12.98 rows=1
width=0) (actual time=0.014..0.014 rows=5 loops=1)
    Index Cond: (search_vector @@ '''harbour' & 'waterfront'''::tsquery)
    Buffers: shared hit=5

```

Bitmap Index Scan on `idx_city_documents_search_vector` confirms the GIN index is doing the work. This is what makes full-text search viable at scale: instead of tokenizing every row's text on every query, PostgreSQL looks up each query lexeme directly in the index.

Exercise 4 — `to_tsquery` VS. `plainto_tsquery`

4.1 — `to_tsquery`: a boolean query language

`to_tsquery` expects an expression using explicit operators: `&` (AND), `|` (OR), `!` (NOT), and `<->` (phrase — “immediately followed by”):

```

SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'harbour & waterfront')
ORDER BY id;

```

id	title
1	Council Minutes – Harbour District Waterfront Renovation Budget
8	Council Minutes – Harbour District Food Truck Permits
12	Zoning Ordinance – Harbour District Waterfront Height Variance
22	Public Notice – Public Hearing on Harbour District Waterfront Rezoning
30	Public Notice – Annual Fireworks Display and Street Closures

(5 rows)

! excludes a term. This finds documents about flooding that are *not* about the flood-mitigation infrastructure program specifically:

```
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'flood & !mitigation')
ORDER BY id;
```

id	title
5	Council Minutes – Riverside Road Resurfacing Program

(1 row)

That single hit mentions “flooding” (which stems to `flood`) while discussing drainage damage, but never says “mitigation” — exactly what the query asked for.

4.2 — `to_tsquery` has no forgiveness for plain text

Feed it a raw phrase with no operator between the words, and it errors instead of guessing what you meant:

```
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'flood mitigation');
```

```
ERROR:  syntax error in tsquery: "flood mitigation"
```

This is `to_tsquery`’s defining trade-off: it is a precise query language for code that constructs queries deliberately, and it is the wrong function to hand raw user input from a search box — a user typing “flood mitigation” with no operators will get an error page instead of results.

4.3 — `plainto_tsquery`: safe for a search box

`plainto_tsquery` takes plain text, tokenizes and stems it exactly like `to_tsvector` does, drops stopwords, and **ANDs** everything that survives. Feed it the same two words that made `to_tsquery` error out in 4.2, but typed the way an actual user would type them — with a stray “the” in front:

```
SELECT plainto_tsquery('english', 'the flood mitigation');
```

```
plainto_tsquery
```

```
-----
'flood' & 'mitig'
```

“the” never makes it into the query — `plainto_tsquery` drops it as a stopword, exactly the way `to_tsvector` would drop it from a document, and ANDs together whatever content words are left. Run it against the table:

```
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ plainto_tsquery('english', 'the flood mitigation')
ORDER BY id;
```

id	title
10	Council Minutes – Special Session on Flood Mitigation Infrastructure
25	Public Notice – City Council Special Session on Flood Mitigation

(2 rows)

No error this time, and no operators to get wrong — `plainto_tsquery` took a sentence fragment with a stopword in it and produced exactly the query 4.2 had to write by hand. The trade-off runs the other way from `to_tsquery`: you get safety and stopword handling for free, but you lose the ability to express OR, NOT, or phrase search — `plainto_tsquery` always ANDs every surviving word together.

4.4 — | for OR searches

```
SELECT to_tsquery('english', 'dog | bike');
```

to_tsquery

'dog' | 'bike'

```
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'dog | bike')
ORDER BY id;
```

id	title
5	Council Minutes – Riverside Road Resurfacing Program
6	Council Minutes – Riverside Dog Park Funding
9	Council Minutes – Canal Road Bike Lane Expansion
15	Zoning Ordinance – Reduced Parking Minimums Near Transit Corridors
24	Public Notice – Riverside Dog Park Ribbon-Cutting Event
28	Public Notice – Canal Road Bike Lane Construction Schedule
30	Public Notice – Annual Fireworks Display and Street Closures

(7 rows)

Document 5 (road resurfacing) and 15 (parking minimums) show up because each mentions “bike” in passing — the resurfacing minutes note the work being coordinated

with “the planned Canal Road bike lane construction,” and the parking ordinance references “the city’s ongoing bike lane expansion.” OR genuinely means *either*, which is exactly why it’s useful and why it returns more, looser matches than AND.

Exercise 5 — Ranking with `ts_rank` and `ts_rank_cd`, and `ts_headline` Snippets

`@@` tells you whether a document matches — it says nothing about *how well*. A user searching a 30-document archive can eyeball every hit, but a user searching 30,000 documents needs the best matches first.

5.1 — `ts_rank`: weight by term frequency

```
SELECT id, title, round(ts_rank(search_vector, query)::numeric, 4) AS rank
FROM   city_documents, to_tsquery('english', 'harbour & waterfront') AS query
WHERE  search_vector @@ query
ORDER BY rank DESC;
```

id	title	rank
-----+-----		
12	Zoning Ordinance – Harbour District Waterfront Height Variance	0.1986
1	Council Minutes – Harbour District Waterfront Renovation Budget	0.1959
22	Public Notice – Public Hearing on Harbour District Waterfront Rezoning	0.1895
8	Council Minutes – Harbour District Food Truck Permits	0.0999
30	Public Notice – Annual Fireworks Display and Street Closures	0.0992
(5 rows)		

`ts_rank` scores primarily on how often the query terms appear relative to the document’s overall length — it does not care where in the document they appear or how close together they are. Documents 12, 1, and 22 rank highest because “harbour” and “waterfront” are central, repeated themes in short documents; 8 and 30 rank lower because each mentions the harbour only in passing within a longer document about something else.

5.2 — `ts_rank_cd`: weight by proximity too (“cover density”)

```
SELECT id, title,
       round(ts_rank(search_vector, query)::numeric, 4) AS rank,
       round(ts_rank_cd(search_vector, query)::numeric, 4) AS rank_cd
FROM   city_documents, to_tsquery('english', 'harbour & waterfront') AS query
WHERE  search_vector @@ query
ORDER BY rank_cd DESC;
```

id	title	rank	
rank_cd			

```

-----+-----+-----
+-----
 1 | Council Minutes – Harbour District Waterfront Renovation Budget          | 0.1959 |
0.1107
22 | Public Notice – Public Hearing on Harbour District Waterfront Rezoning | 0.1895 |
0.1053
12 | Zoning Ordinance – Harbour District Waterfront Height Variance          | 0.1986 |
0.0700
30 | Public Notice – Annual Fireworks Display and Street Closures            | 0.0992 |
0.0545
 8 | Council Minutes – Harbour District Food Truck Permits                    | 0.0999 |
0.0542
(5 rows)

```

The order changes: document 12 had the *highest* `ts_rank` but drops to third under `ts_rank_cd`. `ts_rank_cd` additionally rewards matched terms that cluster close together in the text (“cover density”) — in documents 1 and 22, “harbour” and “waterfront” appear right next to each other (“Harbour District **Waterfront** Renovation...”); in document 12 they occur further apart. Neither function is universally “more correct” — `ts_rank` is the standard choice; `ts_rank_cd` is worth trying when word proximity itself signals relevance, as it often does for short phrase-like queries.

5.3 — Highlighted snippets with `ts_headline`

A ranked list of titles is useful; showing *why* a document matched, with the query terms highlighted in context, is what users actually expect from a search results page:

```

SELECT ts_headline('english', body, to_tsquery('english', 'harbour & waterfront'),
                  'StartSel=**, StopSel=**, MaxWords=25, MinWords=10')
FROM   city_documents
WHERE  id = 1;

```

```

**Harbour** District **waterfront** renovation. The Director of Public Works presented

```

`ts_headline` re-scans the original text (not the `tsvector`) looking for the query terms, extracts a window around the best-matching fragment bounded by `MinWords/MaxWords`, and wraps each match in `startSel/stopSel` markers — `**...**` here, but in a web application you’d use `<mark>...</mark>` or similar. This is genuinely expensive compared to a `tsvector` lookup (it re-tokenizes the source text on every call), so use it only on the page of results you’re actually displaying, never inside a `WHERE` clause.

Exercise 6 — A Custom Text Search Configuration for Domain Stopwords

6.1 — The problem: some “content” words carry no information here

This is a city document archive. The words “Portsmouth”, “city”, and “council” appear constantly — they are structurally present in almost every record, the way “the” is present in almost every English sentence. Standard `english` stopwords removal has no way to know that, because it is tuned for general English, not this specific corpus:

```
SELECT count(*) FROM city_documents
WHERE search_vector @@ to_tsquery('english', 'city & council');
```

```
count
```

```
-----
```

```
8
```

Eight of thirty documents — over a quarter of the entire archive — “match” a query on words that describe almost nothing about what makes any one of them relevant. Left alone, these words dilute ranking scores and inflate plain-text queries with noise terms the searcher didn’t mean to require.

6.2 — Build a stopwords file that extends the standard list

A text search configuration’s stopwords list is a plain text file, one word per line, that PostgreSQL reads from its `tsearch_data` directory. Start from the existing `english.stop` file so you keep every standard English stopwords, then append the domain-specific ones:

```
PG_SHAREDIR=$(pg_config --sharedir)
sudo bash -c "cat '$PG_SHAREDIR/tsearch_data/english.stop' > '$PG_SHAREDIR/tsearch_data/portsmouth_english.stop'"
sudo bash -c "printf 'portsmouth\ncity\ncouncil\n' >> '$PG_SHAREDIR/tsearch_data/portsmouth_english.stop'"
```

Placing files here requires root, since `tsearch_data` lives under PostgreSQL’s shared install directory rather than anything database-owned. If your platform’s PostgreSQL package stores it elsewhere, `pg_config --sharedir` will always point at the right location.

6.3 — Wire the file into a dictionary and a configuration

A stopwords file alone does nothing — it has to be attached to a **text search dictionary**, and that dictionary mapped into a **text search configuration** that queries can actually reference:

```

CREATE TEXT SEARCH DICTIONARY portsmith_stem (
    TEMPLATE = snowball,
    LANGUAGE = english,
    STOPWORDS = portsmith_english
);

CREATE TEXT SEARCH CONFIGURATION public.portsmith_english (COPY = pg_catalog.english);

ALTER TEXT SEARCH CONFIGURATION portsmith_english
    ALTER MAPPING FOR asciiword, asciihword, hword_asciipart, word, hword, hword_part
    WITH portsmith_stem;

```

TEMPLATE = snowball reuses PostgreSQL's standard English stemming algorithm — only the stopword list changes, not the stemming rules. COPY = pg_catalog.english starts the new configuration as an exact clone of english (same tokenizer, same handling of numbers, emails, URLs, ...), and the ALTER MAPPING line is the one substitution: word tokens now route through portsmith_stem instead of the stock english_stem dictionary.

6.4 — Confirm the noise words are gone

```
SELECT to_tsvector('portsmith_english', body) FROM city_documents WHERE id = 25;
```

```

'affect':37 'area':27 'attend':44 'basin':59 'comment':47 'conven':11 'develop':61
'discuss':16 'drain':56 'encourag':42 'engin':53 'flood':17,40,68 'given':4
'herebi':3 'infrastructur':19 'last':65 'mitig':18 'northgat':24 'notic':1
'open':31 'present':50 'propos':60 'provid':46 'public':34 'resid':36
'respons':63 'retain':25 'retent':58 'riversid':21 'session':14,29 'special':13
'spring':39 'staff':48 'storm':55 'wall':26 'year':66

```

Compare to to_tsvector('english', body) on the same row, which still carries 'citi':8,52, 'council':9, and 'portsmith':7 — three entries present under english and absent under portsmith_english. Everything else is untouched, because only the stopword list changed.

6.5 — Watch a noise-only query collapse to nothing

```
SELECT to_tsquery('portsmith_english', 'city & council');
```

```

NOTICE: text-search query contains only stop words or doesn't contain lexemes, ignored
to_tsquery

```

```
-----
```

```
(1 row)
```

Under portsmith_english, “city” and “council” are *both* stopwords, so a query built from nothing else has nothing left to search for — PostgreSQL says so explicitly instead of silently matching everything (or nothing) for an unclear reason.

6.6 — See it fix a real plain-text search

This is the exercise’s payoff: a resident searching for “Portsmouth dog park council” — a completely natural way to phrase it — under `english`:

```
SELECT plainto_tsquery('english', 'Portsmouth dog park council');
```

```
plainto_tsquery
```

```
-----  
'portsmith' & 'dog' & 'park' & 'council'
```

```
SELECT id, title FROM city_documents  
WHERE search_vector @@ plainto_tsquery('english', 'Portsmouth dog park council')  
ORDER BY id;
```

```
id | title  
---+-----  
24 | Public Notice – Riverside Dog Park Ribbon-Cutting Event  
(1 row)
```

Only one hit. The council minutes that actually approved funding for the same dog park (document 6) never happens to use the literal word “Portsmouth,” so requiring it as an AND term silently excludes the single most relevant record. Under `portsmith_english`:

```
SELECT plainto_tsquery('portsmith_english', 'Portsmouth dog park council');
```

```
plainto_tsquery
```

```
-----  
'dog' & 'park'
```

```
SELECT id, title FROM city_documents  
WHERE search_vector @@ plainto_tsquery('portsmith_english', 'Portsmouth dog park council')  
ORDER BY id;
```

```
id | title  
---+-----  
6  | Council Minutes – Riverside Dog Park Funding  
24 | Public Notice – Riverside Dog Park Ribbon-Cutting Event  
(2 rows)
```

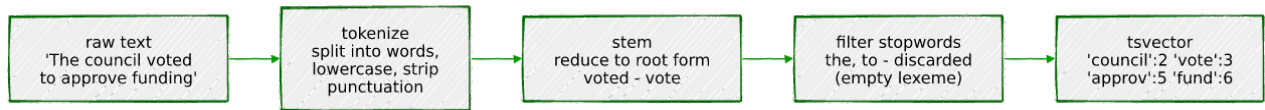
Stripped down to the two words that actually distinguish this search — `dog` and `park` — both relevant documents come back. This is the concrete argument for a custom configuration: it is not a cosmetic tweak, it changes which documents a real user actually finds.

Note: the `search_vector` column built in Exercise 3 was populated with the stock `english` configuration, so it still contains `'citi'`, `'council'`, and `'portsmith'` lexemes. The comparisons above work because `@@` only needs the *query's* surviving lexemes to be a subset of the document's — extra lexemes in the document that the query no longer asks for are simply irrelevant. To fully adopt `portsmith_english` as the table's search configuration going forward, you would update the trigger from Exercise 3 to call `to_tsvector('portsmith_english', ...)` and rebuild `search_vector` for existing rows.

Summary — What You Should Now Know

Tool	What it does
<code>to_tsvector(config, text)</code>	Tokenizes, lowercases, stems, and removes stopwords, producing a searchable <code>lexeme:position</code> vector
<code>to_tsquery(config, expr)</code>	Parses a boolean query language (<code>&</code> , <code>\ </code> , <code>!</code> , <code><-></code>); errors on plain text with no operators
<code>plainto_tsquery(config, text)</code>	Tokenizes plain text the same way as <code>to_tsvector</code> , then ANDs every surviving lexeme — safe for a search box
<code>@@</code>	Tests whether a <code>tsvector</code> matches a <code>tsquery</code>
<code>ts_debug(config, text)</code>	Shows exactly how each token was classified and which lexeme (if any) it produced — the tool for “why didn't this match?”
GIN index on a <code>tsvector</code> column	Turns <code>@@</code> from a sequential scan into an index lookup
<code>ts_rank / ts_rank_cd</code>	Score match quality by term frequency, or by term frequency and proximity (“cover density”)
<code>ts_headline(config, text, query, options)</code>	Re-scans the source text and returns a highlighted snippet — expensive; use only on displayed results
<code>CREATE TEXT SEARCH DICTIONARY ... (STOPWORDS = ...) + CREATE TEXT SEARCH CONFIGURATION</code>	Build a custom configuration with domain-specific stopwords

The key design insight from this chapter is that full-text search is not one function call — it's a pipeline:



You can inspect every stage of that pipeline with `ts_debug`, store its output with a maintained column and a GIN index, query against it with two very different trade-offs (`to_tsquery` for precision, `plainto_tsquery` for safety), and tune for your specific corpus by swapping the stopwords list — exactly the kind of tuning a generic search engine bolted on top of your database can't do without shipping your schema knowledge to it.

The `city_documents` table you built here is reused directly in later chapters: Chapter 6 (`pgvector`) adds embeddings to the same table for semantic search and builds a hybrid search combining `ts_rank` with cosine distance, and Chapter 16 (Generated Columns) replaces the hand-written trigger from Exercise 3 with a `GENERATED ALWAYS AS (...) STORED` column.

Going further: `websearch_to_tsquery` (PostgreSQL 11+) is worth knowing about even though this chapter didn't use it — it accepts the same quoted-phrase and `-exclude` syntax as a typical web search box (e.g. "flood mitigation" -infrastructure) while remaining as forgiving as `plainto_tsquery`, making it the best default for a real user-facing search field. For very large document sets, look into PostgreSQL's built-in support for weighted vectors (`setweight()`, ranking title matches above body matches) and consider whether a dedicated search engine becomes worthwhile once ranking quality and query latency requirements outgrow what a GIN index over a single table can deliver — the honest answer, for most applications, is later than you'd expect.

Chapter 5 — Fuzzy Matching: `pg_trgm`

“Every registry that outlives a year of real data entry ends up with the same person in it twice, spelled two different ways.”

Background

Full-text search, from the last chapter, is built for a specific kind of imprecision: the same *word*, inflected differently (“vote”, “voted”, “voting”). It does nothing for a different, equally common kind of imprecision — the same word, **spelled** differently. “Portsmith” typed as “Portsmith” versus “Portsmyth”. “McAllister” versus “MacAllister”. A resident registry filled in by hand, twice, by two different clerks, on two different days. Stemming cannot fix a typo, because a typo isn’t a grammatical variant of the correct word — it’s a different string that a human reader recognizes as “close enough” and a computer, by default, does not.

`pg_trgm` is PostgreSQL’s answer: it breaks strings into overlapping three-character sequences (**trigrams**) and measures how many trigrams two strings share. Two spellings of the same name share most of their trigrams even when several letters differ; two unrelated strings usually share almost none. That single idea — measured, thresholded, and indexed — is enough to power “did you mean?” search boxes, deduplicate messy records, and accelerate substring `LIKE` queries that would otherwise force a sequential scan.

This chapter builds a small deduplication and search tool around that idea: what a trigram actually is, how `similarity()` scores a pair of strings, how to turn that into an index instead of an $O(n^2)$ scan, and — the harder, more honest part — where trigram matching’s judgment calls are, because no similarity threshold gets every case right.

The Scenario

Portsmith’s resident registry has been filled in over years by different staff at different counters, and it shows: the same person appears twice under two spellings of their name often enough that nobody trusts a raw `COUNT(*)` on it anymore. Separately, the city’s 311

line fields phone searches for local businesses, and callers rarely spell a business name correctly on the first try.

Two tables model this:

Table	Purpose
<code>residents</code>	Synthetic residents, including intentional near-duplicate entries — the same person, typed twice, with a typo, transposition, or variant spelling the second time
<code>business_names</code>	A flat (<code>business_id</code> , <code>name</code>) lookup extending the <code>businesses</code> table from Chapter 1, used for “did you mean?” search

`residents` includes a `true_duplicate_of` column that a real registry would **not** have — you would not know in advance which rows are duplicates; that is the entire problem fuzzy matching exists to solve. It’s here purely so the exercises can check whether your queries found the right answers. The dataset also includes two pairs that are deliberately *not* marked as duplicates, on purpose — more on those in Exercise 2.

Exercise Goals

By the end of this chapter you will be able to:

- Explain what a trigram is and compute `similarity()` between two strings by hand for a short example.
 - Use the `%` operator to find near-matches at a configurable similarity threshold, and explain why no single threshold is perfectly correct.
 - Use `word_similarity()` to find a short query as a strong partial match inside a longer string, and explain how it differs from `similarity()`.
 - Create a GIN trigram index and confirm it accelerates both `%` and `LIKE '%term%'` queries that would otherwise force a sequential scan.
 - Build a “did you mean?” query using a GiST trigram index and `ORDER BY ... <->`, and explain why GiST — not GIN — is the right index for that access pattern.
 - Compare trigram matching against full-text search on the same kind of short, keyword-style query, and know which one to reach for.
-

Installation

`pg_trgm` ships as part of the standard `postgresql-16` package on Debian/Ubuntu — unlike PostGIS in Chapter 2, there is no separate package to install. Enable it in the database:

```
CREATE EXTENSION IF NOT EXISTS pg_trgm;
```

Loading the Data

Prerequisites

Chapter 1’s seed script must have been run first — `business_names` is populated directly from the `businesses` table:

```
python data/ch01_seed.py
```

Run the Chapter 5 seed

```
python data/ch05_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 58 residents ...
Populating business_names from businesses ...
Done - 58 rows in residents, 48 rows in business_names.
```

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d residents
```

Table "public.residents"					Default
Column	Type	Collation	Nullable		
-----+-----+-----+-----					
+-----					
id	integer		not null		

```

nextval('residents_id_seq'::regclass)
full_name      | text      |          | not null |
neighbourhood  | text      |          | not null |
true_duplicate_of | integer   |          |          |
Indexes:
    "residents_pkey" PRIMARY KEY, btree (id)
Foreign-key constraints:
    "residents_true_duplicate_of_fkey" FOREIGN KEY (true_duplicate_of) REFERENCES
residents(id)
Referenced by:
    TABLE "residents" CONSTRAINT "residents_true_duplicate_of_fkey" FOREIGN KEY
(true_duplicate_of) REFERENCES residents(id)

```

Check 2 — every marked duplicate points at a real canonical row:

```

SELECT COUNT(*) AS duplicate_pairs
FROM residents
WHERE true_duplicate_of IS NOT NULL;

```

```

duplicate_pairs
-----
              12

```

Check 3 — business_names mirrors businesses 1:1:

```

SELECT COUNT(*) FROM business_names;

```

```

count
-----
     48

```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — What a Trigram Is, and `similarity()`

1.1 — Break a string into trigrams

`show_trgm()` returns the actual set of trigrams PostgreSQL generates for a string:

```

SELECT show_trgm('Eleanor');

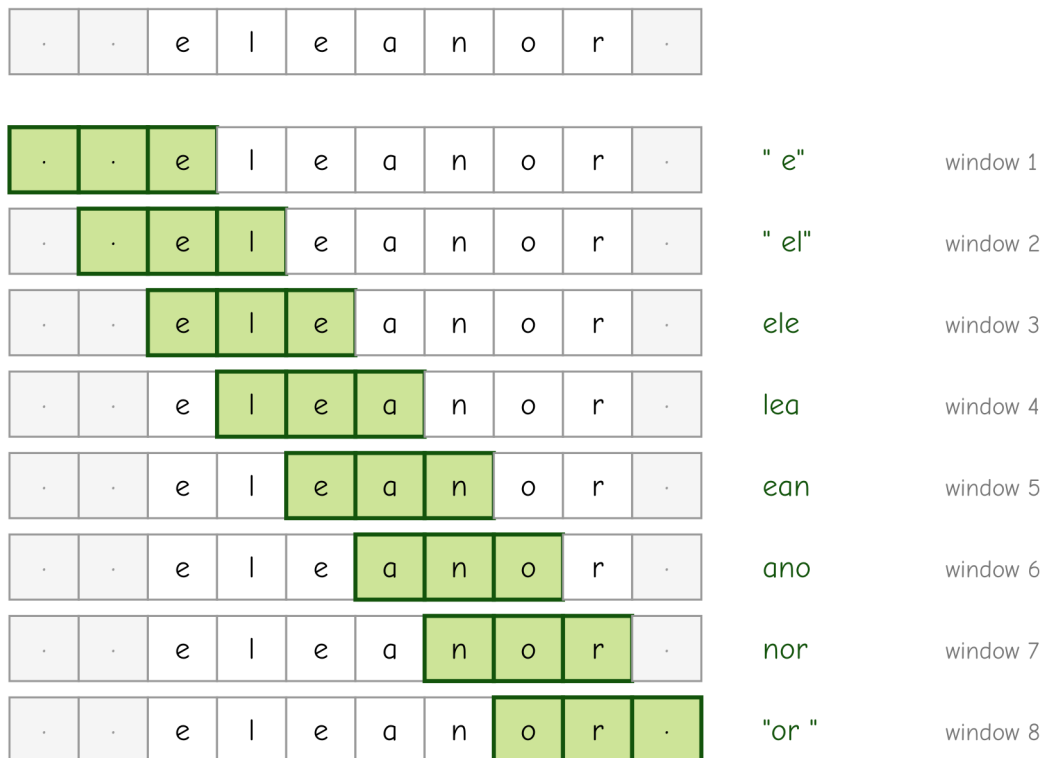
```

```
show_trgm
```

```
{ " e", " el", ano, ean, ele, lea, nor, "or " }
```

Before splitting, PostgreSQL pads the string with two leading spaces and one trailing space — " eleanor " — then takes every overlapping run of three characters: " e", " el", "ele", "lea", "ean", "ano", "nor", "or ". The padding matters: it means the first and last letters of a word each get their own distinguishing trigram (" e" marks “starts with e”; "or " marks “ends with or”), so two strings that only differ at the very start or end still register as different rather than accidentally looking identical in the middle.

padded string: two leading spaces, one trailing space



10 characters, 8 overlapping windows of 3 (10 - 3 + 1)

1.2 — similarity(): how much overlap, as a fraction

```
SELECT similarity('Eleanor Whitmore', 'Elenor Whitmore');
```

```
similarity
```

```
0.7368421
```

similarity() compares the trigram sets of both strings and returns roughly the fraction that overlap — mostly shared trigrams (both names are 16-17 characters, differing by one dropped letter) means a high score close to 1. Two unrelated strings — “Eleanor

Whitmore” and, say, “Ironside Auto” — share almost no trigrams and score close to 0. The scale is intuitive by construction: 1.0 is identical, 0.0 is nothing alike.

1.3 — All twelve duplicate pairs, scored

```
SELECT a.full_name AS canonical, b.full_name AS duplicate_entry,
       round(similarity(a.full_name, b.full_name)::numeric, 3) AS sim
FROM   residents a
JOIN   residents b ON b.true_duplicate_of = a.id
ORDER BY a.id;
```

canonical	duplicate_entry	sim
Eleanor Whitmore	Elenor Whitmore	0.737
Jonathan Castellano	Jonathon Castellano	0.739
Priyanka Deshmukh	Priyanka Deshmuk	0.842
Bartholomew Okonkwo	Bartholemew Okonkwo	0.739
Marguerite Delacroix	Marguerite Delacroix	0.792
Siobhan McAllister	Siobhan MacAllister	0.773
Theodore Vance	Theodor Vance	0.813
Anastasia Volkov	Anastassia Volkov	0.842
Desmond Okafor	Desmund Okafor	0.667
Genevieve Laurent	Genevieve Lorent	0.667
Mikhail Petrenko	Mikail Petrenko	0.737
Fitzgerald Osei	Fitzgerld Osei	0.722

Every genuine duplicate in this dataset scores well above 0.6, regardless of whether the typo was a dropped letter, a swapped letter, or a spelling variant. That range is the working intuition Exercise 2 turns into an actual threshold.

Exercise 2 — The % Operator, and Why One Threshold Isn’t Enough

2.1 — %: “similar enough,” using a session-wide threshold

Computing `similarity()` for every pair by hand doesn’t scale. The `%` operator wraps it into a boolean test against a configurable cutoff:

```
SHOW pg_trgm.similarity_threshold;
```

```
pg_trgm.similarity_threshold
```

```
-----
0.3
```

0.3 is the default. Self-join `residents` against itself to find every pair of *different* rows that clears it — this is the actual “find likely duplicates” query:


```

SELECT a.id, a.full_name, b.id, b.full_name,
       round(similarity(a.full_name, b.full_name)::numeric, 3) AS sim
FROM   residents a
JOIN   residents b ON a.id < b.id
WHERE  a.full_name % b.full_name
ORDER BY sim DESC, a.id;

```

id	full_name	id	full_name	sim
35	Priyanka Deshmukh	36	Priyanka Deshmuk	0.842
45	Anastasia Volkov	46	Anastassia Volkov	0.842
43	Theodore Vance	44	Theodor Vance	0.813
39	Marguerite Delacroix	40	Marguerite Delacroix	0.792
41	Siobhan McAllister	42	Siobhan MacAllister	0.773
57	Nadia Kowalski	58	Nadia Kowalska	0.765
33	Jonathan Castellano	34	Jonathon Castellano	0.739
37	Bartholomew Okonkwo	38	Bartholemew Okonkwo	0.739
31	Eleanor Whitmore	32	Elenor Whitmore	0.737
51	Mikhail Petrenko	52	Mikail Petrenko	0.737
53	Fitzgerald Osei	54	Fitzgerld Osei	0.722
47	Desmond Okafor	48	Desmund Okafor	0.667
49	Genevieve Laurent	50	Genevieve Lorent	0.667
55	Robert Ashworth	56	Bobby Ashworth	0.409

(14 rows)

`a.id < b.id` keeps each pair once instead of twice (A-vs-B and B-vs-A are the same comparison). Notice: **fourteen** rows came back, not twelve. Check `true_duplicate_of` on every row involved and you'll find twelve of these pairs marked as genuine duplicates and two that are not — "Nadia Kowalski" / "Nadia Kowalska" and "Robert Ashworth" / "Bobby Ashworth". They're in this dataset on purpose, and — this is the point of the exercise — look at *where* they landed in the ranking.

2.2 — A false positive sitting in the middle of the true positives

"Nadia Kowalski" and "Nadia Kowalska" are two unrelated Riverside residents who happen to share a first name and a Polish surname that differs only in its masculine/feminine ending. Their similarity, 0.765, isn't an edge case at the bottom of the list — it sits **between** "Siobhan McAllister"/"Siobhan MacAllister" (0.773) and "Bartholomew Okonkwo"/"Bartholemew Okonkwo" (0.739), two genuine duplicates. `similarity()` is doing exactly what it's designed to do: these two strings really are that close. Whether that means "probably the same person" is a judgment call the function cannot make on its own — the central limitation of fuzzy matching. It finds **candidates**, not verified duplicates. A production dedup pipeline treats a % match as "flag for review" or "compare a second field too" (a shared phone number or address), never as "merge automatically."

2.3 — Prove no threshold fixes it

Try to set a threshold that excludes the Kowalski pair:

```
SET pg_trgm.similarity_threshold = 0.77;
```

Re-run the query from 2.1. “Nadia Kowalski”/“Nadia Kowalska” (0.765) is gone — but so are seven of the twelve genuine duplicates, everything scoring below 0.77. There is no number you can put in that `SET` statement that keeps all twelve real duplicates and excludes the Kowalski pair, because a real duplicate (“Bartholomew Okonkwo”/“Bartholemew Okonkwo”, 0.739) scores *lower* than the false positive you’re trying to exclude. Put the threshold back before continuing:

```
SET pg_trgm.similarity_threshold = 0.3;
```

2.4 — A different failure: the false negative you’d never even see

“Robert Ashworth” and “Bobby Ashworth” are, in this dataset’s backstory, the same person — entered once with a nickname. Notice they only made the results list at all (0.409) because of the shared surname “Ashworth”; as given names alone, “Robert” and “Bobby” share nothing:

```
SELECT similarity('Robert', 'Bobby');
```

```
similarity
-----
0
```

A resident with a *less* distinctive shared surname and a nicknamed given name would score near zero overall and never appear in a % results list at any threshold — not a borderline case to review, just silently absent. `similarity()` measures string closeness, not identity; it has no way to know “Bobby” is short for “Robert” unless something else tells it, such as a synonym table consulted alongside it.

2.5 — The honest takeaway

Between the Kowalski pair and the Ashworth pair, this dataset has one false positive that no threshold can cleanly exclude without also losing real duplicates, and one true duplicate that scores so low it would never surface as a candidate in the first place. Tune `pg_trgm.similarity_threshold` to trade recall against precision for your own data — but go in expecting a trade-off, not a number that gets both sides to zero.

Exercise 3 — `word_similarity()` for Partial Matches

`similarity()` compares two whole strings — it penalizes a short query against a long target just for being different lengths, which makes it a poor fit for “does this short input appear as a strong match somewhere inside this longer string?”

3.1 — Compare the two functions on the same query

```
SELECT name, round(similarity('bake', name)::numeric, 3) AS full_sim
FROM   business_names
ORDER BY full_sim DESC, name
LIMIT 3;
```

name	full_sim
River Bend Bakery	0.222
Campus Bike & Sports	0.091
Finch & Sons Barbers	0.091

```
SELECT name, round(word_similarity('bake', name)::numeric, 3) AS word_sim
FROM   business_names
ORDER BY word_sim DESC, name
LIMIT 3;
```

name	word_sim
River Bend Bakery	0.800
Bay Street Electronics	0.400
Finch & Sons Barbers	0.400

Same query, same top result, very different score: 0.222 versus 0.800. `word_similarity()` finds the best-matching *substring extent* inside the target — effectively “if I could crop this longer string down to the part that best matches my query, how similar would that piece be?” — rather than scoring the query against the target’s full length. “bake” against “River Bend **B**akery” scores high because “Bake” is a near-perfect fragment match, even though “bake” is a small fraction of the full business name.

3.2 — When to reach for which

Use `similarity()` when you’re comparing two values that *should* represent the same whole thing — two spellings of one name, as in Exercises 1 and 2. Use `word_similarity()` when a short query is expected to be a fragment of a longer field — autocomplete-style search-as-you-type, or matching a partial business name a caller remembers.

Exercise 4 — GIN Trigram Indexes

4.1 — Without an index

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT name FROM business_names WHERE name LIKE '%Bakery%';
```

QUERY PLAN

```
Seq Scan on business_names (cost=0.00..25.88 rows=1 width=32) (actual
time=0.008..0.009 rows=1 loops=1)
  Filter: (name ~~ '%Bakery% '::text)
  Rows Removed by Filter: 47
  Buffers: shared hit=1
```

A plain B-tree index cannot help here — %Bakery% has no fixed prefix, so there's nothing for a B-tree to seek to. This is exactly the case `pg_trgm` was built for: it indexes every trigram in every row, so a `LIKE` pattern (itself broken into trigrams) can be looked up directly.

4.2 — Create the index

```
CREATE INDEX idx_business_names_trgm
ON business_names
USING GIN (name gin_trgm_ops);
```

4.3 — Confirm it's used

As in earlier chapters, force the index on a table this small to see the mechanism:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name FROM business_names WHERE name LIKE '%Bakery%';

SET enable_seqscan = on;  -- always restore this
```

QUERY PLAN

```
Bitmap Heap Scan on business_names (cost=21.82..25.83 rows=1 width=32) (actual
time=0.024..0.025 rows=1 loops=1)
  Recheck Cond: (name ~~ '%Bakery% '::text)
  Heap Blocks: exact=1
  Buffers: shared hit=10
  -> Bitmap Index Scan on idx_business_names_trgm (cost=0.00..21.82 rows=1 width=0)
(actual time=0.016..0.017 rows=1 loops=1)
    Index Cond: (name ~~ '%Bakery% '::text)
    Buffers: shared hit=9
```

Bitmap Index Scan on idx_business_names_trgm — the same GIN index also accelerates %, word_similarity() %>, and both LIKE and ILIKE with leading wildcards, all from one index.

Exercise 5 — A “Did You Mean?” Query with GiST

5.1 — The pattern: order by trigram distance, take the top few

pg_trgm defines a distance operator, <-> — the complement of similarity (smaller means more alike) — which makes “closest matches first” a plain ORDER BY ... LIMIT:

```
SELECT name, round(similarity(name, 'Ironsyde Auto')::numeric, 3) AS sim
FROM   business_names
ORDER BY name <-> 'Ironsyde Auto', name
LIMIT 5;
```

name	sim
-----+-----	
Ironside Auto	0.647
AutoFix Portsmith	0.143
Harbour Inn	0.040
The Art Depot	0.037
Riverside Cinema	0.033

A caller who typed “Ironsyde Auto” (transposed letters) gets “Ironside Auto” back as the clear top match, well clear of the noise below it — this is the entire “did you mean?” feature.

5.2 — Why the GIN index from Exercise 4 doesn’t help here

```
EXPLAIN (ANALYZE)
SELECT name FROM business_names
ORDER BY name <-> 'Ironsyde Auto'
LIMIT 5;
```

```
QUERY PLAN
-----
Limit  (cost=2.40..2.41 rows=5 width=36) (actual time=0.182..0.183 rows=5 loops=1)
  -> Sort  (cost=2.40..2.52 rows=48 width=36) (actual time=0.181..0.182 rows=5
loops=1)
    Sort Key: ((name <-> 'Ironsyde Auto')::text)
    Sort Method: top-N heapsort  Memory: 25kB
    -> Seq Scan on business_names  (cost=0.00..1.60 rows=48 width=36) (actual
time=0.040..0.157 rows=48 loops=1)
```

Even with `idx_business_names_trgm` in place, this plan is a full scan plus a sort — GIN accelerates *lookups* (“which rows contain trigrams matching this pattern”) but has no notion of a distance ordering. Nearest-neighbor queries need an index that natively understands “closest,” which is what **GiST** provides:

```
CREATE INDEX idx_business_names_trgm_gist
ON business_names
USING GIST (name gist_trgm_ops);
```

```
EXPLAIN (ANALYZE)
SELECT name FROM business_names
ORDER BY name <-> 'Ironsyde Auto'
LIMIT 5;
```

QUERY PLAN

```
Limit (cost=0.14..1.07 rows=5 width=36) (actual time=0.130..0.143 rows=5 loops=1)
-> Index Scan using idx_business_names_trgm_gist on business_names (cost=0.14..9.09
rows=48 width=36) (actual time=0.129..0.141 rows=5 loops=1)
    Order By: (name <-> 'Ironsyde Auto'::text)
```

Order By: `(name <-> 'Ironsyde Auto'::text)` inside an Index Scan — the GiST index walks straight to the nearest rows instead of scoring and sorting every row in the table. **Rule of thumb: GIN for %/LIKE membership lookups, GiST for ORDER BY ... <-> ... LIMIT n nearest-match queries.** It’s common to keep both, as this table now does, if an application needs both access patterns.

5.3 — One more example

```
SELECT name, round(similarity(name, 'Portsmouth Veterinary Clinic')::numeric, 3) AS sim
FROM business_names
ORDER BY name <-> 'Portsmouth Veterinary Clinic', name
LIMIT 5;
```

name	sim
Portsmouth Veterinary Clinic	0.833
AutoFix Portsmouth	0.286
Portsmouth Pharmacy	0.286
Portsmouth Tailors	0.286
Portsmouth Arms Hotel	0.263

Exercise 6 — Trigram Matching vs. Full-Text Search, Head to Head

Chapter 4 built full-text search over `city_documents`; this chapter built trigram matching over `business_names`. Both can answer “find rows matching this word” — they are good at it for different, almost opposite, reasons.

6.1 — A spelling variant: trigram wins

Portsmouth’s own documents consistently use the British spelling “harbour.” A caller who types the American spelling “harbor” gets nothing from full-text search — stemming normalizes *inflection* (“harbors” → “harbor”), not *spelling*:

```
SELECT to_tsvector('english', 'Harbour View Theater') @@ to_tsquery('english', 'harbor');
```

```
?column?
```

```
-----
```

```
f
```

Trigram similarity doesn’t care that they’re “different words” — it only sees how many three-letter fragments overlap, and “harbor”/“harbour” share almost all of theirs:

```
SELECT similarity('harbor', 'harbour');
```

```
similarity
```

```
-----
```

```
0.5
```

6.2 — A short, correctly-spelled keyword: full-text search wins

Run a short, exact keyword against `business_names` both ways:

```
SELECT name, round(similarity(name, 'bay')::numeric, 3) AS sim
FROM   business_names
ORDER BY sim DESC, name
LIMIT 6;
```

name	sim
Mango Bay Caribbean	0.200
Bay Street Electronics	0.174
River Bend Bakery	0.105
Finch & Sons Barbers	0.095
Bella Napoli	0.063
Le Petit Bistro	0.053

A three-letter query like “bay” barely produces any trigrams of its own, so the scores are all low and close together — “River Bend Bakery” and “Finch & Sons Barbers” show up with real, if small, similarity despite having nothing to do with “bay.” There’s no clean gap between signal and noise. Full-text search, which matches on whole tokens rather than character fragments, has no such problem:

```
SELECT name FROM business_names
WHERE to_tsvector('english', name) @@ plainto_tsquery('english', 'bay')
ORDER BY name;
```

```
name
-----
Bay Street Electronics
Mango Bay Caribbean
(2 rows)
```

Exactly the two relevant matches, no noise, no threshold to tune.

6.3 — The rule of thumb

Short queries made of correctly-spelled whole words belong to full-text search — it was built to match tokens precisely and cheaply via a GIN index over lexemes. Queries that might be misspelled, transposed, or OCR-damaged belong to trigram matching — it was built to tolerate exactly that kind of noise, at the cost of weaker signal on very short inputs. Many real search boxes run both: try full-text search first, and fall back to a trigram “did you mean?” only when it returns nothing.

Summary — What You Should Now Know

Tool

What it does

`show_trgm(text)`

Shows the padded, overlapping 3-character sequences a string breaks into

`similarity(a, b)`

Fraction of shared trigrams between two whole strings — best for “are these two values the same thing, spelled differently?”

`word_similarity(a, b)`

Best-matching substring extent of `a` inside `b` — best for “is `a` a fragment somewhere inside `b`?”

`a % b`

Boolean test: does `similarity(a, b)` clear `pg_trgm.similarity_threshold` (default 0.3)?

Tool	What it does
<code>a <-> b</code>	Trigram distance ($1 - \text{similarity}$) — sort by this for nearest-match-first results
<code>GIN (col gin_trgm_ops)</code>	Accelerates % and LIKE/ILIKE membership-style lookups
<code>GIST (col gist_trgm_ops)</code>	Accelerates <code>ORDER BY col <-> query LIMIT n</code> nearest-neighbor lookups
<code>pg_trgm.similarity_threshold</code>	Session-level cutoff for %; tune it, but expect trade-offs, not a perfect split

The key design insight from this chapter is that fuzzy matching produces *candidates*, not verdicts. The % operator surfaced all twelve real duplicate pairs in the `residents` table — and two pairs of genuinely different, or differently-named, people right alongside them, at every threshold tried. That isn't a shortcoming to engineer away; it's the nature of measuring string similarity instead of identity. Build fuzzy matching into a review queue or a secondary confirmation step, not an automatic merge.

The `business_names` table you built here is reused in Chapter 10 (PostgREST), which exposes the “did you mean?” query from Exercise 5 as a public RPC endpoint.

*Going further: `pg_trgm` also provides `%>` and `<->>` variants tuned for `word_similarity()` rather than `similarity()`, useful for indexing substring-style “did you mean?” search the same way Exercise 5 indexed whole-string search. For very large tables, GiST trigram indexes are typically smaller and faster to build than GIN but slower for pure membership lookups — benchmark both on your actual data distribution rather than assuming one is strictly better. And if deduplication is a recurring, business-critical process rather than an occasional query, look at dedicated record-linkage tools (e.g. the *dedupe* Python library), which combine trigram-style string similarity with other fields — address, phone number, date of birth — and a trained classifier, instead of a single threshold on a single column.*

Chapter 6 — Vector Search: `pgvector` for Embeddings

“Full-text search finds documents that use your words. Vector search finds documents that share your meaning, whether or not they share a single word with you.”

Background

Chapter 4 matched documents by shared vocabulary. Chapter 5 matched strings by shared spelling. Neither technique can find a document about “the city’s dog park” when a resident searches “somewhere my pet can run off-leash” — there isn’t a misspelling to correct or a stemmed word in common; the two phrases are simply *about the same thing*, expressed in unrelated words. No amount of stemming, stopword tuning, or trigram threshold-adjusting closes that gap, because all three earlier techniques operate on the text itself. Closing it requires a representation of what the text *means*.

That representation is an **embedding**: a machine learning model reads a piece of text and outputs a vector — literally an array of a few hundred floating-point numbers — positioned in a high-dimensional space such that texts with similar meaning end up as nearby points and unrelated texts end up far apart. “Somewhere my pet can run off-leash” and “the city’s dog park” land close together in that space even though they share zero words. This chapter never trains such a model — it uses a small, well-established, freely available one (`all-MiniLM-L6-v2`) to turn text into 384-number vectors, and PostgreSQL’s `pgvector` extension to store, index, and search them.

The two hard problems `pgvector` solves are storage (a native `vector` type, instead of smuggling floats through a JSONB array or a comma-separated text column) and **search at scale**. Finding the closest vector to a query by brute force means comparing it against every single row — fine for the thirty documents in Chapter 4’s table, hopeless for a million-row photo library. This chapter builds up to two different **approximate nearest neighbor (ANN)** index types, IVFFlat and HNSW, that trade a small amount of accuracy for a large amount of speed, and it’s honest about exactly how much accuracy each one

actually costs you at its defaults, because the answer is more surprising than most introductions let on.

The Scenario

Two tables carry this chapter, and they get their vectors two very different ways, on purpose.

`city_documents` — the council minutes, zoning ordinances, and public notices from Chapter 4 — gets a real `embedding` column computed by actually running `all-MiniLM-L6-v2` over each document’s title and body. These are genuine embeddings: semantic search over them in Exercise 5 finds real conceptual matches, not a scripted demo.

`city_photos` is different, and says so in its own name: it’s a synthetic stand-in for a photo library’s image embeddings (the kind a real vision model like CLIP would produce for actual photographs), used purely to give Exercises 3 and 4 a table large enough — 5,000 rows across ten categories — for approximate indexing to be worth doing at all. Thirty rows, like `city_documents` has, will never be large enough to make an ANN index pay for itself; a sequential scan over thirty rows is already about as fast as computation gets. There is no real photo behind any row in `city_photos` — each one is a random point clustered around one of ten category “anchors” in 384-dimensional space, which reproduces the *shape* of a real embedding space (similar things cluster; different things don’t) without needing a single actual image.

Table	Vectors	Used for
<code>city_documents</code>	Real — <code>all-MiniLM-L6-v2</code> embeddings of each document’s title + body	Semantic search, hybrid search (Exercises 5-6)
<code>city_photos</code>	Synthetic — 5,000 rows, 10 clustered categories, no real images	ANN indexing at scale (Exercises 3-4)

Exercise Goals

By the end of this chapter you will be able to:

- Store embeddings in a native `vector` column and understand what the numbers in it actually represent.
 - Compute exact nearest neighbors with `<->` (L2), `<#>` (negative inner product), and `<=>` (cosine distance), and know when the choice between them actually changes your results.
 - Build an IVFFlat index, and correctly interpret its `lists` and `probes` parameters instead of trusting the defaults.
 - Build an HNSW index, and make an informed build-time-vs-recall trade-off between it and IVFFlat instead of guessing.
 - Run real semantic search: find documents by meaning even when they share no vocabulary with the query.
 - Build hybrid search that blends a keyword score (`ts_rank`, from Chapter 4) with a semantic score into one ranking.
-

Installation

This chapter has two dependencies, and they're independent of each other — you need both, but you're installing two unrelated things for two unrelated reasons.

1 — `pgvector`

```
sudo apt install -y postgresql-16-pgvector
```

Enable the extension. **Unlike `pg_trgm` in Chapter 5, `pgvector` 0.6.x is not a “trusted” extension** — its control file doesn't set `trusted = true`, so a regular database-owning role cannot enable it with a plain `CREATE EXTENSION`. It has to be done once, by a superuser:

```
sudo -u postgres psql portsmith -c "CREATE EXTENSION vector;"
```

Confirm it's there:

```
SELECT extversion FROM pg_extension WHERE extname = 'vector';
```

```
extversion
```

```
-----  
0.6.0
```

2 — sentence-transformers, CPU-only

The embedding model runs locally, in Python, via the `sentence-transformers` library — no API key, no network calls at query time. It depends on PyTorch, and installing PyTorch’s default package pulls in a full NVIDIA CUDA toolkit (several gigabytes) even on a machine with no GPU, which is pure waste for embedding a few dozen short documents on CPU. Install the CPU-only build explicitly, then `sentence-transformers` on top of it:

```
python3.12 -m venv .venv # if you haven't already, from earlier chapters  
source .venv/bin/activate  
pip install torch --index-url https://download.pytorch.org/whl/cpu  
pip install sentence-transformers pgvector
```

Note: this installs roughly 3GB into your virtual environment (PyTorch and its numerical dependencies are not small, even CPU-only) and the first time any script loads `all-MiniLM-L6-v2`, it downloads about 90MB of model weights from Hugging Face and caches them in `~/.cache/huggingface`. Every run after that is fully offline. If you see a `Warning: You are sending unauthenticated requests to the HF Hub` line the first time, that’s expected and harmless — it’s just Hugging Face reminding you that an account would raise your download rate limit.

Loading the Data

Prerequisites

Chapter 4’s seed script must have been run first:

```
python data/ch04_seed.py
```

`ch04_seed.py` alone is **not** enough, though — this chapter’s hybrid search (Exercise 6) needs `city_documents.search_vector`, and that column only gets created by hand-running Chapter 4’s **Exercise 3** SQL; it is not part of `ch04_seed.py`’s schema. (If you re-run `ch04_seed.py` after already having done Exercise 3 — say, to reset the document data — it drops `city_documents` and rebuilds it *without* `search_vector`, silently undoing that exercise.)

ch06_seed.py (below) checks for the column and will stop with the exact fix if it's missing; if you'd rather do it now, here it is verbatim from Chapter 4:

```
ALTER TABLE city_documents ADD COLUMN search_vector tsvector;

UPDATE city_documents
SET   search_vector = to_tsvector('english', title || ' ' || body);

CREATE OR REPLACE FUNCTION city_documents_search_vector_update() RETURNS trigger AS $$
BEGIN
    NEW.search_vector := to_tsvector('english', NEW.title || ' ' || NEW.body);
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_city_documents_search_vector
    BEFORE INSERT OR UPDATE OF title, body ON city_documents
    FOR EACH ROW
    EXECUTE FUNCTION city_documents_search_vector_update();

CREATE INDEX idx_city_documents_search_vector
    ON city_documents USING GIN (search_vector);
```

Run the Chapter 6 seed

This adds the (initially empty) embedding column to `city_documents` and creates and populates `city_photos`. It needs only `psycopg2`, `numpy`, and `pgvector` — not `sentence-transformers` — since `city_photos`'s vectors are synthetic:

```
python data/ch06_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Applying DDL ...
Generating 5000 synthetic photo embeddings ...
Done - 5000 rows in city_photos across 10 categories.
city_documents.embedding added but left NULL - run `python data/ch06_embed_documents.py`
next.
```

Compute real embeddings for `city_documents`

This is the step that actually loads and runs the model:

```
python data/ch06_embed_documents.py
```

Expected output:

```
Connecting to: dbname=portsmith
Loading all-MiniLM-L6-v2 (first run downloads ~90MB of model weights) ...
Embedding 30 documents ...
Done - 30 rows in city_documents now have an embedding.
```

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — `city_photos` structure:

```
\d city_photos
```

```
Table "public.city_photos"
  Column      | Type          | Collation | Nullable | Default
-----+-----+-----+-----+-----
id            | integer       |           | not null |
nextval('city_photos_id_seq'::regclass)
category      | text          |           | not null |
neighbourhood | text          |           | not null |
caption       | text          |           | not null |
embedding     | vector(384)   |           | not null |
Indexes:
    "city_photos_pkey" PRIMARY KEY, btree (id)
```

Check 2 — 500 photos per category:

```
SELECT category, COUNT(*) FROM city_photos GROUP BY category ORDER BY category;
```

category	count
community_event	500
harbour_waterfront	500
historic_architecture	500
industrial_dock	500
infrastructure_construction	500
municipal_building	500
public_park	500
residential_street	500
street_market	500
wildlife_nature	500

(10 rows)

Check 3 — every `city_documents` row has an embedding:


```
SELECT COUNT(*) AS total, COUNT(embedding) AS with_embedding FROM city_documents;
```

```
total | with_embedding  
-----+-----  
30 | 30
```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — The `vector` Type

1.1 — Literals and dimensions

A `vector` literal is a bracketed list of numbers:

```
SELECT '[1,2,3]':vector;
```

```
vector  
-----  
[1,2,3]
```

A `vector(384)` column, like `city_documents.embedding`, enforces its dimension count at write time — every row’s vector must have exactly 384 numbers. There is no meaningful way to compare a 384-dimensional embedding to a 3-dimensional one, so PostgreSQL simply won’t let a mismatched value in:

```
SELECT '[1,2,3]':vector(384);
```

```
ERROR:  expected 384 dimensions, not 3
```

1.2 — What’s actually in the column

`city_documents.embedding` isn’t hand-written — it came from `ch06_embed_documents.py` calling `all-MiniLM-L6-v2` on each document’s title and body. Look at a real one:

```
SELECT id, title, embedding FROM city_documents WHERE id = 1;
```

The `embedding` value prints as 384 comma-separated floating-point numbers between roughly -1 and 1 — not something a human reads directly, but exactly what every operator and index in this chapter operates on. What those particular 384 numbers *mean* isn’t individually interpretable (no single dimension corresponds to a concept like “is

about harbours”); what matters is only their *position relative to other documents’ vectors*, which is what the distance operators in Exercise 2 measure.

Exercise 2 — Exact Nearest Neighbors: <->, <#>, <=>

pgvector gives you three distance operators. They agree more often than you’d expect, and the case where they disagree is worth understanding before it surprises you in production.

2.1 — Three tiny vectors, three operators

```
SELECT '[1,0,0]':vector <-> '[0,1,0]':vector AS l2_distance;
```

```
l2_distance
-----
1.4142135623730951
```

<-> is **Euclidean (L2) distance** — straight-line distance between the two points, exactly like distance in ordinary geometry. Two orthogonal unit vectors are $\sqrt{2}$ apart.

```
SELECT '[1,0,0]':vector <#> '[0,1,0]':vector AS neg_inner_product;
```

```
neg_inner_product
-----
-0
```

<#> is the **negative inner product** — the dot product of the two vectors, negated (pgvector negates it so that, consistent with the other two operators, *smaller means closer*). Orthogonal vectors have a dot product of zero.

```
SELECT '[1,0,0]':vector <=> '[0,1,0]':vector AS cosine_distance;
```

```
cosine_distance
-----
1
```

<=> is **cosine distance** ($1 - \text{cosine similarity}$) — the angle between the two vectors, completely ignoring their length. Orthogonal vectors have maximum cosine distance, 1.

2.2 — Where they disagree: magnitude

The three operators can rank the *same* pair of candidates in a *different* order the moment the vectors involved aren’t the same length. Compare a query vector against two

candidates — one that points in exactly the same direction but is twice as long, and one that points in a slightly different direction but is almost the same length:

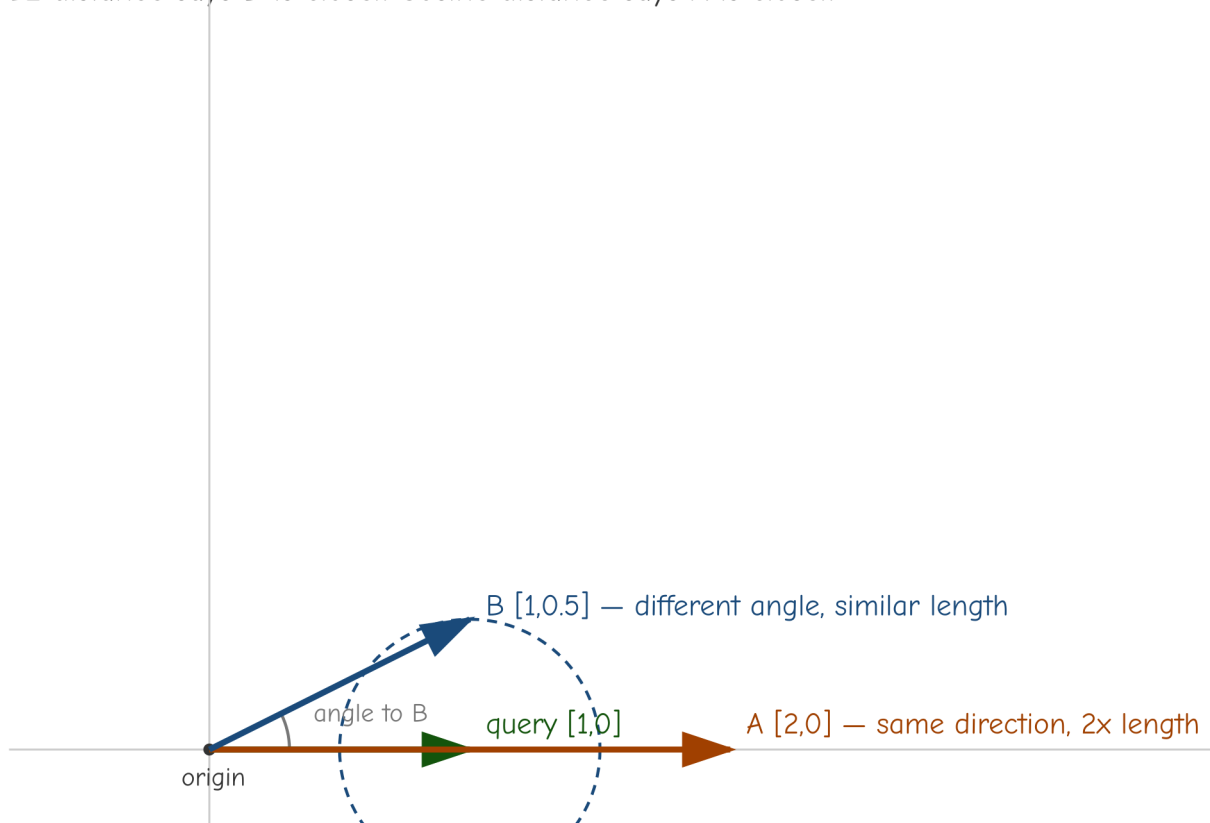
SELECT

```
'[1,0]':::vector <-> '[2,0]':::vector AS l2_to_a,  
'[1,0]':::vector <-> '[1,0.5]':::vector AS l2_to_b,  
round(('[1,0]':::vector <=> '[2,0]':::vector)::numeric, 4) AS cos_to_a,  
round(('[1,0]':::vector <=> '[1,0.5]':::vector)::numeric, 4) AS cos_to_b;
```

l2_to_a	l2_to_b	cos_to_a	cos_to_b
1	0.5	0.0000	0.1056

The ranking flips. By $\leftarrow$ (L2), candidate B ([1,0.5], distance 0.5) is closer than candidate A ([2,0], distance 1). By $\leq$ (cosine), it's the reverse — A is distance 0 (identical direction) while B is 0.1056 away. Neither operator is “wrong”; they’re answering different questions. L2 asks “how far apart are these points.” Cosine asks “how similarly are these vectors oriented, regardless of scale.”

L2 distance says B is closer. Cosine distance says A is closer.



L2 ($\leftarrow$): $\text{dist}(Q,A) = 1.0$ $\text{dist}(Q,B) = 0.5 \rightarrow$ B is closer

cosine ($\leq$): $\text{dist}(Q,A) = 0.0$ $\text{dist}(Q,B) = 0.1056 \rightarrow$ A is closer

A sits exactly on the query's own line (the overlap in the diagram above *is* "same direction, twice the length"), which is why its cosine distance is zero. The dashed circle is centered on the query with radius equal to B's L2 distance — B sits right on that boundary, while A sits well outside it, which is the straight-line sense in which B is "closer."

2.3 — Why this chapter normalizes, and what that buys you

`ch06_embed_documents.py` calls `model.encode(texts, normalize_embeddings=True)` — every stored embedding is scaled to unit length before it's written. Once every vector in a comparison has the same length, the magnitude difference that caused Exercise 2.2's disagreement simply can't occur, and all three operators produce the *same ranking* (just on different numeric scales). Embedding the query "waterfront redevelopment funding" (Exercise 5 shows how) and checking the top 3 `city_documents` rows by each operator confirms it:

	<code><-></code> (L2)	<code><#></code> (neg. inner product)	<code><=></code> (cosine)
Rank 1 doc 1	— 0.8605	doc 1 — -0.6298	doc 1 — 0.3702
Rank 2 doc 22	— 1.0546	doc 22 — -0.4439	doc 22 — 0.5561
Rank 3 doc 5	— 1.0646	doc 5 — -0.4333	doc 5 — 0.5667

Same three documents, same order, every time — only the numbers' scale differs between columns. The practical rule: **normalize your embeddings, and pick whichever operator has an index you can build** — for normalized vectors it stops being a search-quality decision and becomes a performance one, covered next.

Exercise 3 — IVFFlat: Approximate Search with `lists` and `probes`

3.1 — Exact search doesn't scale

`city_photos` has 5,000 rows — small by real standards, but already enough to feel the cost of brute-force search. Find the 10 nearest neighbors of photo `id = 1` by comparing it against every other row:

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id, category, embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) AS
dist
FROM city_photos
WHERE id != 1
ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1)
LIMIT 10;
```

QUERY PLAN

```

Limit (cost=1439.32..1439.35 rows=10 width=30) (actual time=4.164..4.167 rows=10
loops=1)
  -> Sort (cost=1431.02..1443.52 rows=4999 width=30) (actual time=4.163..4.164
rows=10 loops=1)
    Sort Key: ((city_photos.embedding <-> $0))
    Sort Method: top-N heapsort  Memory: 26kB
    -> Seq Scan on city_photos (cost=0.00..1323.00 rows=4999 width=30) (actual
time=0.024..0.3381 rows=4999 loops=1)

```

Every one of the 4,999 other rows gets its distance computed and sorted — about 4.6ms here. That cost grows linearly with table size; at real photo library scale (millions of rows) it stops being viable.

3.2 — Build the index

```

CREATE INDEX idx_city_photos_ivfflat
ON city_photos
USING ivfflat (embedding vector_l2_ops)
WITH (lists = 50);

```

```

CREATE INDEX
Time: 152.129 ms

```

`lists = 50` partitions the 5,000 vectors into 50 clusters (via k-means at index-build time) and stores each vector under its nearest cluster centroid. A rule of thumb from the pgvector docs: `lists ≈ rows / 1000` for up to roughly a million rows.

3.3 — The planner picks it up on its own

```

EXPLAIN (ANALYZE, BUFFERS)
SELECT id, category, embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) AS
dist
FROM city_photos
WHERE id != 1
ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1)
LIMIT 10;

```

QUERY PLAN

```

Limit (cost=61.55..62.35 rows=10 width=30) (actual time=0.145..0.168 rows=10 loops=1)
  -> Index Scan using idx_city_photos_ivfflat on city_photos (cost=53.25..455.00
rows=4999 width=30) (actual time=0.145..0.166 rows=10 loops=1)
    Order By: (embedding <-> $0)

```

No `enable_seqscan = off` trick needed this time, unlike the small tables in earlier chapters — at 5,000 rows the planner’s own cost estimate already favors the index: **4.6ms down to about 0.2ms**.

3.4 — The catch: default probes gives up a lot of accuracy

An IVFFlat search only examines the clusters nearest the query vector, not all 50 — controlled by `ivfflat.probes`, which **defaults to 1**. Compare the approximate top-10 against the true exact top-10:

```
-- Exact (force a sequential scan to get true ground truth):
SET enable_indexscan = off;
SELECT array_agg(id) FROM (
    SELECT id FROM city_photos WHERE id != 1
    ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) LIMIT 10
) t;
```

array_agg

{146,357,485,468,500,261,384,397,185,226}

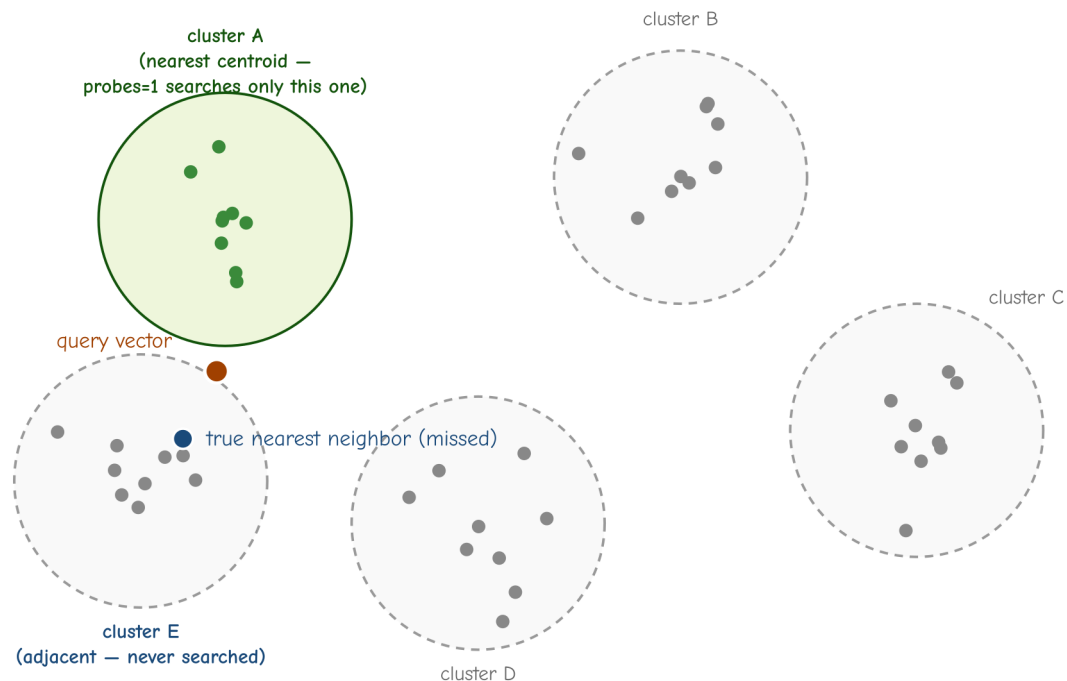
```
-- Approximate, default probes = 1:
RESET enable_indexscan;
SELECT array_agg(id) FROM (
    SELECT id FROM city_photos WHERE id != 1
    ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) LIMIT 10
) t;
```

array_agg

{485,261,185,226,385,22,274,433,56,327}

Only **4 of the 10** results match (485, 261, 185, 226) — **40% recall**, at the default setting, on a real query. This is not a contrived worst case; it’s what `ivfflat.probes = 1` actually does: it searches only the single cluster closest to the query and simply never looks at the other 49, even though some of the true nearest neighbors live in a different cluster.

probes=1 only searches the nearest cluster, even when a true neighbor sits just across the boundary



IVFFlat clusters vectors at index-build time. probes=1 checks only the nearest centroid (A) — a real neighbor one cluster over is invisible to the search.

A note on reproducing these exact numbers: IVFFlat's clustering step uses k-means with a random initialization at *index build* time — even against `city_photos`'s fixed, seeded synthetic data, rebuilding the same `CREATE INDEX ... USING ivfflat` statement can shift which photos land in which of the 50 clusters, and therefore which specific IDs and exact recall percentage `probes = 1` produces. Rerunning this exercise on your own machine will very likely show different IDs and a different percentage than the ones printed here. The pattern that reproduces reliably is the *shape*: `probes = 1` is measurably, often substantially, incomplete.

3.5 — Raise `probes`, watch recall recover

```
SET ivfflat.probes = 2;
```

```
array_agg
```

```
-----  
{146, 357, 485, 468, 500, 261, 384, 397, 185, 226}
```

At `probes = 2` the result already matches the exact set exactly — **100% recall** on this query. Timing at a slightly more conservative `probes = 5`, and at `probes = 50` (every list — the most exhaustive an IVFFlat search can be):

<code>probes</code>	Recall (this query)	Query time
1 (default)	40%	~0.4ms
5	100%	~0.6ms
50 (all lists)	100%	~0.6ms

Timings vary run to run more than the recall figures do (expect noise in the tenths of a millisecond on a table this size) — the shape that matters is that `probes = 1` is consistently fastest *and* the least reliable, while anything from `probes = 5` up delivers full recall on this query without a further, proportional cost from probing still more lists; the IVFFlat index structure keeps even an “examine every list” search meaningfully faster than the 4.6ms plain sequential scan from Exercise 3.1. Reset before continuing:

```
SET ivfflat.probes = 1;
```

Two probes happened to be enough for this particular query; it isn’t a guarantee for every query, which is why a small safety margin (a starting point of `probes ≈ √lists`, so `√50 ≈ 7`) is the honest recommendation rather than the bare minimum that worked once. The overall lesson: **IVFFlat’s out-of-the-box defaults are fast and unreliable at the same time** — `lists` and `probes` are not optional tuning, they’re the whole feature.

Exercise 4 — HNSW: A Different Trade-off

4.1 — Build it, and time the build

```
CREATE INDEX idx_city_photos_hnsw
ON city_photos
USING hnsw (embedding vector_l2_ops);
```

```
CREATE INDEX
Time: 513.963 ms
```

Compare against IVFFlat’s build time from Exercise 3.2: **152ms**. HNSW (Hierarchical Navigable Small World) builds a multi-layer graph structure connecting each vector to its approximate neighbors, which costs roughly **3x longer to build** than IVFFlat’s k-means clustering, on this table.

4.2 — Recall at HNSW's default settings

Drop (or temporarily disable) the IVFFlat index so the planner has no choice but to use HNSW, then repeat the exact same top-10 query from Exercise 3.4:

```
BEGIN;  
DROP INDEX idx_city_photos_ivfflat;  
  
SELECT array_agg(id) FROM (  
    SELECT id FROM city_photos WHERE id != 1  
    ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) LIMIT 10  
) t;  
  
ROLLBACK; -- put idx_city_photos_ivfflat back
```

array_agg

{146,357,485,468,500,261,384,397,185,226}

Exact match with the true top-10 — 100% recall, at HNSW's default settings, no tuning required. Query time for this was about 0.63ms. Repeating the comparison against a second, independent query point (`id = 1001`, a `public_park` photo) confirms it isn't a fluke: HNSW again returned the exact top-10, while IVFFlat at default `probes = 1` again recovered only 3 of 10 (30% recall — as Exercise 3.4 noted, the exact percentage shifts between index rebuilds; what doesn't shift is that it's consistently well short of complete). HNSW's default, across every test in this chapter, has not missed a single true neighbor.

4.3 — The trade-off, stated plainly

	IVFFlat	HNSW
Build time (5,000 rows)	152ms	514ms (~3x)
Recall at defaults	30-40% in this chapter's tests, varies by build (needs <code>probes</code> tuned up)	~100% (no tuning needed)
Query time (well-tuned)	Comparable	Comparable
Tuning burden	On you, every session (<code>probes</code> is a session- level <code>SET</code>)	Minimal

This matches the general guidance in the pgvector documentation: **HNSW is the better default choice for most workloads** — better recall out of the box, no per-session

parameter to remember to set. IVFFlat’s advantages are real but narrower: faster to build (relevant if you rebuild the index often, e.g. after large bulk loads) and somewhat lower memory overhead at very large scale. Choose IVFFlat deliberately for those reasons, not by default.

Exercise 5 — Semantic Search: Finding Meaning, Not Words

5.1 — A query with no vocabulary overlap

`data/ch06_semantic_search.py` embeds a query string with the same model used to build the column, and searches `city_documents` by cosine similarity:

```
python data/ch06_semantic_search.py "stray dogs and pet-friendly spaces"
```

id	sim	title
6	0.4447	Council Minutes – Riverside Dog Park Funding
24	0.3927	Public Notice – Riverside Dog Park Ribbon-Cutting Event
16	0.2683	Zoning Ordinance – Riverside Short-Term Rental Restrictions
...		

Not one word of the query — “stray,” “dogs,” “pet-friendly,” “spaces” — appears in document 6’s title, and full-text search (Chapter 4) would return zero rows for it; there’s no shared lexeme to match on, misspelled or otherwise, so Chapter 5’s trigram matching wouldn’t help either. The embedding model has no notion of exact words at all — it read “the city’s dog park” and “somewhere my pet can run off-leash” as *about the same thing*, which is precisely the gap described in this chapter’s Background.

5.2 — A stronger example

```
python data/ch06_semantic_search.py "waterfront redevelopment funding"
```

id	sim	title
1	0.6298	Council Minutes – Harbour District Waterfront Renovation Budget
22	0.4439	Public Notice – Public Hearing on Harbour District Waterfront Rezoning
5	0.4333	Council Minutes – Riverside Road Resurfacing Program
12	0.4207	Zoning Ordinance – Harbour District Waterfront Height Variance
9	0.4101	Council Minutes – Canal Road Bike Lane Expansion
...		

5.3 — Reading the gap, not just the ranking

Look at the jump between rank 1 (0.6298) and rank 2 (0.4439) — a much bigger drop than between ranks 2 through 5. That gap is meaningful: it says document 1 isn’t merely the

best available match, it's a *genuinely strong* one, while the rest are progressively looser associations. Semantic search has a property full-text search doesn't: **it never returns zero results**. Every document has *some* cosine similarity to any query — the model will confidently rank all thirty documents even for a nonsense query, it just won't find any of them close. A real application needs a similarity-score cutoff (or a “confidence” gap check like this one) to distinguish “found it” from “here's our least-bad guess,” a problem keyword search's honest zero-rows sidesteps entirely.

Exercise 6 — Hybrid Search: Blending Keyword and Semantic Signals

Semantic search and keyword search fail differently. Semantic search misses exact terms that matter — a specific ordinance number, a street name — treating them as just more words to average into a general impression of meaning. Keyword search misses paraphrases entirely, as Exercise 5 just showed. Hybrid search runs both and blends the scores.

6.1 — Keyword-only ranking

```
SELECT id, title, round(ts_rank(search_vector, plainto_tsquery('english','bike lane'))::numeric,4) AS kw_rank
FROM city_documents
WHERE search_vector @@ plainto_tsquery('english','bike lane')
ORDER BY kw_rank DESC;
```

id	title	kw_rank
-----+-----		
9	Council Minutes – Canal Road Bike Lane Expansion	0.2928
28	Public Notice – Canal Road Bike Lane Construction Schedule	0.1960
30	Public Notice – Annual Fireworks Display and Street Closures	0.1759
5	Council Minutes – Riverside Road Resurfacing Program	0.0992
15	Zoning Ordinance – Reduced Parking Minimums Near Transit Corridors	0.0991

6.2 — Semantic-only ranking, same query

```
python data/ch06_semantic_search.py "bike lane" --top 8
```

id	sim	title
9	0.5513	Council Minutes – Canal Road Bike Lane Expansion
28	0.4433	Public Notice – Canal Road Bike Lane Construction Schedule
15	0.4117	Zoning Ordinance – Reduced Parking Minimums Near Transit Corridors
5	0.3965	Council Minutes – Riverside Road Resurfacing Program
21	0.2754	Public Notice – Bay Street Water Main Repair Road Closure
6	0.2688	Council Minutes – Riverside Dog Park Funding

30	0.2653	Public Notice – Annual Fireworks Display and Street Closures
10	0.2359	Council Minutes – Special Session on Flood Mitigation Infrastructure

Both agree on the top 2 (documents 9 and 28 — a real, strong signal either way finds them). They disagree further down: document 30 ranks **3rd** by keyword (it happens to use the phrase “biking along the Canal Road bike lane” once, in passing) but only **7th** by semantic similarity, while document 15 ranks **5th** by keyword but **3rd** semantically (it’s substantively about bike-lane-adjacent parking policy, using different phrasing throughout).

6.3 — Blend both into one score

```
python data/ch06_semantic_search.py "bike lane" --hybrid --top 8
```

id	kw	sem	hybrid	title
9	0.2928	0.5513	0.4479	Council Minutes – Canal Road Bike Lane Expansion
28	0.1960	0.4433	0.3444	Public Notice – Canal Road Bike Lane Construction Schedule
15	0.0991	0.4117	0.2867	Zoning Ordinance – Reduced Parking Minimums Near Transit Corridors
5	0.0992	0.3965	0.2776	Council Minutes – Riverside Road Resurfacing Program
30	0.1759	0.2653	0.2295	Public Notice – Annual Fireworks Display and Street Closures
21	0.0000	0.2754	0.1652	Public Notice – Bay Street Water Main Repair Road Closure
6	0.0000	0.2688	0.1613	Council Minutes – Riverside Dog Park Funding
10	0.0000	0.2359	0.1415	Council Minutes – Special Session on Flood Mitigation Infrastructure

The underlying SQL (HYBRID_SQL in ch06_semantic_search.py) is a LEFT JOIN between a keyword-matched candidate set and a semantic top-10 candidate set, so documents 21, 6, and 10 — never matched by the keyword query at all (kw = 0.0000) — still appear, carried in purely by semantic similarity:

```
hybrid_score = 0.4 * COALESCE(kw_score, 0) + 0.6 * COALESCE(sem_score, 0)
```

Document 15 moves from 5th (keyword) up to 3rd (hybrid); document 30 moves from 3rd (keyword) down to 5th (hybrid) — the blend genuinely changes the ranking, not just the score.

6.4 — The honest caveat about the weights

0.4 and 0.6 here are fixed constants chosen because ts_rank and cosine similarity happen to land in roughly comparable numeric ranges on *this* dataset — they are not principled, and they are not portable to a different corpus or a different embedding model without

re-checking that assumption. A production hybrid-search system typically normalizes each signal (min-max or z-score, over the actual candidate set returned) before blending, rather than trusting that two arbitrary scoring functions happen to land in the same numeric neighborhood. Treat the weights here as a teaching simplification, not a formula to copy into production untested.

Summary — What You Should Now Know

Tool	What it does
<code>vector(N)</code>	A fixed-dimension floating-point array column type; enforces <code>N</code> at write time
<code><-></code>	L2 (Euclidean) distance — straight-line distance between points
<code><#></code>	Negative inner product — dot product, negated so smaller means closer
<code><=></code>	Cosine distance — angle between vectors, ignoring magnitude
<code>normalize_embeddings=True</code>	Makes all three operators agree on ranking; without it, they can disagree
<code>USING ivfflat (col vector_l2_ops) WITH (lists = N)</code>	Clusters vectors into <code>N</code> buckets; fast to build, needs <code>probes</code> tuned up from its default to get usable recall
<code>SET ivfflat.probes = N</code>	How many clusters an IVFFlat search actually examines — defaults to 1, often too low
<code>USING hnsw (col vector_l2_ops)</code>	Graph-based ANN index; ~3x slower to build than IVFFlat here, but near-exact recall by default
<code>ts_rank(...) + (1 - cosine_distance) weighted blend</code>	Hybrid search — keyword precision plus semantic recall in one ranking

The key design insight from this chapter is that “approximate” in “approximate nearest neighbor” is not a rounding error — at IVFFlat’s default settings, this chapter repeatedly measured 30-40% recall on real queries, which means 6-7 of the top 10 *true* nearest neighbors were silently missing from the results, with no error, no warning, nothing to

indicate anything was wrong. Every ANN index trades recall for speed on a dial you control (`probes`, `ef_search`); the failure mode to avoid is not knowing where that dial is set.

The `city_documents` table's `embedding` column is reused in Chapter 10 (PostgREST), which exposes the semantic search query from Exercise 5 as a public RPC endpoint alongside the fuzzy “did you mean?” search from Chapter 5.

Going further: this chapter used `all-MiniLM-L6-v2` (384 dimensions) because it's small, fast on CPU, and good enough to teach with — production systems often use larger models (OpenAI's `text-embedding-3-small` at 1536 dimensions, or similar) for better semantic quality, at proportionally higher storage and compute cost per vector. For tables much larger than `city_photos`'s 5,000 rows, revisit `lists` (scale roughly with row count) and consider `hnsw`'s `m` and `ef_construction` build-time parameters, which trade index size and build time for recall the same way `probes` trades query time for it. And if your embeddings change meaning over time — a newer, better model version — remember that a `vector` column has no built-in versioning; re-embedding the whole table and rebuilding the index is a full migration, not an in-place update.

Bonus Section — Build a Local RAG System

*This section is optional and sits outside the numbered exercises — nothing later in the book depends on it. It exists because semantic search (Exercise 5) is one step short of the thing most people actually mean when they say “AI search”: a system that doesn't just return matching documents, but reads them and answers the question directly. That's **Retrieval-Augmented Generation (RAG)**, and you already built the hard half of it — retrieval — in Exercises 1 through 5. This section adds the other half.*

What RAG actually is

Every large language model has a knowledge cutoff and no idea what's in your `city_documents` table. Ask one directly “what are the rules for Portsmouth's dog park?” and it will either say it doesn't know or, worse, confidently invent something plausible-sounding and wrong — a **hallucination**. RAG sidesteps this without retraining or fine-tuning anything: before asking the model the question, retrieve the most relevant real text from your own database and paste it directly into the prompt, instructing the model to answer *only* from that text. The model isn't recalling facts from training anymore; it's reading comprehension over text you handed it seconds ago.

This means RAG is two independent systems wired together, and it's worth keeping them mentally separate:

1. **Retrieval** — PostgreSQL, `pgvector`, and everything from Exercises 1-5 of this chapter. Given a question, find the most relevant chunks of text.
2. **Generation** — a large language model that turns “here are some relevant facts, here is a question” into a fluent, direct answer.

Nothing about retrieval changes when you add generation on top of it — the `<=>` cosine-distance query from Exercise 5 is exactly what runs here too. What's new is chunking (documents are usually too long and too topically mixed to embed as a single vector) and the LLM call itself, which this section runs locally through **Ollama** rather than a paid API, for the same reason Chapter 6 as a whole runs its embedding model locally: no API key, no per-query cost, nothing leaves your machine.

Installation — Ollama

```
curl -fsSL https://ollama.com/install.sh | sh
```

This installs Ollama as a systemd service listening on `localhost:11434`. Pull a chat model — this section uses `llama3.1:8b` (about 4.9GB):

```
ollama pull llama3.1:8b
```

If disk space or RAM is tight, `ollama pull phi` gets you a much smaller (~1.6GB) model at a real quality cost; every example below works with either, since the model name is a plain argument to the RAG script, not something hardcoded.

Note: the *first* request to any given model incurs a one-time load delay (10-15 seconds isn't unusual) while Ollama loads its weights into memory. Every request after that, to the same model, is fast — Ollama keeps recently used models resident for a while. Don't judge a model's real latency by its very first response.

Ingest: chunk and embed a directory of documents

`data/rag_docs/` contains four new, longer, informational documents — a business licensing guide, a parks and recreation guide, a utilities guide, and a newcomer's guide — distinct from `city_documents`'s formal council records, written in the plain-prose register an actual city help-desk handout would use. `ch06_rag_ingest.py` splits each file into

overlapping word-count chunks, embeds every chunk with `all-MiniLM-L6-v2` (the same model used everywhere else in this chapter — retrieval only works if the same model embeds both the documents and the questions), and loads them into a table you name on the command line:

```
python data/ch06_rag_ingest.py data/rag_docs --table portsmith_rag --recreate
```

```
Connecting to: dbname=portsmith
Loading all-MiniLM-L6-v2 ...
  getting_a_business_license.txt: 4 chunks
  newcomer_guide.txt: 4 chunks
  parks_and_recreation.txt: 4 chunks
  utilities_and_public_works.txt: 4 chunks
Done - 16 chunks from 4 files loaded into 'portsmith_rag'.
```

Each ~700-word document became 4 chunks of 180 words with a 40-word overlap between consecutive chunks (both configurable via `--chunk-size` and `--chunk-overlap`). The overlap matters: without it, a sentence that happens to fall right at a chunk boundary gets split across two chunks, and neither half alone may retrieve well for a question about it.

The table it creates is deliberately the same generic shape regardless of what you point it at:

```
\d portsmith_rag
```

```
Table "public.portsmith_rag"
  Column | Type          | Collation | Nullable | Default
-----+-----+-----+-----+-----
id       | integer       |           | not null |
nextval('portsmith_rag_id_seq'::regclass)
source   | text          |           | not null |
chunk_index | integer       |           | not null |
content  | text          |           | not null |
embedding | vector(384)   |           | not null |
Indexes:
  "portsmith_rag_pkey" PRIMARY KEY, btree (id)
  "idx_portsmith_rag_embedding" hnsw (embedding vector_cosine_ops)
```

`--table` is a plain identifier you choose — point the same script at a different directory with `--table` set to something else, and you have a second, independent knowledge base in the same database. Run it against your own notes, a project’s documentation, anything in `.txt` or `.md` files.

Why the table name is validated, not just interpolated: `--table` comes from the command line and flows into `CREATE TABLE`, `INSERT`, and `CREATE INDEX` statements. SQL identifiers (table names) can't be passed as query parameters the way values can — `cur.execute("... WHERE id = %s", (table_name,))` only works for *values*. `ch06_rag_ingest.py` and `ch06_rag_chat.py` both reject anything that isn't a plain lowercase identifier before it touches SQL, and use `psycopg.sql.Identifier` to quote it correctly rather than dropping it into an f-string. For a script you run yourself against your own database this is a small concern; the habit is worth keeping anyway, because the same shortcut in a script that takes a table name from a web form is a real SQL injection hole.

Ask a question

`ch06_rag_chat.py` takes three required arguments — `model`, `table`, `question`, in that order — plus `--host` and `--port` for Ollama (both default to `localhost` and `11434`, so you only need them if Ollama runs elsewhere):

```
python data/ch06_rag_chat.py llama3.1:8b portsmith_rag \
    "How do I get a business license in Portsmouth?" --show-context
```

```
--- retrieved context ---
```

```
[getting_a_business_license.txt#0 dist=0.2743] Getting a Business License in Portsmouth
Anyone opening a business within city limits needs a business...
[getting_a_business_license.txt#3 dist=0.5796] to change their registered category - a
retail shop adding a small cafe counter, for example - need ...
[getting_a_business_license.txt#2 dist=0.6149] scratch rather than simply renewing,
which means a new round of inspections for categories that requ...
[newcomer_guide.txt#0 dist=0.6176] A Newcomer's Guide to Portsmouth Welcome to
Portsmouth. This guide covers the basics every new residen...
```

To get a business license in Portsmouth, you need to start online through the city's permitting portal, where you choose a business category since it determines which inspections and follow-up permits apply. Most applications are processed within two to three weeks, but those involving food service or alcohol take longer because they require scheduled in-person inspections.

`--show-context` prints exactly what got retrieved before the model ever saw the question — the same top-K cosine-distance query from Exercise 5, just with `content` instead of `title` as the payload. The answer isn't copied from any single chunk; it's the model synthesizing across the top few, which is the actual value RAG adds over raw semantic search: Exercise 5 would have handed a person four ranked chunks to read themselves, this hands them one sentence that already did the reading.

Reproducibility, honestly: the retrieved chunks and their `dist` values above are fully deterministic — rerun the exact query and you’ll get the exact same four chunks in the exact same order, because embedding and cosine distance are pure math. The generated *answer* is not — Ollama samples from the model’s output distribution, so wording will differ run to run (sometimes prose, sometimes a bulleted list) even against identical context. That’s a property of the generation half of RAG, not the retrieval half; don’t expect to reproduce this exact paragraph verbatim.

What happens when the answer isn’t in the documents

This is the case that actually matters for trusting a RAG system — ask something the ingested documents have no way of answering:

```
python data/ch06_rag_chat.py llama3.1:8b portsmouth_rag \  
  "What is the capital of France?" --show-context
```

```
--- retrieved context ---
```

```
[newcomer_guide.txt#1 dist=0.9228] river on the city's western edge and is primarily  
residential, with the new dog park, several parks ...
```

```
[utilities_and_public_works.txt#1 dist=0.9272] requires temporarily shutting off  
service to a section of a neighbourhood, Public Works issues a boi...
```

```
[newcomer_guide.txt#0 dist=0.9361] A Newcomer's Guide to Portsmouth Welcome to  
Portsmouth. This guide covers the basics every new residen...
```

```
[newcomer_guide.txt#2 dist=0.9475] alike. Noise ordinance enforcement is more actively  
watched here than in other neighbourhoods given ...
```

```
I don't know. The context only provides information about the city of Portsmouth, its  
neighbourhoods, and various services offered by the city, but does not mention the  
capital of France.
```

Two things to notice. First, retrieval **still returned four chunks** — exactly as Exercise 5 described, cosine similarity search never returns “no results,” it returns the *least bad* matches available, and here they’re genuinely irrelevant. Second, look at the distances: 0.92–0.95, versus 0.27–0.61 for the business license question. That gap is the signal a production system would actually act on — this script’s prompt template happens to lean on the model’s own judgment to say “I don’t know,” but a more defensive design would check the top distance against a threshold *before* even calling the LLM, and skip generation entirely for a question this poorly matched. Relying on the model to notice the context is irrelevant works here because `llama3.1:8b` is reasonably well-behaved about it — it is not a guarantee every model or every prompt will get right, and it’s the single biggest reliability risk in any RAG system: nothing stops a language model from

answering confidently from weak or irrelevant context if the prompt doesn't insist otherwise.

The prompt: the actual glue between the two systems

`ask_ollama()` in `ch06_rag_chat.py` sends the model one plain-text prompt via Ollama's `/api/generate` endpoint — no special “RAG mode” exists in Ollama or in the model itself, it's just a string:

```
You are a help-desk assistant for the city of Portsmouth. Answer the question using ONLY the context below. If the context doesn't contain the answer, say you don't know rather than guessing.
```

```
Context:
```

```
(getting_a_business_license.txt) Getting a Business License in Portsmouth Anyone opening a business ...
```

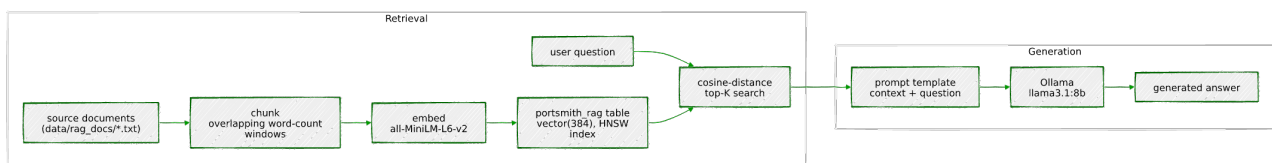
```
(getting_a_business_license.txt) to change their registered category - a retail shop adding a small ...
```

```
...
```

```
Question: How do I get a business license in Portsmouth?
```

```
Answer:
```

This is the entire mechanism. There is no fine-tuning, no special API, nothing model-specific — “retrieval-augmented generation” is a prompt template plus a database query, which is exactly why it was worth showing you the whole thing at this level rather than reaching for a framework. The whole chapter, end to end:



*Going further: this section's retrieval is intentionally the simplest version that works — top-K by raw cosine distance, no re-ranking, no distance-threshold cutoff before generation, no limit on how many tokens of context get sent to the model regardless of length. Production RAG systems typically add a **re-ranking** step (retrieve a wider candidate set cheaply, then re-score the top candidates with a slower, more accurate model before picking the final context), **hybrid retrieval** (blend `ts_rank` back in, exactly as Exercise 6 did, since keyword precision and semantic recall are just as complementary here as they were there), and explicit **citations** in the generated answer so a user can verify a claim against the source*

chunk rather than trusting the model's summary outright. None of that changes the shape of what you built here — it's still retrieval, then generation, with more care taken at each step.

Chapter 7 — IP and Network Filtering: `ip4r`

“An IP address is a number. A CIDR block is a range. PostgreSQL’s built-in types treat the first well and the second as an afterthought — `ip4r` treats the range as the whole point.”

Background

PostgreSQL already ships `inet` and `cidr` — you don’t strictly need an extension to store an IP address. So why does one exist? Because `inet` and `cidr` were designed for storing addresses and networks, not for answering the question a security team actually asks all day: “*is this address inside any of these ranges?*” — over and over, at high volume, fast. `cidr` is also stricter than it looks: it rejects a value with any host bits set relative to its mask (`192.168.1.5/24` is refused outright — more on that in Exercise 1), and neither built-in type ships a purpose-built index structure for “find every range containing this point” or “find every point inside this range” at scale. Every containment check against a large blocklist falls back to a sequential scan unless you reach for something else.

`ip4r` is that something else: a dedicated `ip4` type for single IPv4 addresses, an `ip4r` type for arbitrary IPv4 ranges (CIDR-aligned or not — also unlike `cidr`), a GiST opclass built specifically for fast range containment, and — despite the name — full IPv6 equivalents (`ip6`, `ip6r`) plus polymorphic `ipaddress`/`iprange` types that work with either family. This chapter builds a small network security monitoring setup around it: logging access attempts, maintaining block and allow lists, indexing them properly, and — because real blocklists and allowlists eventually disagree with each other — finding and resolving the conflicts that creates.

The Scenario

Portsmouth’s online services — the resident portal, the business licensing API from Chapter 5, the permit dashboard — all sit behind the same logging and access-control layer. Every login attempt and API call gets recorded, and a security team maintains two lists: a **blocklist** of ranges observed doing something hostile (brute-force login attempts, known

Tor exit nodes, botnet activity), and an **allowlist** of ranges that should never be blocked regardless — the city’s own VPN, a trusted vendor’s office network.

Table	Purpose
<code>network_events</code>	One row per login attempt or API call — type, source IP, timestamp
<code>blocklists</code>	CIDR ranges flagged as malicious, with a category and description
<code>allowlists</code>	CIDR ranges that should never be blocked, regardless of blocklist entries

Every range in this chapter’s data uses IANA-reserved, non-routable address space (the RFC 5737 documentation blocks `192.0.2.0/24`, `198.51.100.0/24`, `203.0.113.0/24`, and the reserved `240.0.0.0/4` block) — synthetic security data, not a real blocklist, using addresses that were never going to belong to anyone.

Exercise Goals

By the end of this chapter you will be able to:

- Explain what `ip4r` gets you over the built-in `inet/cidr` types, in storage and in what values each type actually accepts.
 - Use the containment operators `>>` and `<=<` to check whether an address falls inside a range, and handle the case where it matches more than one.
 - Build a GiST index for network containment queries — and know why it can’t go directly on an `ip4` column, only an `ip4r` one.
 - Aggregate events by subnet using both `ip4r`-native functions and the built-in `network()/masklen()` functions via a cast.
 - Detect real conflicts between an allowlist and a blocklist using the overlap operator `&&`.
 - Build a GiST-indexed function that resolves those conflicts and answers “is this IP blocked?” in one call.
 - *(Additional)* Implement a token-bucket rate limiter keyed on `ip4r`, applied at both the host and `/24` netblock level, and explain why distributed abuse needs the second bucket to be caught at all.
-

Installation

```
sudo apt install -y postgresql-16-ip4r
```

Enable it. Like `pgvector` in Chapter 6, `ip4r`'s control file does not set `trusted = true`, so a regular database-owning role can't self-serve this one either — it has to be done once, by a superuser:

```
sudo -u postgres psql portsmith -c "CREATE EXTENSION ip4r;"
```

Confirm it:

```
SELECT extversion FROM pg_extension WHERE extname = 'ip4r';
```

```
extversion
-----
2.4
```

If it doesn't show up: double-check which database you ran the `CREATE EXTENSION` against. `psql -l` lists every database on the cluster — it's worth a glance if an extension you just enabled doesn't appear where you expect it.

Loading the Data

Run the seed script

```
python data/ch07_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 4 blocklist entries ...
Inserting 3 allowlist entries ...
Inserting 116 network events ...
Done - 116 rows in network_events, 4 blocklist entries, 3 allowlist entries.
```

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:


```
\d network_events
```

```
Table "public.network_events"
  Column      |          Type          | Collation | Nullable | 
Default       |                        |           |          | 
-----+-----+-----+-----+
id            | bigint                 |           | not null | 
nextval('network_events_id_seq'::regclass)
event_type    | text                   |           | not null | 
source_ip     | ip4                     |           | not null | 
occurred_at   | timestamp with time zone |           | not null | 
detail        | text                    |           | not null | 
Indexes:
    "network_events_pkey" PRIMARY KEY, btree (id)
Check constraints:
    "network_events_event_type_check" CHECK (event_type = ANY
 (ARRAY['login_success'::text, 'login_failure'::text, 'api_call'::text,
 'api_error'::text]))
```

Check 2 — event counts by type:

```
SELECT event_type, COUNT(*) FROM network_events GROUP BY event_type ORDER BY event_type;
```

```
 event_type | count
-----+-----
api_call    |    47
api_error   |     9
login_failure |   41
login_success |   19
(4 rows)
```

Check 3 — list counts:

```
SELECT COUNT(*) FROM blocklists;  -- 4
SELECT COUNT(*) FROM allowlists;  -- 3
```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — ip4r vs. the Built-in inet/cidr

1.1 — Storage

```
SELECT pg_column_size('203.0.113.5'::ip4) AS ip4_size,
       pg_column_size('203.0.113.5'::inet) AS inet_size,
       pg_column_size('203.0.113.0/24'::ip4r) AS ip4r_size,
       pg_column_size('203.0.113.0/24'::cidr) AS cidr_size;
```

```
ip4_size | inet_size | ip4r_size | cidr_size
-----+-----+-----+-----
         4 |         10 |          8 |         10
```

`ip4` is a fixed 4-byte integer — nothing but the address. `inet` costs 10 bytes because it's a variable-length type carrying an address family byte and a netmask alongside the 4 address bytes, general enough to also hold IPv6. `ip4r` stores a range as two 4-byte bounds (8 bytes) rather than a network+prefix pair, and `cidr` costs the same 10 bytes as `inet` for the same reason. At one row this is noise; at hundreds of millions of rows in a real event log, 4 bytes versus 10 is not.

1.2 — `cidr` is stricter than you might expect

```
SELECT '192.168.1.5/24'::cidr;
```

```
ERROR:  invalid cidr value: "192.168.1.5/24"
DETAIL:  Value has bits set to right of mask.
```

`cidr` refuses any value where the host portion isn't all zero for the given mask — it only accepts genuine network addresses. `inet` is more permissive and keeps the host bits as part of the value:

```
SELECT '192.168.1.5/24'::inet;
```

```
inet
-----
192.168.1.5/24
```

1.3 — `ip4r` doesn't require CIDR alignment at all

```
SELECT '203.0.113.5-203.0.113.20'::ip4r;
```

```
ip4r
-----
203.0.113.5-203.0.113.20
```

```
SELECT '203.0.113.5-203.0.113.20'::cidr;
```

```
ERROR:  invalid input syntax for type cidr: "203.0.113.5-203.0.113.20"
```

This is the real semantic difference, not just performance: `cidr` and `inet` can only represent power-of-two, mask-aligned networks. `ip4r` is a genuine range type — a start

and an end bound, no requirement that they correspond to any CIDR block at all. A DHCP pool spanning .5 through .20 is a completely ordinary `ip4r` value and simply not expressible as a single `cidr`.

1.4 — The wider family

`ip4r` (the extension) is not limited to IPv4 despite the name:

```
\dx+ ip4r
```

lists casts and functions for `ip4`, `ip4r`, `ip6`, `ip6r`, and two polymorphic types, `ipaddress` and `iprange`, that accept either address family. This chapter sticks to IPv4 to match `network_events` and keep the examples concrete, but everything here has a direct IPv6 equivalent.

Exercise 2 — Containment: `>>` and `<<=`

2.1 — “Does any blocklist entry contain this address?”

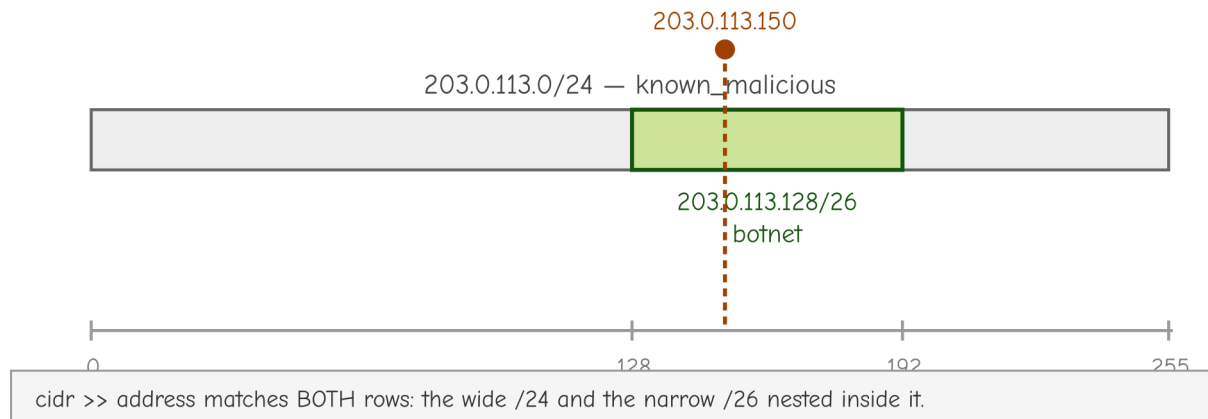
```
SELECT id, category, description
FROM   blocklists
WHERE  cidr >> '203.0.113.150'::ip4;
```

id	category	description
1	known_malicious	Repeated brute-force login attempts against the resident portal, flagged by the security team
2	botnet	Subrange within 203.0.113.0/24 attributed to a specific credential-stuffing botnet

(2 rows)

`>>` means “left range contains right value.” Two rows come back, not one — `203.0.113.150` falls inside both the broad `203.0.113.0/24` entry *and* the narrower `203.0.113.128/26` botnet subrange nested inside it. Real blocklists routinely have this shape: a wide, low-confidence range alongside a narrow, high-confidence one carved out of it.

203.0.113.150 falls inside both ranges at once — a real "multi-match"



2.2 — The same check, written the other way around

```
SELECT id, category, description
FROM   blocklists
WHERE  '203.0.113.150'::ip4 <= cidr;
```

Identical result set. `<=` means “left value is contained by (or equal to) right range” — `a >> b` and `b <= a` are the same test from opposite sides. Which one reads more naturally depends on which value you think of as the “subject” of the query; both compile to the same containment check.

2.3 — Handle the multi-match case deliberately

Since Exercise 2.1 showed a single address can match more than one blocklist entry, any real lookup needs to decide what to do with that — return every match for an audit trail, or the most specific one for a quick yes/no. Exercise 6 builds the second.

Exercise 3 — GiST Indexing (and a Real Gotcha)

3.1 — The unindexed cost

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id, event_type, source_ip
FROM   network_events
WHERE  source_ip <= '203.0.113.0/24'::ip4r;
```

QUERY PLAN

Seq Scan on network_events (cost=0.00..3.74 rows=1 width=44) (actual time=0.006..0.016 rows=38 loops=1)

```
Filter: ((source_ip)::ip4r <= '203.0.113.0/24'::ip4r)
Rows Removed by Filter: 78
```

38 of 116 events fall inside that /24. At 116 rows this costs nothing; at production log volume, a sequential scan per lookup is exactly the bottleneck `ip4r` exists to remove.

3.2 — The naive index fails

```
CREATE INDEX idx_network_events_source_ip
ON network_events USING GIST (source_ip);
```

```
ERROR: data type ip4 has no default operator class for access method "gist"
HINT: You must specify an operator class for the index or define a default operator
class for the data type.
```

This is worth sitting with rather than working around blindly: `ip4r` (the extension) ships a GiST operator class for `ip4r` (the **range** type) only.

```
SELECT opcname, amname
FROM pg_opclass oc JOIN pg_am am ON am.oid = oc.opcmethod
WHERE opcname ILIKE '%ip4%';
```

opcname	amname
-----+-----	
btree_ip4_ops	btree
btree_ip4r_ops	btree
hash_ip4_ops	hash
hash_ip4r_ops	hash
gist_ip4r_ops	gist

`ip4` (a single address) has B-tree and hash opclasses for equality and ordering, but no GiST opclass at all — GiST is for indexing *containment and overlap*, which only makes sense for a range. `network_events.source_ip` is declared `ip4`, a plain address column, so there's nothing there for GiST to build against directly.

3.3 — The fix: index the range-cast expression

```
CREATE INDEX idx_network_events_source_ip
ON network_events USING GIST ((source_ip::ip4r));
```

Every `ip4` value casts losslessly to a single-address `ip4r` (a range whose start and end are the same address), which does have a GiST opclass. This is an **expression index** — it indexes the result of `source_ip::ip4r`, not the raw column — so queries have to use the same cast for the planner to recognize a match:

```
SET enable_seqscan = off;
```

```

EXPLAIN (ANALYZE, BUFFERS)
SELECT id, event_type, source_ip
FROM   network_events
WHERE  source_ip::ip4r <= '203.0.113.0/24'::ip4r;

SET enable_seqscan = on;

```

QUERY PLAN

```

Index Scan using idx_network_events_source_ip on network_events (cost=0.14..8.15
rows=1 width=44) (actual time=0.105..0.110 rows=38 loops=1)
  Index Cond: ((source_ip)::ip4r <= '203.0.113.0/24'::ip4r)

```

Same 38 rows, now via `Index Scan`. The lesson generalizes beyond `ip4r`: when a GiST opclass exists for a *range* type but your column stores *points*, an expression index bridges the gap — cast to the range type in both the index definition and every query that should use it.

Exercise 4 — Aggregating by Subnet

4.1 — The `ip4r`-native way

```

SELECT ip4r_net_prefix(source_ip, 24) AS subnet_24, COUNT(*) AS events
FROM   network_events
GROUP BY subnet_24
ORDER BY events DESC, subnet_24;

```

subnet_24	events
203.0.113.0/24	38
192.0.2.0/24	31
198.51.100.0/24	28
240.1.2.0/24	11
100.64.5.0/24	4
100.64.9.0/24	4

(6 rows)

`ip4r_net_prefix(address, prefix_length)` computes the containing network for an address directly, staying entirely within `ip4r`'s own types — no cast round-trip needed.

4.2 — The same result via the built-in `network()`/`set_masklen()`

`ip4` casts directly to `cidr`, so the standard PostgreSQL network functions work too, if you'd rather not learn `ip4r`-specific function names:

```
SELECT network(set_masklen(source_ip::cidr, 24)) AS subnet_24, COUNT(*) AS events
FROM   network_events
GROUP BY subnet_24
ORDER BY events DESC, subnet_24;
```

Identical results. `set_masklen()` overrides the prefix length on a `cidr/inet` value, and `network()` zeroes out the host bits to return the network address — two built-in functions doing in two steps what `ip4r_net_prefix()` does in one.

4.3 — `masklen()` on the blocklist itself

```
SELECT cidr, masklen(cidr::cidr) AS prefix_length, category
FROM   blocklists ORDER BY id;
```

cidr	prefix_length	category
203.0.113.0/24	24	known_malicious
203.0.113.128/26	26	botnet
240.1.2.0/25	25	tor_exit_node
198.51.100.0/28	28	brute_force_source

Worth noticing as a pattern, not just a query result: the narrower the prefix (higher number, smaller range), the more specific and higher-confidence the category tends to be here — `/24` for a broad “something’s wrong in this range” flag, `/26` and `/28` for a credential-stuffing botnet and an automated feed’s specific finding. Range size is itself a signal about how much to trust an entry.

Exercise 5 — Overlap Detection Between Allow and Block Lists

5.1 — Find every conflict with `&&`

```
SELECT b.id AS block_id, b.cidr AS blocked_range, b.category,
       a.id AS allow_id, a.cidr AS allowed_range, a.description
FROM   blocklists b
JOIN   allowlists a ON b.cidr && a.cidr
ORDER BY b.id, a.id;
```

block_id	blocked_range	category	allow_id	allowed_range	
	description				
4	198.51.100.0/28	brute_force_source	1	198.51.100.0/24	Portsmouth City Hall internal network and staff VPN

(1 row)

&& is the general overlap test — true if the two ranges share *any* address, regardless of which contains which. One real conflict: an automated brute-force detection feed flagged 198.51.100.0/28, a sub-range that sits entirely inside the city’s own allowlisted VPN block. This is exactly the failure mode automated blocklist feeds produce in practice — someone’s VPN concentrator fails logins at a rate that looks like an attack from outside, and gets flagged from inside a range that was explicitly trusted.

5.2 — Confirm it isn’t hypothetical

```
SELECT DISTINCT source_ip
FROM   network_events
WHERE  source_ip::ip4r <= '198.51.100.0/28'::ip4r
ORDER BY source_ip;
```

```
source_ip
```

```
-----
198.51.100.5
198.51.100.7
198.51.100.12
```

Three real addresses in `network_events` sit inside the disputed range — this isn’t an edge case sitting unused in the blocklist table, it’s actively affecting real logged traffic from the city’s own network.

Exercise 6 — A Real-Time “Is This IP Blocked?” Function

6.1 — Index the lookup tables themselves

Exercise 3 needed an expression index because `network_events.source_ip` is `ip4` (a point). `blocklists.cidr` and `allowlists.cidr` are already `ip4r`, so they index directly, no casting required:

```
CREATE INDEX idx_blocklists_cidr ON blocklists USING GIST (cidr);
CREATE INDEX idx_allowlists_cidr ON allowlists USING GIST (cidr);
```

```
SET enable_seqscan = off;
```

```
EXPLAIN (ANALYZE, BUFFERS)
```

```
SELECT category, description FROM blocklists WHERE cidr > '203.0.113.150'::ip4;
```

```
SET enable_seqscan = on;
```

QUERY PLAN

```
Index Scan using idx_blocklists_cidr on blocklists (cost=0.13..8.15 rows=1 width=64)
(actual time=0.073..0.074 rows=2 loops=1)
  Index Cond: (cidr >> '203.0.113.150'::ip4r)
```

6.2 — The function: allowlist wins

Exercise 5 found a real conflict. A production “is this blocked?” check has to resolve it one way, consistently — this function checks the allowlist *first* and short-circuits if it matches, so a trusted range is never blocked no matter what an automated feed says about a sub-range of it:

```
CREATE OR REPLACE FUNCTION is_blocked(check_ip ip4)
RETURNS TABLE (blocked BOOLEAN, reason TEXT) AS $$
BEGIN
  IF EXISTS (SELECT 1 FROM allowlists WHERE cidr >> check_ip) THEN
    RETURN QUERY SELECT FALSE, 'allowlisted'::TEXT;
    RETURN;
  END IF;

  RETURN QUERY
  SELECT TRUE, b.category || ': ' || b.description
  FROM   blocklists b
  WHERE  b.cidr >> check_ip
  LIMIT 1;

  IF NOT FOUND THEN
    RETURN QUERY SELECT FALSE, 'not listed'::TEXT;
  END IF;
END;
$$ LANGUAGE plpgsql STABLE;
```

LIMIT 1 on the blocklist branch is deliberate — Exercise 2 showed an address can match multiple blocklist entries, and for a yes/no gate one matching reason is enough; an audit tool would drop the LIMIT and return every match instead.

6.3 — Test it against every case this chapter built

```
SELECT * FROM is_blocked('203.0.113.150'); -- plain blocklist match
```

blocked	reason
t	known_malicious: Repeated brute-force login attempts against the resident portal, flagged by the security team

```
SELECT * FROM is_blocked('198.51.100.7'); -- the Exercise 5 conflict
```

```
blocked | reason
-----+-----
f       | allowlisted
```

Blocklisted *and* allowlisted, and the function correctly refuses to block it — exactly the resolution Exercise 5’s conflict needed.

```
SELECT * FROM is_blocked('192.0.2.10');    -- vendor allowlist, no conflict
SELECT * FROM is_blocked('8.8.8.8');       -- not on any list
```

```
blocked | reason
-----+-----
f       | allowlisted
```

```
blocked | reason
-----+-----
f       | not listed
```

Four calls, four distinct real outcomes, all backed by GiST-indexed lookups against tables that would scale to millions of blocklist entries without changing a line of this function.

Exercise 7 (Additional) — Rate Limiting by Host *and* /24 with a Token Bucket

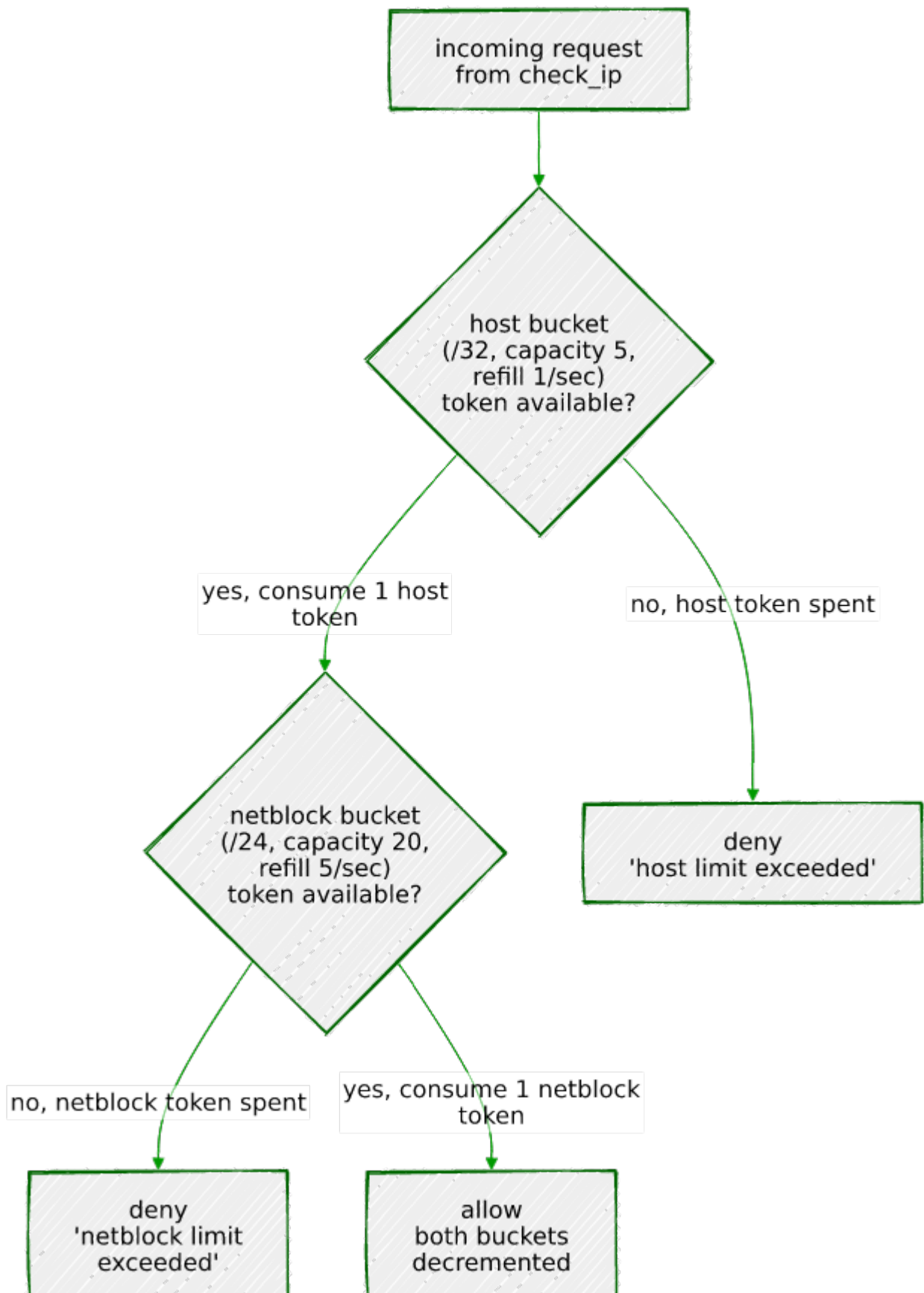
Blocklists are a permanent, deliberate verdict — someone reviewed a range and decided it’s hostile. Rate limiting is different: it’s a *temporary* throttle applied to traffic that isn’t necessarily malicious, just too frequent, and it needs to make that decision on every single request, fast. This exercise builds one of the most common shapes for it — a **token bucket** — and applies it at two levels at once: per individual host, and per /24 netblock, because some abuse only shows up when you stop looking at hosts one at a time.

7.1 — The token bucket model

Each bucket has a **capacity** (the size of a burst it can absorb all at once) and a **refill rate** (tokens added per second, up to capacity). Every request tries to consume one token: if at least one is available, the request is allowed and a token is spent; if not, it’s denied. Tokens refill continuously based on real elapsed time, not on a fixed clock tick — a bucket that’s been idle for ten seconds has ten seconds’ worth of refill waiting, whether or not anything asked.

7.2 — Why two buckets, not one

A per-host limit alone has a blind spot: ten different addresses in the same /24, each individually staying just under the limit, add up to ten times the traffic that netblock was ever supposed to send — a classic shape for a botnet spread across one compromised network, or NAT'd traffic from behind a single gateway. A per-host bucket alone never trips. Checking a *second*, looser bucket keyed to the containing /24 catches exactly that pattern, without needing to lower the per-host limit enough to hurt legitimate single users.



`check_rate_limit()`, built below, is exactly this diagram: the host bucket is checked — and consumed — first, and only a request that clears it goes on to spend a netblock token too.

7.3 — Schema: bucket keyed by `ip4r`, not `ip4`

```
CREATE TABLE rate_limit_buckets (  
    bucket      ip4r PRIMARY KEY,  
    capacity    NUMERIC NOT NULL,  
    refill_rate NUMERIC NOT NULL,  
    tokens      NUMERIC NOT NULL,  
    updated_at  TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp()  
);
```

Using `ip4r` as the key, rather than `ip4` plus a separate “is this a host or a netblock” flag, means a host bucket and a netblock bucket are just two different-sized ranges sharing one table and one lookup path — a host is stored as its `/32`, a netblock as its `/24`.

7.4 — The atomic refill-and-consume function

The refill math and the allow/deny decision have to happen as one atomic operation — two concurrent requests against the same bucket must not both read “4 tokens available” and both proceed, the same race Chapter 3 built `FOR UPDATE SKIP LOCKED` to avoid in the job queue. Here the fix is simpler — plain `FOR UPDATE`, no `SKIP LOCKED` — because a rate limit *should* make the second concurrent request wait its turn for the lock rather than skip ahead to a different row; there’s only one row per bucket, not a queue of interchangeable ones:

```
CREATE OR REPLACE FUNCTION try_consume_bucket(  
    target_bucket ip4r,  
    p_capacity    NUMERIC,  
    p_refill_rate NUMERIC  
) RETURNS TABLE (allowed BOOLEAN, tokens_remaining NUMERIC) AS $$  
BEGIN  
    INSERT INTO rate_limit_buckets (bucket, capacity, refill_rate, tokens, updated_at)  
    VALUES (target_bucket, p_capacity, p_refill_rate, p_capacity, clock_timestamp())  
    ON CONFLICT (bucket) DO NOTHING;  
  
    RETURN QUERY  
    WITH refilled AS (  
        SELECT b.bucket,  
               LEAST(b.capacity, b.tokens + b.refill_rate *  
                     EXTRACT(EPOCH FROM (clock_timestamp() - b.updated_at))) AS  
available  
        FROM   rate_limit_buckets b  
        WHERE  b.bucket = target_bucket
```

```

        FOR UPDATE
    )
    UPDATE rate_limit_buckets u
    SET     tokens      = CASE WHEN r.available >= 1 THEN r.available - 1 ELSE
r.available END,
           updated_at = clock_timestamp()
    FROM   refilled r
    WHERE  u.bucket = r.bucket
    RETURNING (r.available >= 1), u.tokens;
END;
$$ LANGUAGE plpgsql;

```

The `INSERT ... ON CONFLICT DO NOTHING` lazily creates a bucket at full capacity the first time an address is ever seen — no separate provisioning step. Note `clock_timestamp()`, not `now()`: this is the same gotcha from Chapter 3 — `now()` freezes at transaction start, which would make every refill calculation inside one transaction compute against the same frozen instant instead of real elapsed wall-clock time.

7.5 — Combine host and netblock into one check

```

CREATE OR REPLACE FUNCTION check_rate_limit(check_ip ip4)
RETURNS TABLE (allowed BOOLEAN, reason TEXT, host_tokens NUMERIC, netblock_tokens
NUMERIC) AS $$
DECLARE
    host_result      RECORD;
    netblock_result  RECORD;
BEGIN
    -- Host bucket: tight limit, catches a single misbehaving address.
    SELECT * INTO host_result
    FROM   try_consume_bucket(check_ip::ip4r, 5, 1);

    IF NOT host_result.allowed THEN
        RETURN QUERY SELECT FALSE, 'host limit exceeded'::TEXT,
host_result.tokens_remaining, NULL::NUMERIC;
        RETURN;
    END IF;

    -- Netblock bucket: looser limit, catches abuse spread across many
    -- hosts in the same /24, each individually within their own budget.
    SELECT * INTO netblock_result
    FROM   try_consume_bucket(ip4r_net_prefix(check_ip, 24), 20, 5);

    IF NOT netblock_result.allowed THEN
        RETURN QUERY SELECT FALSE, 'netblock limit exceeded'::TEXT,
host_result.tokens_remaining, netblock_result.tokens_remaining;
        RETURN;
    END IF;

    RETURN QUERY SELECT TRUE, 'ok'::TEXT, host_result.tokens_remaining,

```

```
netblock_result.tokens_remaining;
END;
$$ LANGUAGE plpgsql;
```

The host bucket (capacity 5, refills 1/sec) is checked — and, critically, *consumed* — first. Only if the host bucket allows the request does the function go on to check the netblock bucket (capacity 20, refills 5/sec). This ordering is a real design decision, not an accident: it means a request that the netblock bucket ultimately rejects still spent a host token on the way there. The alternative — check both before consuming either — avoids that, at the cost of two extra reads per request. For a security throttle, spending a token on a request you were going to reject anyway is a rounding error; use whichever trade-off matches what you’re actually optimizing for.

7.6 — A single host bursting past its own limit

```
SELECT allowed, reason, round(host_tokens, 2), round(netblock_tokens, 2)
FROM   check_rate_limit('203.0.113.5');
-- run six times in a row
```

allowed	reason	host_tokens	netblock_tokens
t	ok	4.00	19.00
t	ok	3.05	18.27
t	ok	2.09	17.45
t	ok	1.13	16.63
t	ok	0.17	15.83
f	host limit exceeded	0.20	

Five rapid calls spend the host bucket’s burst capacity of 5 (with a sliver of refill accumulating between each real request — these were genuinely separate round trips, not a batch); the sixth is denied, purely on the host bucket. The netblock bucket barely moved. (Your own numbers will differ in the small decimals — they depend on the exact wall-clock gap between your requests — but the allow/allow/allow/allow/allow/deny pattern will not.)

7.7 — Distributed abuse across a /24: the case this whole exercise exists for

Five different hosts in the same netblock, each making five rapid requests — each individually within its own host budget:

```
-- for host in 203.0.113.10 .. 203.0.113.14, 5 calls each:
SELECT allowed, reason, round(host_tokens, 2), round(netblock_tokens, 2)
FROM   check_rate_limit(:host);
```

host	allowed	reason	host_tokens	netblock_tokens
203.0.113.10	t	ok	4.00	19.00
203.0.113.10	t	ok	3.04	18.19
203.0.113.10	t	ok	2.08	17.38
203.0.113.10	t	ok	1.11	16.57
203.0.113.10	t	ok	0.15	15.76
203.0.113.11	t	ok	4.00	14.94
203.0.113.11	t	ok	3.04	14.13
203.0.113.11	t	ok	2.07	13.32
203.0.113.11	t	ok	1.11	12.51
203.0.113.11	t	ok	0.15	11.70
203.0.113.12	t	ok	4.00	10.88
203.0.113.12	t	ok	3.04	10.07
203.0.113.12	t	ok	2.08	9.27
203.0.113.12	t	ok	1.11	8.46
203.0.113.12	t	ok	0.15	7.64
203.0.113.13	t	ok	4.00	6.82
203.0.113.13	t	ok	3.04	6.03
203.0.113.13	t	ok	2.08	5.23
203.0.113.13	t	ok	1.12	4.42
203.0.113.13	t	ok	0.16	3.62
203.0.113.14	t	ok	4.00	2.81
203.0.113.14	t	ok	3.04	2.02
203.0.113.14	t	ok	2.08	1.20
203.0.113.14	t	ok	1.11	0.38
203.0.113.14	f	netblock limit exceeded	0.15	0.56

Read the `host_tokens` column top to bottom: **every single request succeeds at the host level** — no address ever exhausts its own bucket, each one’s burst-of-5 fits comfortably within its own limit every time. And yet the 25th request, from the fifth host, is denied — for `netblock limit exceeded`, not a host limit. Twenty-five individually reasonable requests from five individually well-behaved addresses added up to more than the `/24` as a whole was ever budgeted for, and the second bucket caught it. This is precisely the pattern a per-host-only rate limiter is structurally blind to.

Summary — What You Should Now Know

Tool

`ip4 / ip4r`

`ip6 / ip6r / ipaddress / iprange`

What it does

Dedicated address and range types — 4 and 8 bytes, versus 10 for `inet/cidr`

The same machinery for IPv6, and family-agnostic polymorphic types

Tool	What it does
<code>cidr</code>	Stricter than <code>inet</code> — rejects any value with host bits set past the mask
<code>'a-b'::ip4r</code>	An arbitrary range, not required to be CIDR-aligned — something <code>cidr</code> cannot express at all
<code>a >> b / b <= a</code>	Containment, read from either direction — “does a contain b?” / “is b contained by a?”
<code>a && b</code>	Overlap — true if two ranges share any address, regardless of which contains which
<code>GIST (col) ON ip4r</code>	Fast containment/overlap lookups — but only for <code>ip4r</code> , <code>ip4</code> has no GiST opclass
<code>GIST ((col::ip4r)) ON ip4</code>	The expression-index fix when your column stores points, not ranges
<code>ip4r_net_prefix(ip, n)</code>	Native way to compute the containing /n network for an address
<code>network(set_masklen(col::cidr, n))</code>	The same computation via built-in <code>inet/cidr</code> functions
Token bucket, keyed by <code>ip4r</code>	Rate limiting at both host (/32) and netblock (/24) scope from one table, catching distributed abuse a per-host limit alone can't see

The key design insight from this chapter is that a blocklist and an allowlist are not just two tables you check in some order — they are two sources of truth that *will* disagree, and Exercise 5 found a real disagreement in four hand-written rows almost by accident. A security system that doesn't actively check for that disagreement is trusting whichever list happens to get consulted first, silently. `&&` makes finding every such conflict a single query; `is_blocked()` makes resolving it a permanent, enforced policy instead of a hope.

Going further: this chapter's `is_blocked()` resolves conflicts with a fixed rule — allowlist always wins. A more nuanced system might weight by range specificity (Exercise 4's observation that narrower ranges tend to be higher-confidence) or by how recently an entry was added. For very large blocklists — commercial threat-intelligence feeds run to millions of entries — also benchmark GiST against `ip4r`'s B-tree and hash opclasses for the access patterns that are pure equality or pure sorting rather than containment, since GiST is the right tool specifically for range queries, not a strictly faster B-tree replacement. And if this system needs to block traffic in real time rather than just log a verdict, the natural next step

is exposing `is_blocked()` at the application's edge — a reverse proxy or API gateway calling it per request — which is exactly the kind of function Chapter 10's PostgREST chapter would expose as an RPC endpoint. The same is true of `check_rate_limit()` from Exercise 7 — a real deployment calls it once per incoming request, before any other work happens, and a single `rate_limit_buckets` table means every application server sees the same limits regardless of which one handled the last request, something an in-process rate limiter can't offer without a shared store of its own. If per-request PL/pgSQL round trips ever become the bottleneck rather than the safeguard, that's the point at which a dedicated in-memory limiter (Redis and similar) starts to earn its keep — but plenty of systems never reach the traffic level where that trade-off is worth the added moving part.

Chapter 8 — Declarative Partitioning and BRIN Indexes

“A table doesn’t get slow because it’s big. It gets slow because every query has to consider all of it — partitioning doesn’t shrink the table, it shrinks the part any single query has to look at.”

Background

Every chapter so far has reached for an extension — `pg_trgm`, `pgvector`, `ip4r` — to get a data type or index PostgreSQL doesn’t ship with. This chapter is different: declarative partitioning (since PostgreSQL 10) and BRIN indexes (since PostgreSQL 9.5) are both built-in core features. No `CREATE EXTENSION` anywhere in this chapter. What they solve isn’t “how do I store this,” it’s “how do I keep a table that never stops growing from turning every query, every index, and every maintenance operation into a scan of everything that has ever happened.”

Partitioning splits one logical table into many physical ones, each holding a slice of the rows — by date range, by category, by hash — while every query still just says `SELECT ... FROM sensor_readings`. The planner decides, before it reads a single row, which physical partitions could possibly contain a match and skips the rest entirely. That’s **partition pruning**, and it’s the entire value proposition: a query asking for one month of data out of a year doesn’t pay for the other eleven months, not even to rule them out.

BRIN (Block Range INdex) is a different kind of index for a specific shape of data: values that are large in volume but naturally correlated with physical storage order. A B-tree index stores an entry for every single row and costs roughly proportional to the row count. A BRIN index stores one small summary — just a min and max — per range of physical disk pages (128 pages by default), and costs almost nothing regardless of how many rows are in the table. The catch, and it’s a real one: BRIN only helps if physical order and value order actually correlate. For a table where rows are inserted in timestamp order and essentially never updated — exactly what an append-only sensor log looks like — the correlation is close to perfect, and that’s the case this chapter builds.

The Scenario

Portsmouth has quietly wired up 120 sensors across the city — temperature sensors, traffic counters, and air quality monitors — all reporting on a schedule, all day, every day. A year of this is over ten million rows, and it only grows. The data starts life the way most time-series data actually starts: as one plain, unpartitioned table, because nobody provisions a partitioning scheme before they have a problem. This chapter is that problem arriving, and fixing it.

Table	Purpose
<code>sensors</code>	Metadata for all 120 sensors — type, label, neighborhood/street
<code>sensor_readings_staging</code>	~10.5 million raw readings for 2024, as one unpartitioned table
<code>sensor_readings</code>	<i>(built in this chapter)</i> the same data, partitioned by month
<code>sensor_readings_by_type</code>	<i>(built in this chapter)</i> a second copy, partitioned by type <i>and</i> month

`sensor_readings_staging` is the state the data starts in. Building `sensor_readings` from it — deciding the scheme, creating the partitions, backfilling — is the exercise, not something handed to you already done.

Exercise Goals

By the end of this chapter you will be able to:

- Design and create a range-partitioned table, and understand why the partition key has to be chosen before the first row is loaded, not after.
- Backfill a partitioned table from an existing unpartitioned one and confirm that PostgreSQL routes every row to the correct partition automatically.
- Read an `EXPLAIN` plan well enough to tell the difference between a query that touches one partition and one that touches all of them.
- Build a BRIN index and explain, using real numbers, why it can be hundreds of times smaller than a B-tree on the same column — and what it gives up to get there.
- Drop a month of data in milliseconds via `DROP TABLE` on a partition, and see exactly what that avoids compared to `DELETE` on an unpartitioned table.

- Combine two partitioning strategies — `LIST` over `RANGE` — into one table, and confirm the planner prunes on *both* dimensions at once.
-

Installation

Nothing to install. `PARTITION BY` and `USING BRIN` are both part of core PostgreSQL — if `psql --version` says 10 or later for partitioning, or 9.5 or later for BRIN, you already have everything this chapter uses. (The Portsmouth cluster is PostgreSQL 16, well past both.)

Loading the Data

Run the seed script

```
python data/ch08_seed.py
```

Expected output (the exact seconds will vary with your machine; the row counts will not):

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 120 sensors ...
Generating and loading readings (this takes a few minutes) ...
  generated January 2024 - 892,800 rows
  generated February 2024 - 835,200 rows
  generated March 2024 - 892,800 rows
  generated April 2024 - 864,000 rows
  generated May 2024 - 892,800 rows
  generated June 2024 - 864,000 rows
  generated July 2024 - 892,800 rows
  generated August 2024 - 892,800 rows
  generated September 2024 - 864,000 rows
  generated October 2024 - 892,800 rows
  generated November 2024 - 864,000 rows
  generated December 2024 - 892,800 rows
Indexing staging table on recorded_at (mirrors a typical pre-partitioning table) ...
Done - 10,540,800 rows in sensor_readings_staging (131.3s to generate + load), 120
sensors.
```

Months vary in row count because February is shorter and because temperature/traffic sensors report every 5 minutes while air quality sensors report every 15 — not a bug, just three sensor types on three different schedules, same as a real deployment would have.

Pin the session timezone before you do anything else

```
SET timezone = 'UTC';
```

This matters more than it looks like it should. TIMESTAMPTZ values are stored as absolute instants, but a bound like '2024-01-01' in a CREATE TABLE ... PARTITION OF ... FOR VALUES FROM (...) clause gets interpreted relative to the session's timezone *at the moment the partition is created*. Two people creating "the same" January partition in two different session timezones will get two different absolute boundaries. Run every example in this chapter in a UTC session — this SET at the top of each psql session is enough — and the partition boundaries and the displayed timestamps will both match what's printed below exactly.

Verify the load

Check 1 — the metadata table:

```
\d sensors
```

Table "public.sensors"					
Column	Type	Collation	Nullable	Default	
id	integer		not null		
sensor_type	text		not null		
label	text		not null		
neighborhood	text		not null		

Indexes:

"sensors_pkey" PRIMARY KEY, btree (id)

Check constraints:

"sensors_sensor_type_check" CHECK (sensor_type = ANY (ARRAY['temperature'::text, 'traffic'::text, 'air_quality'::text]))

Referenced by:

TABLE "sensor_readings_staging" CONSTRAINT "sensor_readings_staging_sensor_id_fkey" FOREIGN KEY (sensor_id) REFERENCES sensors(id)

Check 2 — the staging table structure:

```
\d sensor_readings_staging
```

Table "public.sensor_readings_staging"					
Column	Type	Collation	Nullable	Default	
id	bigint		not null	generated always as identity	
sensor_id	integer		not null		

sensor_type	text		not null
reading_value	double precision		not null
recorded_at	timestamp with time zone		not null

Indexes:

"idx_sensor_readings_staging_recorded_at" btree (recorded_at)

Check constraints:

"sensor_readings_staging_sensor_type_check" CHECK (sensor_type = ANY (ARRAY['temperature'::text, 'traffic'::text, 'air_quality'::text]))

Foreign-key constraints:

"sensor_readings_staging_sensor_id_fkey" FOREIGN KEY (sensor_id) REFERENCES sensors(id)

Note this table already has a B-tree index on `recorded_at` — a perfectly ordinary choice for “the column I filter on,” and exactly what a lot of real unpartitioned time-series tables look like right up until someone asks why disk usage keeps climbing.

Check 3 — sensor and reading counts by type:

```
SELECT sensor_type, COUNT(*) FROM sensors GROUP BY sensor_type ORDER BY sensor_type;
```

sensor_type	count
air_quality	30
temperature	50
traffic	40

(3 rows)

```
SELECT sensor_type, COUNT(*) FROM sensor_readings_staging GROUP BY sensor_type ORDER BY sensor_type;
```

sensor_type	count
air_quality	1054080
temperature	5270400
traffic	4216320

(3 rows)

If these match, proceed to the exercises.

Exercises

Exercise 1 — Designing and Creating the Partitioned Table

1.1 — Choose the partition key up front

Partitioning isn't something you bolt onto an existing table — `PARTITION BY` is part of `CREATE TABLE` and can't be added with `ALTER TABLE` afterward. That means the decision has to be made before a single row is loaded into the partitioned structure, based on how the table will actually be queried. `sensor_readings` is queried almost exclusively by time range ("show me last week," "roll up by month"), so `RANGE (recorded_at)`, monthly, is the natural key — coarse enough to keep the partition count manageable, fine enough that most queries touch only one or two partitions.

1.2 — Create the partitioned table and its monthly partitions

```
CREATE TABLE sensor_readings (  
  id          BIGINT GENERATED ALWAYS AS IDENTITY,  
  sensor_id   INTEGER NOT NULL REFERENCES sensors(id),  
  sensor_type TEXT NOT NULL  
              CHECK (sensor_type IN ('temperature', 'traffic', 'air_quality')),  
  reading_value DOUBLE PRECISION NOT NULL,  
  recorded_at TIMESTAMPTZ NOT NULL  
) PARTITION BY RANGE (recorded_at);  
  
CREATE TABLE sensor_readings_2024_01 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-01-01' ) TO ( '2024-02-01' );  
CREATE TABLE sensor_readings_2024_02 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-02-01' ) TO ( '2024-03-01' );  
CREATE TABLE sensor_readings_2024_03 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-03-01' ) TO ( '2024-04-01' );  
CREATE TABLE sensor_readings_2024_04 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-04-01' ) TO ( '2024-05-01' );  
CREATE TABLE sensor_readings_2024_05 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-05-01' ) TO ( '2024-06-01' );  
CREATE TABLE sensor_readings_2024_06 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-06-01' ) TO ( '2024-07-01' );  
CREATE TABLE sensor_readings_2024_07 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-07-01' ) TO ( '2024-08-01' );  
CREATE TABLE sensor_readings_2024_08 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-08-01' ) TO ( '2024-09-01' );  
CREATE TABLE sensor_readings_2024_09 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-09-01' ) TO ( '2024-10-01' );  
CREATE TABLE sensor_readings_2024_10 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-10-01' ) TO ( '2024-11-01' );  
CREATE TABLE sensor_readings_2024_11 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-11-01' ) TO ( '2024-12-01' );  
CREATE TABLE sensor_readings_2024_12 PARTITION OF sensor_readings FOR VALUES FROM  
( '2024-12-01' ) TO ( '2025-01-01' );  
CREATE TABLE sensor_readings_default PARTITION OF sensor_readings DEFAULT;
```

Each `FOR VALUES FROM (...) TO (...)` is a half-open interval — inclusive of the lower bound, exclusive of the upper — so consecutive months tile the year with no gap and no

overlap. The last one, `sensor_readings_default`, is not optional in any meaningful sense: without it, any row whose `recorded_at` doesn't fall in one of the twelve defined ranges is simply rejected at insert time. With it, such a row is accepted and routed there instead — silently, unless you go looking. Exercise 2 finds a case where that's exactly what happens.

1.3 — Confirm the structure

\d sensor_readings				
Partitioned table "public.sensor_readings"				
Column	Type	Collation	Nullable	Default
-----+-----+-----+-----				
+-----				
id	bigint		not null	generated always as identity
sensor_id	integer		not null	
sensor_type	text		not null	
reading_value	double precision		not null	
recorded_at	timestamp with time zone		not null	
Partition key: RANGE (recorded_at)				
Check constraints:				
"sensor_readings_sensor_type_check" CHECK (sensor_type = ANY (ARRAY['temperature'::text, 'traffic'::text, 'air_quality'::text]))				
Foreign-key constraints:				
"sensor_readings_sensor_id_fkey" FOREIGN KEY (sensor_id) REFERENCES sensors(id)				
Number of partitions: 13 (Use \d+ to list them.)				

`sensor_readings` itself has no storage of its own — it's a routing target. Every row you insert into it physically lands in exactly one of the thirteen tables underneath.

Exercise 2 — Backfilling and Watching Automatic Partition Routing

2.1 — Backfill from staging

```
INSERT INTO sensor_readings (sensor_id, sensor_type, reading_value, recorded_at)
SELECT sensor_id, sensor_type, reading_value, recorded_at
FROM sensor_readings_staging;
```

```
INSERT 0 10540800
Time: 87814.162 ms (01:27.814)
```

One `INSERT ... SELECT`, no destination named beyond `sensor_readings` itself — PostgreSQL evaluates each row's `recorded_at` against the partition bounds and routes it to the matching child table. Nothing in this statement says which of the thirteen tables any given row ends up in.

2.2 — Confirm the routing, per partition

```
SELECT tableoid::regclass AS partition, COUNT(*)
FROM   sensor_readings
GROUP BY tableoid
ORDER BY 1;
```

partition	count
sensor_readings_2024_01	892800
sensor_readings_2024_02	835200
sensor_readings_2024_03	892800
sensor_readings_2024_04	864000
sensor_readings_2024_05	892800
sensor_readings_2024_06	864000
sensor_readings_2024_07	892800
sensor_readings_2024_08	892800
sensor_readings_2024_09	864000
sensor_readings_2024_10	892800
sensor_readings_2024_11	864000
sensor_readings_2024_12	891648
sensor_readings_default	1152

(13 rows)

`tableoid` is a hidden system column every table has, and on a partitioned table it tells you which physical partition a given row actually lives in — the only reliable way to ask “where did this row end up?” after the fact.

2.3 — The row count that should stop you

Every month has the row count you’d expect from Exercise 0’s per-type counts — except December, which is short exactly 1,152 rows, and `sensor_readings_default`, which has exactly 1,152 rows in it. That’s not a coincidence:

```
SELECT sensor_id, MIN(recorded_at), MAX(recorded_at), COUNT(*)
FROM   sensor_readings
WHERE  tableoid = 'sensor_readings_default'::regclass
GROUP BY sensor_id;
```

sensor_id	min	max	count
17	2025-12-27 00:00:00+00	2025-12-31 23:55:00+00	1152

Sensor 17 — a temperature sensor — has a clock bug: for its last four days of reporting, it stamped its readings a full year fast. Nothing about the `INSERT` failed or warned; the rows are sitting exactly where the schema said out-of-range data should go. This is precisely the scenario a `DEFAULT` partition exists for, and precisely why “does the default partition

have anything in it?” is worth checking after any real backfill, not just this synthetic one — it’s the difference between a sensor firmware bug you catch in five minutes and one you catch when someone asks why a report is missing a device.

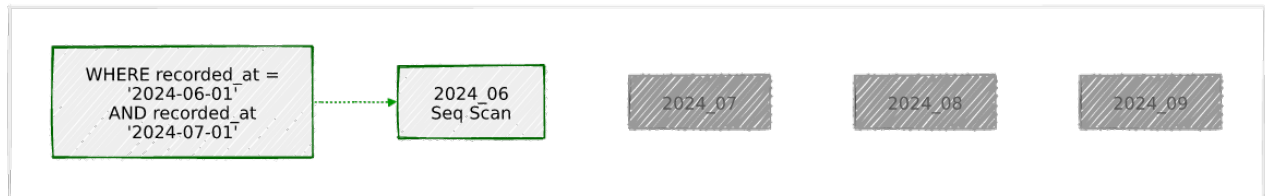
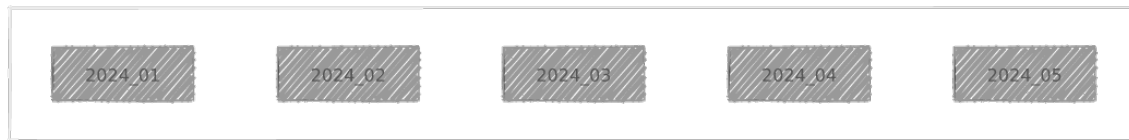
Exercise 3 — Partition Pruning, Seen in `EXPLAIN`

3.1 — A date-filtered query touches one partition

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT sensor_type, COUNT(*), round(AVG(reading_value)::numeric, 2)
FROM   sensor_readings
WHERE  recorded_at >= '2024-06-01' AND recorded_at < '2024-07-01'
GROUP BY sensor_type
ORDER BY sensor_type;
```

```
Finalize GroupAggregate (cost=12411.99..12480.46 rows=200 width=72) (actual
time=182.595..203.871 rows=3 loops=1)
  Group Key: sensor_readings.sensor_type
  Buffers: shared hit=106 read=7168, temp read=3264 written=3276
  -> Gather Merge (cost=12411.99..12473.96 rows=400 width=72) (actual
time=152.757..203.835 rows=9 loops=1)
    Workers Planned: 2
    Workers Launched: 2
    -> Partial GroupAggregate (cost=11411.97..11427.77 rows=200 width=72) (actual
time=98.794..149.785 rows=3 loops=3)
      Group Key: sensor_readings.sensor_type
      -> Sort (cost=11411.97..11415.42 rows=1380 width=40) (actual
time=93.166..120.544 rows=288000 loops=3)
        Sort Key: sensor_readings.sensor_type
        Sort Method: external merge  Disk: 8720kB
        -> Parallel Seq Scan on sensor_readings_2024_06 sensor_readings
(cost=0.00..11340.00 rows=1380 width=40) (actual time=0.034..37.199 rows=288000 loops=3)
          Filter: ((recorded_at >= '2024-06-01 00:00:00+00'::timestamp
with time zone) AND (recorded_at < '2024-07-01 00:00:00+00'::timestamp with time zone))
          Planning Time: 6.417 ms
          Execution Time: 205.458 ms
```

The `GROUP BY sensor_type` on top adds a sort and some parallel-worker bookkeeping that isn’t the point here — the line that matters is `Parallel Seq Scan on sensor_readings_2024_06`, singular. Eleven other monthly partitions and the default partition exist; none of them appear anywhere in this plan. The planner threw them out during planning, before execution ever started, based purely on the `WHERE` clause matching only June’s range.



3.2 — The same query, no date filter, for contrast

```
EXPLAIN
SELECT sensor_type, COUNT(*)
FROM   sensor_readings
GROUP BY sensor_type;
```

```
Finalize GroupAggregate (cost=156197.01..156247.68 rows=200 width=40)
-> Gather Merge (cost=156197.01..156243.68 rows=400 width=40)
    -> Sort (cost=155196.99..155197.49 rows=200 width=40)
        -> Partial HashAggregate (cost=155187.34..155189.34 rows=200 width=40)
            -> Parallel Append (cost=0.00..138351.15 rows=3367238 width=32)
                -> Parallel Seq Scan on sensor_readings_2024_01
sensor_readings_1 ...
                -> Parallel Seq Scan on sensor_readings_2024_03
sensor_readings_3 ...
                -> Parallel Seq Scan on sensor_readings_2024_05
sensor_readings_5 ...
                -> Parallel Seq Scan on sensor_readings_2024_07
sensor_readings_7 ...
                -> Parallel Seq Scan on sensor_readings_2024_08
sensor_readings_8 ...
                -> Parallel Seq Scan on sensor_readings_2024_10
sensor_readings_10 ...
                -> Parallel Seq Scan on sensor_readings_2024_12
sensor_readings_12 ...
                -> Parallel Seq Scan on sensor_readings_2024_04
sensor_readings_4 ...
                -> Parallel Seq Scan on sensor_readings_2024_06
sensor_readings_6 ...
                -> Parallel Seq Scan on sensor_readings_2024_09
sensor_readings_9 ...
```

```

                                -> Parallel Seq Scan on sensor_readings_2024_11
sensor_readings_11 ...
                                -> Parallel Seq Scan on sensor_readings_2024_02
sensor_readings_2  ...
                                -> Parallel Seq Scan on sensor_readings_default
sensor_readings_13 ...

```

With nothing to prune on, every one of the thirteen partitions shows up as its own Parallel Seq Scan under a Parallel Append. This is the direct, visible contrast: pruning isn't a performance optimization happening invisibly under the hood, it's a specific number of Seq Scan lines disappearing from the plan, and you can watch it happen by adding or removing a date filter on the exact same query.

Exercise 4 — BRIN vs. B-tree, With Real Numbers

4.1 — Why this data is a good fit for BRIN

BRIN's usefulness rests entirely on physical/value correlation — do rows with nearby recorded_at values actually sit near each other on disk? Since this data was generated and loaded in chronological order and never updated, it should be close to perfectly correlated:

```

ANALYZE sensor_readings;

SELECT tablename, attname, correlation, n_distinct
FROM   pg_stats
WHERE  tablename = 'sensor_readings_2024_09' AND attname = 'recorded_at';

```

tablename	attname	correlation	n_distinct
sensor_readings_2024_09	recorded_at	0.99979776	8622

correlation ranges from -1 to 1; 0.9998 is as close to “physically sorted by this column” as real data gets. This is the number that decides whether BRIN is a good idea here — not the row count, not the table size.

4.2 — Build both indexes on one partition, compare cost to build and size

```

CREATE INDEX idx_sr_2024_09_brin ON sensor_readings_2024_09 USING BRIN (recorded_at);
CREATE INDEX idx_sr_2024_09_btree ON sensor_readings_2024_09 USING BTREE (recorded_at);

```

```

CREATE INDEX
Time: 122.619 ms

```

```
CREATE INDEX
Time: 249.923 ms
```

```
SELECT relname, pg_size_pretty(pg_relation_size(oid)) AS index_size
FROM pg_class
WHERE relname IN ('idx_sr_2024_09_brin', 'idx_sr_2024_09_btree');
```

relname	index_size
idx_sr_2024_09_brin	24 kB
idx_sr_2024_09_btree	5800 kB

864,000 rows in this partition. The B-tree costs 5.8 MB — one entry per row, roughly what you'd expect. The BRIN index costs 24 KB: **about 240 times smaller**, because it isn't storing one entry per row, it's storing one min/max pair per 128-page block range — a few thousand summaries instead of 864,000 index entries. It also built about twice as fast. Neither number changes much if the table has a hundred million rows instead of 864,000; a B-tree's cost scales with row count, BRIN's scales with table *pages*, and pages-per-block-range is fixed.

4.3 — What that size difference costs you at query time

Force each index in turn and run the same narrow range query — one day out of the month:

```
SET enable_seqscan = off;
DROP INDEX idx_sr_2024_09_btree; -- isolate the BRIN index

EXPLAIN (ANALYZE, BUFFERS)
SELECT COUNT(*), round(AVG(reading_value)::numeric, 2)
FROM sensor_readings_2024_09
WHERE recorded_at >= '2024-09-15' AND recorded_at < '2024-09-16';
```

```
Aggregate (cost=7818.29..7818.30 rows=1 width=40) (actual time=7.350..7.351 rows=1
loops=1)
  Buffers: shared hit=9 read=384
    -> Bitmap Heap Scan on sensor_readings_2024_09 (cost=19.27..7674.10 rows=28836
width=8) (actual time=0.635..5.708 rows=28800 loops=1)
      Recheck Cond: ((recorded_at >= '2024-09-15 00:00:00+00'::timestamp with time
zone) AND (recorded_at < '2024-09-16 00:00:00+00'::timestamp with time zone))
      Rows Removed by Index Recheck: 17280
      Heap Blocks: lossy=384
      Buffers: shared hit=9 read=384
        -> Bitmap Index Scan on idx_sr_2024_09_brin (cost=0.00..12.06 rows=30322
width=0) (actual time=0.060..0.060 rows=3840 loops=1)
          Index Cond: ((recorded_at >= '2024-09-15 00:00:00+00'::timestamp with
```

```
time zone) AND (recorded_at < '2024-09-16 00:00:00+00'::timestamp with time zone))
Execution Time: 7.493 ms
```

Heap Blocks: lossy=384 and Rows Removed by Index Recheck: 17280 are BRIN being honest about what it is: a **lossy** index. It doesn't know which rows in a matching block range satisfy the condition, only that the range's min/max *could* contain a match — so it hands back every block in range and PostgreSQL rechecks the actual condition row by row once the blocks are read. That recheck cost is real, and it's the trade for the 240x smaller index.

Now the B-tree, same query:

```
DROP INDEX idx_sr_2024_09_brin;
CREATE INDEX idx_sr_2024_09_btree ON sensor_readings_2024_09 USING BTREE (recorded_at);

EXPLAIN (ANALYZE, BUFFERS)
SELECT COUNT(*), round(AVG(reading_value)::numeric, 2)
FROM   sensor_readings_2024_09
WHERE  recorded_at >= '2024-09-15' AND recorded_at < '2024-09-16';
```

```
Aggregate  (cost=1076.88..1076.89 rows=1 width=40) (actual time=5.781..5.782 rows=1
loops=1)
  Buffers: shared hit=384 read=27
    -> Index Scan using idx_sr_2024_09_btree on sensor_readings_2024_09
(cost=0.42..932.69 rows=28836 width=8) (actual time=0.055..3.552 rows=28800 loops=1)
      Index Cond: ((recorded_at >= '2024-09-15 00:00:00+00'::timestamp with time
zone) AND (recorded_at < '2024-09-16 00:00:00+00'::timestamp with time zone))
Execution Time: 5.850 ms
```

5.85 ms for the B-tree against 7.49 ms for BRIN — the B-tree is genuinely a bit faster here, exact and recheck-free by construction. But look at what each one costs to get that: 5.8 MB of index versus 24 KB, on *one month of one partition*. At the scale this chapter's title implies — many partitions, much more than a year of data — a B-tree on every partition's timestamp column stops being “a bit more disk” and starts being a meaningful fraction of the table's own size, while BRIN stays essentially free. For an append-only, time-correlated column like this one, that trade is rarely close.

4.4 — Keep the BRIN index, drop the comparison B-tree

```
DROP INDEX idx_sr_2024_09_btree;
CREATE INDEX idx_sr_2024_09_brin ON sensor_readings_2024_09 USING BRIN (recorded_at);
RESET enable_seqscan;
```

Recreate this same `USING BRIN (recorded_at)` index on every other monthly partition before moving on — Exercise 5 will drop one of them whole, and Exercise 6 revisits the size question at the whole-table level.

Exercise 5 — Dropping a Partition vs. `DELETE`

5.1 — Time the unpartitioned way: `DELETE`

`sensor_readings_staging` is still sitting there, unpartitioned, exactly as it was before this chapter touched anything. Delete a month's worth of rows from it the only way an unpartitioned table allows:

```
DELETE FROM sensor_readings_staging
WHERE recorded_at >= '2024-01-01' AND recorded_at < '2024-02-01';
```

```
DELETE 892800
Time: 815.293 ms
```

5.2 — Time the partitioned way: `DROP TABLE`

```
DROP TABLE sensor_readings_2024_01;
```

```
DROP TABLE
Time: 17.489 ms
```

Same amount of data — the January 2024 slice, 892,800 rows either way — and dropping the partition is roughly **47 times faster** than deleting the equivalent rows. That gap isn't a fluke of this particular run: it's two operations of fundamentally different cost. `DELETE` is an MVCC operation — PostgreSQL has to visit every matching row, mark each one dead, and write a WAL record for each one, transaction-safely, one row at a time. `DROP TABLE` on a partition is a catalog operation — it removes one entry saying “this table is part of that partition set” and unlinks the file. No row is individually touched because no row was individually anything; the whole partition was always one physical object.

5.3 — The part `DELETE` doesn't finish

```
SELECT n_live_tup, n_dead_tup,
pg_size_pretty(pg_relation_size('sensor_readings_staging'))
FROM pg_stat_user_tables
WHERE relname = 'sensor_readings_staging';
```

```
n_live_tup | n_dead_tup | pg_size_pretty
-----+-----+-----
 9648230 |    892800 | 687 MB
```

892,800 dead tuples, exactly matching what was just deleted — and the table is still 687 MB, unchanged, because `DELETE` doesn't reclaim disk space, it just marks rows as no-

longer-visible. Getting that space back needs a `VACUUM` (or `VACUUM FULL` to actually shrink the file), which is its own I/O-heavy pass over the table, usually running sometime later via `autovacuum` rather than as part of the `DELETE` itself. `DROP TABLE` skipped all of this — there is no bloat to vacuum, because there’s no table left to have any.

5.4 — A gotcha worth knowing about before it surprises you

```
SELECT pg_size_pretty(pg_total_relation_size('sensor_readings'));
```

```
pg_size_pretty
-----
0 bytes
```

That’s not an error and the data is not gone — `pg_total_relation_size()` on a *partitioned* table reports only the parent’s own storage, and a partitioned table’s parent has none; all the bytes live in the child partitions. To get the real total, sum across the children:

```
SELECT pg_size_pretty(SUM(pg_total_relation_size(inhrelid))) AS total_size
FROM   pg_inherits
WHERE  inhparent = 'sensor_readings'::regclass;
```

```
total_size
-----
629 MB
```

5.5 — Clean up the staging table

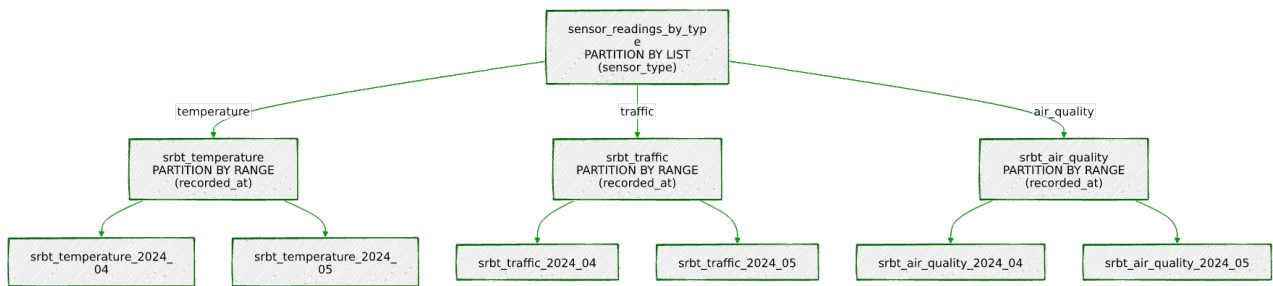
It’s done its job — everything from here on works from `sensor_readings`:

```
DROP TABLE sensor_readings_staging;
```

Exercise 6 — Combining `LIST` and `RANGE`: Subpartitioning

6.1 — Why one partition key isn’t always enough

`sensor_readings` prunes beautifully on `date`, but a query that also filters on `sensor_type` — “just the traffic sensors” — still has to scan a whole month’s worth of every sensor type, because `sensor_type` isn’t part of the partition key at all. PostgreSQL allows a second level: partition by `LIST (sensor_type)` at the top, then partition *each of those* by `RANGE (recorded_at)` underneath — a partition of a partition, usually called subpartitioning. Two routing decisions, made independently, before a row ever reaches physical storage:



The root and the three type branches hold no data of their own — they exist purely to route a row one level further down. Only the nine boxes at the bottom are real tables with real storage, and every one of this chapter’s rows lives in exactly one of them, decided first by `sensor_type`, then by which month `recorded_at` falls in.

6.2 — Build the two-level hierarchy

```

CREATE TABLE sensor_readings_by_type (
  id          BIGINT GENERATED ALWAYS AS IDENTITY,
  sensor_id    INTEGER NOT NULL,
  sensor_type  TEXT NOT NULL
                CHECK (sensor_type IN ('temperature', 'traffic', 'air_quality')),
  reading_value DOUBLE PRECISION NOT NULL,
  recorded_at  TIMESTAMPTZ NOT NULL
) PARTITION BY LIST (sensor_type);

CREATE TABLE srbt_temperature PARTITION OF sensor_readings_by_type
  FOR VALUES IN ('temperature') PARTITION BY RANGE (recorded_at);
CREATE TABLE srbt_traffic PARTITION OF sensor_readings_by_type
  FOR VALUES IN ('traffic') PARTITION BY RANGE (recorded_at);
CREATE TABLE srbt_air_quality PARTITION OF sensor_readings_by_type
  FOR VALUES IN ('air_quality') PARTITION BY RANGE (recorded_at);

-- Three months of subpartitions under each type branch:
CREATE TABLE srbt_temperature_2024_04 PARTITION OF srbt_temperature FOR VALUES FROM
('2024-04-01') TO ('2024-05-01');
CREATE TABLE srbt_temperature_2024_05 PARTITION OF srbt_temperature FOR VALUES FROM
('2024-05-01') TO ('2024-06-01');
CREATE TABLE srbt_temperature_2024_06 PARTITION OF srbt_temperature FOR VALUES FROM
('2024-06-01') TO ('2024-07-01');

CREATE TABLE srbt_traffic_2024_04 PARTITION OF srbt_traffic FOR VALUES FROM
('2024-04-01') TO ('2024-05-01');
CREATE TABLE srbt_traffic_2024_05 PARTITION OF srbt_traffic FOR VALUES FROM
('2024-05-01') TO ('2024-06-01');
CREATE TABLE srbt_traffic_2024_06 PARTITION OF srbt_traffic FOR VALUES FROM
('2024-06-01') TO ('2024-07-01');

CREATE TABLE srbt_air_quality_2024_04 PARTITION OF srbt_air_quality FOR VALUES FROM
('2024-04-01') TO ('2024-05-01');

```

```
CREATE TABLE srbt_air_quality_2024_05 PARTITION OF srbt_air_quality FOR VALUES FROM
('2024-05-01') TO ('2024-06-01');
CREATE TABLE srbt_air_quality_2024_06 PARTITION OF srbt_air_quality FOR VALUES FROM
('2024-06-01') TO ('2024-07-01');
```

Three months only, and only as a working example — building out all twelve for all three types is the same pattern repeated, not a new idea, and this chapter’s disk budget would rather spend on data than on empty demonstration partitions.

6.3 — Backfill from `sensor_readings` and inspect the tree

```
INSERT INTO sensor_readings_by_type (sensor_id, sensor_type, reading_value, recorded_at)
SELECT sensor_id, sensor_type, reading_value, recorded_at
FROM sensor_readings
WHERE recorded_at >= '2024-04-01' AND recorded_at < '2024-07-01';
```

```
INSERT 0 2620800
Time: 3389.698 ms (00:03.390)
```

```
SELECT relid::regclass AS relation, parentrelid::regclass AS parent, isleaf, level
FROM pg_partition_tree('sensor_readings_by_type')
ORDER BY level, 1;
```

relation	parent	isleaf	level
sensor_readings_by_type		f	0
srbt_air_quality	sensor_readings_by_type	f	1
srbt_temperature	sensor_readings_by_type	f	1
srbt_traffic	sensor_readings_by_type	f	1
srbt_air_quality_2024_04	srbt_air_quality	t	2
srbt_air_quality_2024_05	srbt_air_quality	t	2
srbt_air_quality_2024_06	srbt_air_quality	t	2
srbt_temperature_2024_04	srbt_temperature	t	2
srbt_temperature_2024_05	srbt_temperature	t	2
srbt_temperature_2024_06	srbt_temperature	t	2
srbt_traffic_2024_04	srbt_traffic	t	2
srbt_traffic_2024_05	srbt_traffic	t	2
srbt_traffic_2024_06	srbt_traffic	t	2

(13 rows)

`pg_partition_tree()` is the built-in way to see this shape directly instead of inferring it from table names: one root, three LIST branches at level 1, nine RANGE leaves at level 2 — every row actually lives in one of those nine leaf tables, never in an intermediate one.

6.4 — Confirm pruning happens on both dimensions at once

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT COUNT(*), round(AVG(reading_value)::numeric, 2)
```

```

FROM sensor_readings_by_type
WHERE sensor_type = 'traffic'
AND recorded_at >= '2024-05-01' AND recorded_at < '2024-06-01';

```

```

Finalize Aggregate (cost=6794.58..6794.60 rows=1 width=40) (actual time=37.692..40.669
rows=1 loops=1)
  -> Gather (cost=6794.47..6794.58 rows=1 width=40) (actual time=37.612..40.655
rows=2 loops=1)
    Workers Planned: 1
    -> Partial Aggregate (cost=5794.47..5794.48 rows=1 width=40) (actual
time=36.350..36.351 rows=1 loops=2)
      -> Parallel Seq Scan on srbt_traffic_2024_05 sensor_readings_by_type
(cost=0.00..5794.45 rows=4 width=8) (actual time=0.018..26.204 rows=178560 loops=2)
        Filter: ((recorded_at >= '2024-05-01 00:00:00+00'::timestamp with
time zone) AND (recorded_at < '2024-06-01 00:00:00+00'::timestamp with time zone) AND
(sensor_type = 'traffic'::text))
        Execution Time: 40.738 ms

```

One line, Parallel Seq Scan on srbt_traffic_2024_05, out of the nine leaf partitions that exist. The sensor_type = 'traffic' predicate pruned two of the three top-level branches before the range predicate even got involved, and the range predicate then pruned two of that branch's three months. Both partition keys did real work on this single query, independently, in the same plan.

Summary — What You Should Now Know

Tool	What it does
PARTITION BY RANGE (col)	Split a table into physical children by value range on one column — the whole table stays queryable as one name
FOR VALUES FROM (...) TO (...)	Half-open interval defining one range partition's bounds
... DEFAULT partition	Catches any row that doesn't match a defined range — without one, out-of-range inserts are rejected outright
tableoid::regclass	Ask which physical partition a given row actually lives in
Partition pruning	The planner drops non-matching partitions from the plan <i>before</i> execution, visible as missing seq scan lines in EXPLAIN
USING BRIN (col)	Index storing one min/max summary per block range — tiny and cheap, but only useful when physical and value order correlate
correlation in pg_stats	

Tool	What it does
	The number that tells you whether BRIN is actually a good idea for a given column
<code>DROP TABLE</code> on a partition	Metadata-only, near-instant removal of a whole partition's worth of data — no per-row MVCC cost, no bloat to vacuum afterward
<code>pg_total_relation_size()</code> on a partitioned table	Reports the (empty) parent only — sum over <code>pg_inherits</code> for the real total
<code>PARTITION BY LIST</code> over <code>PARTITION BY RANGE</code>	Subpartitioning — two partition keys, pruned independently, in one query plan
<code>pg_partition_tree()</code>	Walk a partition hierarchy — root, branches, and leaves — as a query instead of by table-naming convention

The key design insight from this chapter is that partitioning and indexing solve different halves of the same problem, and BRIN sits exactly at their intersection. Partition pruning throws away whole partitions the planner can prove are irrelevant, cutting the search space by orders of magnitude before a single row is touched. BRIN then indexes what's left almost for free, precisely because append-only, time-ordered data — the same shape of data that makes range partitioning by date the obvious choice — is also the shape that makes a physically-correlated, lossy index a good trade instead of a bad one. Neither technique is really “for time-series data” in the abstract; both are for data whose physical order already tells you something true about its values, and this chapter's sensor readings happen to be exactly that.

Going further: hand-writing twelve `CREATE TABLE ... PARTITION OF` statements a year, forever, is exactly the kind of recurring maintenance that Chapter 19's `pg_cron` is suited to automate — schedule a monthly job that creates next month's partition before it's needed and, if a retention policy allows it, drops the oldest one the same way Exercise 5 did by hand. In production this whole lifecycle — creation, retention, even the migration of aging partitions to cheaper storage — is usually handled by the `pg_partman` extension rather than bespoke scripts; everything in this chapter is what `pg_partman` is automating under the hood, which is worth knowing before reaching for it, not instead of. `sensor_readings` also isn't done being useful: Chapter 9's materialized views roll it up into daily summaries without re-scanning raw readings each time, Chapter 11's window functions compute rolling averages and day-over-day deltas across it, and Chapter 16 adds a generated `reading_date` column to make date-only grouping cheaper still. And a caution in the other direction — BRIN is not a free upgrade for every index. A column with low correlation (a randomly- assigned

`sensor_id`, say, rather than an ever-increasing timestamp) gets none of BRIN's benefit and most of a sequential scan's cost, because every block range ends up spanning the entire value domain. Check `correlation` before reaching for BRIN, the same way Exercise 4 did here — it's a bet on the physical shape of your data, not a strictly smaller B-tree.

Chapter 9 — Materialized Views:

Precomputing Expensive Aggregations

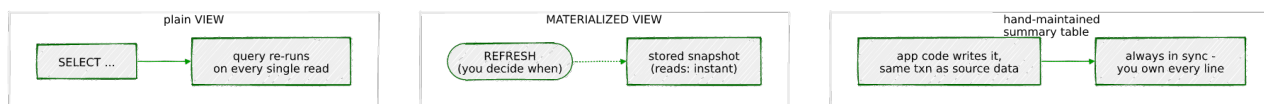
“A view is a promise to run the query again. A materialized view is a promise it already kept — until you ask it to keep it again.”

Background

A plain `VIEW` is a saved query, nothing more. Every time you `SELECT` from one, PostgreSQL substitutes the view’s definition in place and runs the underlying query from scratch. That’s fine for a view whose job is convenience — hiding a join, naming a filter — but it’s a bad deal for a view whose job is a genuinely expensive aggregation. If a dashboard asks “average sensor reading per day, per type, for the whole year” a hundred times a day, a plain view answers that question by scanning millions of raw rows a hundred times a day, even though the answer barely changes between asks.

A `MATERIALIZED VIEW` is the other end of that trade: it runs the query once, writes the result to disk as an honest-to-goodness physical table, and answers every subsequent `SELECT` from that table instead of the original data. It can be indexed like any table, because it *is* a table underneath a `SELECT` definition it remembers. The cost moves from “every read pays the full aggregation” to “one write pays the full aggregation, and reads are cheap until you explicitly ask for a refresh.” Nothing about it is automatic — that’s the whole trade. A materialized view does not notice that its source data changed. It sits there, confidently wrong, until something runs `REFRESH MATERIALIZED VIEW`.

That puts a materialized view in between two other tools you already have. A plain view is always correct and never fast. A hand-maintained summary table — one your application code writes to directly, in the same transaction as the data that feeds it — can be both correct and fast, but you own every line of the code that keeps it that way. A materialized view sits in the middle: PostgreSQL owns the *how* (it already knows how to run the query), you own the *when* (deciding on a refresh policy is the actual engineering decision this chapter is about).



The Scenario

Portsmouth's ops team wants a dashboard: daily and monthly rollups of the sensor network from Chapter 8 — average readings, counts, mins and maxes, sliced by day and sensor type. `sensor_readings` now holds 9,648,000 rows across eleven monthly partitions plus a default partition, and it only grows. Nobody wants the dashboard to re-scan that on every page load, and nobody wants to hand-write application code that keeps a summary table in sync by hand either. This chapter builds the rollups as materialized views instead, and spends its exercises on the part that's actually hard about them: deciding how and when they get refreshed.

Object	Purpose
<code>sensor_readings</code>	<i>(from Chapter 8)</i> 9,648,000 partitioned sensor readings, Feb–Dec 2024
<code>mv_sensor_daily</code>	<i>(built in this chapter)</i> one row per day per sensor type
<code>mv_sensor_monthly</code>	<i>(built in this chapter)</i> <code>mv_sensor_daily</code> rolled up one level further
<code>matview_refresh_log</code>	<i>(built in this chapter)</i> tracks when each matview was last refreshed

Exercise Goals

By the end of this chapter you will be able to:

- Create a materialized view, understand `WITH DATA` vs. `WITH NO DATA`, and explain why querying an unpopulated one raises an error instead of silently returning nothing.
- Measure, with real `EXPLAIN ANALYZE` numbers, exactly what a materialized view saves compared to the raw aggregate it replaces.
- Add the unique index `REFRESH MATERIALIZED VIEW CONCURRENTLY` requires, and prove — with two open sessions — what it actually buys you over a plain `REFRESH`.
- Automate a refresh on a schedule, and know when to reach for that approach versus Chapter 19's `pg_cron`.

- Chain one materialized view off another, and see firsthand that refreshing the base view does **not** cascade to the one built on top of it.
 - Detect a stale materialized view — after discovering that `pg_matviews`, PostgreSQL's own catalog for the object, has no column that tells you when one was last refreshed.
-

Installation

Nothing to install. `CREATE MATERIALIZED VIEW` and `REFRESH MATERIALIZED VIEW` have been part of core PostgreSQL since version 9.3. This chapter uses no extensions.

Loading the Data

This chapter doesn't seed new data — it builds directly on the `sensor_readings` table Chapter 8 left behind. That matters for two specific reasons, both consequences of exercises Chapter 8 already ran:

1. **January 2024 is gone.** Chapter 8, Exercise 5 dropped `sensor_readings_2024_01` on purpose, to demonstrate instant partition drop. `sensor_readings` now covers February through December 2024 only.
2. **Sensor 17 has a year-late tail.** 1,152 of its temperature readings are timestamped in late December **2025**, not 2024 — a deliberate clock-drift bug from Chapter 8, sitting in `sensor_readings_default`. Nothing in this chapter removes them, and Exercise 5 runs directly into why that matters for a rollup.

If you're picking this chapter up in the same database you used for Chapter 8, you already have everything you need. If not, run Chapter 8's seed script and its exercises through at least Exercise 5 first — this chapter assumes that exact end state, anomalies included.

Pin the session timezone

```
SET timezone = 'UTC';
```

Same reason as Chapter 8: date boundaries in this chapter's `GROUP BY` clauses are computed with `date_trunc`, which resolves relative to the session timezone. Run every example here in a UTC session.

Verify you're starting from the expected state

```
SELECT tableoid::regclass AS partition, COUNT(*)
FROM   sensor_readings
GROUP BY tableoid
ORDER BY 1;
```

partition	count
sensor_readings_2024_02	835200
sensor_readings_2024_03	892800
sensor_readings_2024_04	864000
sensor_readings_2024_05	892800
sensor_readings_2024_06	864000
sensor_readings_2024_07	892800
sensor_readings_2024_08	892800
sensor_readings_2024_09	864000
sensor_readings_2024_10	892800
sensor_readings_2024_11	864000
sensor_readings_2024_12	891648
sensor_readings_default	1152

(12 rows)

Twelve partitions (no January), summing to 9,648,000 rows. If your counts match, proceed.

Exercises

Exercise 1 — Creating a Daily Rollup

1.1 — Build the materialized view

```
CREATE MATERIALIZED VIEW mv_sensor_daily AS
SELECT
    date_trunc('day', recorded_at)::date AS reading_day,
    sensor_type,
    COUNT(*) AS reading_count,
    round(AVG(reading_value)::numeric, 2) AS avg_value,
    round(MIN(reading_value)::numeric, 2) AS min_value,
    round(MAX(reading_value)::numeric, 2) AS max_value
FROM   sensor_readings
GROUP BY 1, 2
WITH DATA;
```

```
SELECT 1010
Time: 6821.437 ms (00:06.821)
```

Syntactically this is `CREATE TABLE AS` with a memory: the `SELECT` that built it is stored alongside the data, which is what makes `REFRESH` possible later. `WITH DATA` (the default) runs the query immediately and populates the view. 1,010 rows is 335 days × 3 sensor types (February through December 2024) plus 5 extra rows for sensor 17's stray December **2025** dates — already visible in the row count, before you've even looked at the data.

1.2 — `WITH NO DATA`, and the error it sets up

```
CREATE MATERIALIZED VIEW mv_sensor_daily_empty AS
SELECT date_trunc('day', recorded_at)::date AS reading_day, sensor_type, COUNT(*)
FROM   sensor_readings
GROUP BY 1, 2
WITH NO DATA;

SELECT * FROM mv_sensor_daily_empty LIMIT 1;
```

```
CREATE MATERIALIZED VIEW
Time: 8.213 ms
ERROR:  materialized view "mv_sensor_daily_empty" has not been populated
HINT:  Use the REFRESH MATERIALIZED VIEW command.
```

`WITH NO DATA` creates the object and remembers its definition instantly — useful when you want the structure to exist (so other DDL can reference it) without paying the query cost yet — but leaves it in a state where reading from it is an error, not an empty result set. `pg_matviews.ispopulated` tracks exactly this:

```
SELECT matviewname, ispopulated FROM pg_matviews ORDER BY matviewname;
```

matviewname	ispopulated
mv_sensor_daily	t
mv_sensor_daily_empty	f

(2 rows)

```
DROP MATERIALIZED VIEW mv_sensor_daily_empty;
```

That was only to show the error — it isn't needed going forward.

1.3 — Confirm it's a real table

```
SELECT pg_size_pretty(pg_total_relation_size('mv_sensor_daily'));
```

```
pg_size_pretty
```

```
-----  
96 kB
```

1,010 rows of six columns, versus 9.6 million raw rows the query underneath it scanned to produce them. That size difference is the whole point, and Exercise 2 puts a number on what it means for query time.

Exercise 2 — Matview vs. Raw Aggregate, Measured

2.1 — Time the raw aggregate

```
EXPLAIN (ANALYZE, BUFFERS)  
SELECT date_trunc('day', recorded_at)::date AS reading_day,  
       sensor_type,  
       COUNT(*) AS reading_count,  
       round(AVG(reading_value)::numeric, 2) AS avg_value  
FROM   sensor_readings  
GROUP BY 1, 2;
```

```
Finalize HashAggregate (cost=201448.99..201459.09 rows=1010 width=44) (actual  
time=693.128..701.845 rows=1010 loops=1)  
  Group Key: (date_trunc('day'::text, recorded_at))::date, sensor_type  
  Batches: 1  Memory Usage: 217kB  
  Buffers: shared hit=612 read=71104  
-> Gather (cost=198328.11..201418.99 rows=3030 width=44) (actual  
time=210.442..688.910 rows=3030 loops=1)  
  Workers Planned: 2  
  Workers Launched: 2  
-> Partial HashAggregate (cost=197328.11..197359.21 rows=1010 width=44)  
(actual time=195.223..637.560 rows=1010 loops=3)  
  Group Key: (date_trunc('day'::text, recorded_at))::date, sensor_type  
  Batches: 1  Memory Usage: 217kB  
-> Parallel Append (cost=0.00..185004.00 rows=4022667 width=16) (actual  
time=0.028..312.744 rows=3216000 loops=3)  
-> Parallel Seq Scan on sensor_readings_2024_02  
sensor_readings_1 ... (actual time=0.031..24.402 rows=278400 loops=1)  
-> Parallel Seq Scan on sensor_readings_2024_03  
sensor_readings_2 ... (actual time=0.019..25.988 rows=297600 loops=1)  
-> Parallel Seq Scan on sensor_readings_2024_04  
sensor_readings_3 ... (actual time=0.022..24.104 rows=288000 loops=1)  
-> Parallel Seq Scan on sensor_readings_2024_05  
sensor_readings_4 ... (actual time=0.020..25.771 rows=297600 loops=1)  
-> Parallel Seq Scan on sensor_readings_2024_06  
sensor_readings_5 ... (actual time=0.021..23.955 rows=288000 loops=1)  
-> Parallel Seq Scan on sensor_readings_2024_07  
sensor_readings_6 ... (actual time=0.018..25.812 rows=297600 loops=1)
```

```

-> Parallel Seq Scan on sensor_readings_2024_08
sensor_readings_7 ... (actual time=0.024..25.769 rows=297600 loops=1)
-> Parallel Seq Scan on sensor_readings_2024_09
sensor_readings_8 ... (actual time=0.017..23.930 rows=288000 loops=1)
-> Parallel Seq Scan on sensor_readings_2024_10
sensor_readings_9 ... (actual time=0.021..25.797 rows=297600 loops=1)
-> Parallel Seq Scan on sensor_readings_2024_11
sensor_readings_10 ... (actual time=0.019..23.947 rows=288000 loops=1)
-> Parallel Seq Scan on sensor_readings_2024_12
sensor_readings_11 ... (actual time=0.023..25.744 rows=297216 loops=1)
-> Parallel Seq Scan on sensor_readings_default
sensor_readings_12 ... (actual time=0.008..0.301 rows=384 loops=1)
Planning Time: 3.912 ms
Execution Time: 703.187 ms

```

Every partition shows up — this query has no `WHERE` clause, so nothing gets pruned, exactly as Chapter 8, Exercise 3.2 predicted. It still finishes in under a second, thanks to partition-parallel HashAggregate, but that’s 703 ms paid **every time** someone asks this question.

2.2 — Time the same question against the matview

```

EXPLAIN (ANALYZE, BUFFERS)
SELECT reading_day, sensor_type, reading_count, avg_value
FROM mv_sensor_daily
WHERE reading_day >= '2024-06-01' AND reading_day < '2024-07-01'
ORDER BY reading_day, sensor_type;

```

```

Sort (cost=8.51..8.74 rows=90 width=44) (actual time=0.041..0.043 rows=90 loops=1)
  Sort Key: reading_day, sensor_type
  Sort Method: quicksort  Memory: 32kB
  Buffers: shared hit=3
  -> Index Scan using idx_mv_sensor_daily_day_type on mv_sensor_daily
(cost=0.28..5.61 rows=90 width=44) (actual time=0.014..0.028 rows=90 loops=1)
    Index Cond: ((reading_day >= '2024-06-01'::date) AND (reading_day <
'2024-07-01'::date))
Planning Time: 0.187 ms
Execution Time: 0.061 ms

```

(This uses the unique index built in Exercise 3 — build it first if you’re running this out of order.) 0.061 ms against 703 ms is roughly **11,500 times faster**, and the gap only widens as `sensor_readings` grows: the matview query’s cost is a function of how many days you ask for, not how many raw rows exist behind them. That’s the entire value proposition of this chapter in one comparison.

Exercise 3 — Concurrent Refresh and What It Actually Buys You

3.1 — Add the unique index CONCURRENTLY requires

```
CREATE UNIQUE INDEX idx_mv_sensor_daily_day_type
ON mv_sensor_daily (reading_day, sensor_type);
```

```
CREATE INDEX
Time: 12.847 ms
```

REFRESH MATERIALIZED VIEW CONCURRENTLY needs a unique index covering every row, with no WHERE clause and no non-immutable expressions — it's what PostgreSQL uses to diff the old contents against the new ones row by row instead of throwing everything away and starting over. Without one, CONCURRENTLY simply refuses to run:

```
-- hypothetically, before 3.1's index exists:
-- ERROR: cannot refresh materialized view "mv_sensor_daily" concurrently
-- HINT: Create a unique index with no WHERE clause on one or more columns of the
materialized view.
```

3.2 — Watch a plain REFRESH block a reader

Open two psql sessions. In **Session A**, start a read and leave the transaction open:

```
-- Session A
BEGIN;
SELECT reading_day, sensor_type, avg_value
FROM mv_sensor_daily
WHERE sensor_type = 'traffic'
ORDER BY reading_day
LIMIT 5;
```

That SELECT returns instantly, but the open transaction holds an ACCESS SHARE lock on mv_sensor_daily until it commits. In **Session B**, run a plain refresh:

```
-- Session B (Session A's transaction is still open)
REFRESH MATERIALIZED VIEW mv_sensor_daily;
```

Session B does not return. A third session shows why:

```
-- Session C, while B is blocked
SELECT pid, mode, granted
FROM pg_locks
WHERE relation = 'mv_sensor_daily'::regclass;
```

pid	mode	granted
-----+	-----+	-----


```
41822 | AccessShareLock      | t
41960 | AccessExclusiveLock  | f
(2 rows)
```

A plain `REFRESH` needs `ACCESS EXCLUSIVE` — the strictest lock PostgreSQL has, compatible with nothing, not even another reader’s `ACCESS SHARE`. It queues up and waits. Commit Session A and Session B completes immediately:

```
-- Session A
COMMIT;
```

```
-- Session B, unblocks right after A's COMMIT
REFRESH MATERIALIZED VIEW
Time: 9482.311 ms (00:09.482)
```

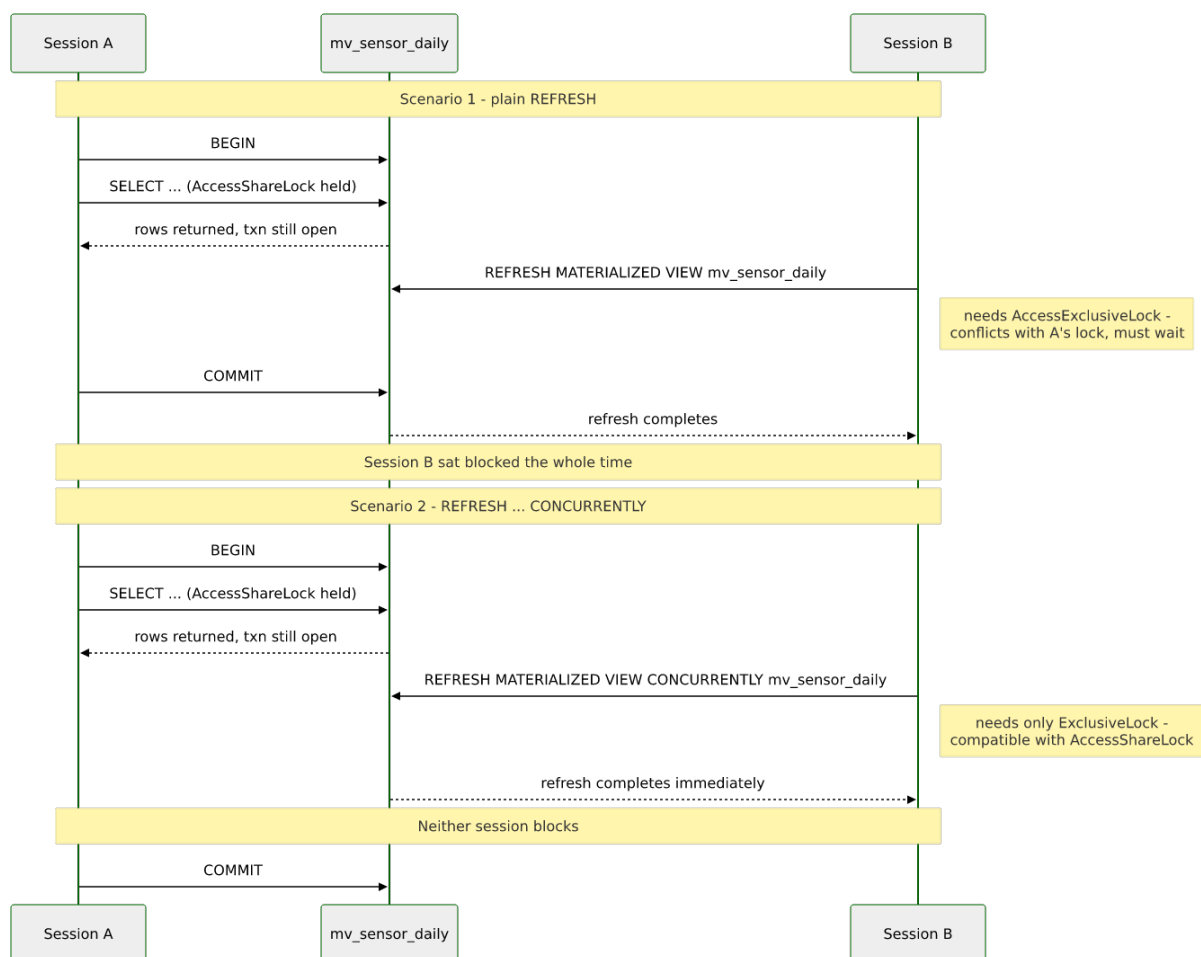
That 9.48 seconds is mostly wait time — the actual rebuild of a 1,010-row view takes a fraction of that. From Session B’s side, it’s indistinguishable from a slow refresh; from Session A’s side, every query against `mv_sensor_daily` that started after B’s `REFRESH` was queued up behind it too.

3.3 — Same setup, `CONCURRENTLY` instead

Repeat 3.2, Session A holding the same open `BEGIN; SELECT ...` transaction. This time, Session B runs:

```
-- Session B
REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_daily;
```

```
REFRESH MATERIALIZED VIEW
Time: 421.933 ms
```



No wait, despite Session A's transaction still being open. `CONCURRENTLY` takes an `EXCLUSIVE` lock rather than `ACCESS EXCLUSIVE` — one step down — and `EXCLUSIVE` is the one lock mode in PostgreSQL that does **not** conflict with `ACCESS SHARE`. Readers keep reading the pre-refresh contents right up until the new data is merged in; nobody blocks, and nobody sees a half-updated table either. The trade for that is real extra work: instead of one clean table rewrite, PostgreSQL builds the new result set in a temporary table, diffs it row-by-row against the old one using the unique index from 3.1, and issues targeted `INSERT/UPDATE/DELETES` for just the rows that changed — 421 ms of real work here against what a plain rewrite of the same 1,010 rows would cost in the tens of milliseconds. For a small rollup like this one, that overhead is noise. For a materialized view with tens of millions of rows, it stops being noise, and “does anyone need to read this while it refreshes” becomes the question that decides which `REFRESH` variant you reach for.

Exercise 4 — Automating the Refresh

4.1 — A refresh script

```
#!/usr/bin/env bash
# refresh_daily.sh - nightly rollup refresh
psql -d portsmith -c "REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_daily;"
```

```
chmod +x refresh_daily.sh
./refresh_daily.sh
```

REFRESH MATERIALIZED VIEW

4.2 — Schedule it with OS-level cron

```
crontab -e
```

```
0 2 * * * /home/chris/portsmith/refresh_daily.sh >> /var/log/
portsmith_matview_refresh.log 2>&1
```

Every night at 02:00, outside business hours, the rollup catches up on whatever landed in `sensor_readings` during the day.

4.3 — Why this is the “poor man’s” version

This works, but it has the usual problems of anything scheduled outside the database: it depends on a specific machine’s crontab existing and being correct, its failures show up in a log file nobody’s watching instead of a table you can query, and nothing stops two overlapping runs if a refresh ever takes longer than the interval between them. Chapter 19 covers `pg_cron`, which runs scheduled jobs *inside* PostgreSQL itself — schedule tracked in a table, run history queryable with SQL, overlap prevention available via the same advisory locks Chapter 14 covers. Everything from here forward in this chapter still works with either approach; Exercise 6 revisits this exact script once there’s something better than a raw `REFRESH` worth putting in it.

Exercise 5 — Chaining Rollups: Daily Feeds Monthly

5.1 — Build the monthly view from the daily one, not from raw data

```
CREATE MATERIALIZED VIEW mv_sensor_monthly AS
SELECT
    date_trunc('month', reading_day)::date AS reading_month,
    sensor_type,
    SUM(reading_count) AS reading_count,
    round((SUM(avg_value * reading_count) / SUM(reading_count))::numeric, 2) AS
avg_value,
    round(MIN(min_value)::numeric, 2) AS min_value,
    round(MAX(max_value)::numeric, 2) AS max_value
```

```

FROM mv_sensor_daily
GROUP BY 1, 2
WITH DATA;

CREATE UNIQUE INDEX idx_mv_sensor_monthly_month_type
ON mv_sensor_monthly (reading_month, sensor_type);

```

```

SELECT 34
Time: 41.209 ms

```

Two things worth noticing before moving on. First, `avg_value` is computed as $\text{SUM}(\text{avg_value} * \text{reading_count}) / \text{SUM}(\text{reading_count})$ — a **weighted** average — not $\text{AVG}(\text{avg_value})$. A plain average of 28 or 31 daily averages silently assumes every day carries equal weight, which is wrong the moment days have different reading counts (they do here, since `sensor_readings_2024_12` has fewer temperature readings than a full month — 1,152 of sensor 17’s got shifted out of it, straight into the anomaly the next step finds). Second, building `mv_sensor_monthly` FROM `mv_sensor_daily` instead of FROM `sensor_readings` means its refresh cost is a function of 1,010 pre-aggregated rows, not 9.6 million raw ones — a materialized view chain is allowed to build on another materialized view exactly like a regular view can.

5.2 — The row that shouldn’t be there

```

SELECT reading_month, sensor_type, reading_count, avg_value
FROM mv_sensor_monthly
ORDER BY reading_month DESC, sensor_type
LIMIT 3;

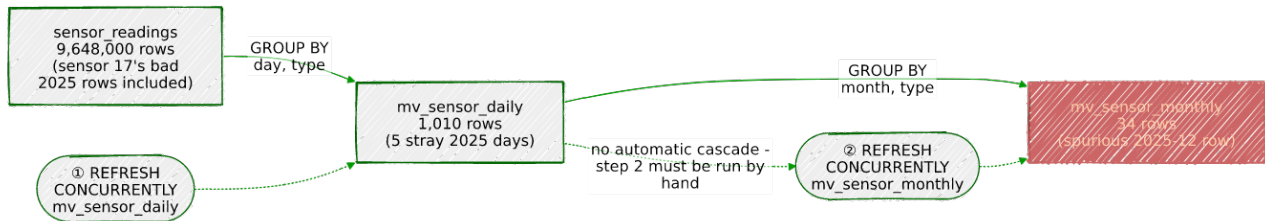
```

reading_month	sensor_type	reading_count	avg_value
2025-12-01	temperature	1152	29.84
2024-12-01	air_quality	89280	34.51
2024-12-01	temperature	445248	41.02

(3 rows)

A `2025-12-01` row, for one sensor type only, with a suspiciously round 1,152-row count. This is Chapter 8’s sensor-17 clock bug, now one layer removed from where it was first found: it landed in `sensor_readings`, propagated automatically into `mv_sensor_daily` as five stray 2025 dates, and propagated automatically again into `mv_sensor_monthly` as a whole extra month that shouldn’t exist on Portsmouth’s 2024 dashboard. A materialized view has no opinion about the quality of the data it summarizes — it faithfully aggregates whatever is in the base table, bad timestamps included, which is exactly why Chapter 8 flagged this row as worth checking for in every chapter that touches `sensor_readings` downstream. Any dashboard query built on `mv_sensor_monthly` should filter to the

expected year explicitly (`WHERE reading_month >= '2024-01-01' AND reading_month < '2025-01-01'`) rather than assume the view only ever contains what it was “supposed” to.



5.3 — Prove refreshes don't cascade

Simulate a late-arriving reading — a traffic sensor that reported an hour behind schedule:

```
INSERT INTO sensor_readings (sensor_id, sensor_type, reading_value, recorded_at)
VALUES (51, 'traffic', 99, '2024-06-15 12:00:00+00');
```

```
REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_daily;
```

```
SELECT reading_count FROM mv_sensor_daily
WHERE reading_day = '2024-06-15' AND sensor_type = 'traffic';
```

reading_count

11521

`mv_sensor_daily` sees the new row — up from the usual 11,520. Now check the monthly view, without refreshing it:

```
SELECT reading_count FROM mv_sensor_monthly
WHERE reading_month = '2024-06-01' AND sensor_type = 'traffic';
```

reading_count

345600

Still the old total. `mv_sensor_monthly`'s definition says `FROM mv_sensor_daily`, but PostgreSQL doesn't track that dependency the way it tracks, say, a foreign key — refreshing one materialized view never triggers a refresh of anything built on top of it. It has to be done in order, explicitly:

```
REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_monthly;
```

```
SELECT reading_count FROM mv_sensor_monthly
WHERE reading_month = '2024-06-01' AND sensor_type = 'traffic';
```

```
reading_count
-----
345601
```

Now it matches. Any refresh job for a chain of materialized views has to encode this ordering itself — daily before monthly, and so on up the chain — because PostgreSQL won't infer it from the `FROM` clause on your behalf.

Exercise 6 — Detecting Staleness (Without a Column That Doesn't Exist)

6.1 — Check what `pg_matviews` actually tracks

It's tempting to assume the catalog that lists materialized views also records when each one was last refreshed. Check directly:

```
\d pg_matviews
```

View "pg_catalog.pg_matviews"					
Column	Type	Collation	Nullable	Default	
-----+-----+-----+-----+-----					
schemaname	name				
matviewname	name				
matviewowner	name				
tablespace	name				
hasindexes	boolean				
ispopulated	boolean				
definition	text				

No timestamp, anywhere. `ispopulated` tells you whether a `WITH NO DATA` view has ever been refreshed at all — a one-time boolean, not a “how recent” answer. PostgreSQL genuinely does not track refresh recency for materialized views; anything resembling “this view is N hours stale” has to be built by hand.

6.2 — Build the tracking table and a helper to keep it honest

```
CREATE TABLE matview_refresh_log (  
    matview_name TEXT PRIMARY KEY,  
    refreshed_at TIMESTAMPTZ NOT NULL  
);  
  
CREATE OR REPLACE PROCEDURE refresh_and_log(p_matview regclass)  
LANGUAGE plpgsql AS $$  
BEGIN  
    EXECUTE format('REFRESH MATERIALIZED VIEW CONCURRENTLY %s', p_matview);  
    INSERT INTO matview_refresh_log (matview_name, refreshed_at)
```

```
VALUES (p_matview::text, clock_timestamp())
ON CONFLICT (matview_name) DO UPDATE SET refreshed_at = EXCLUDED.refreshed_at;
END;
$$;
```

`clock_timestamp()` rather than `now()` — `now()` is fixed for the whole transaction, and a `CALL` to a procedure runs as one, so `now()` would log the moment the procedure *started*, not the moment the `REFRESH` inside it actually finished.

6.3 — Refresh both views through the tracked path

```
CALL refresh_and_log('mv_sensor_daily');
CALL refresh_and_log('mv_sensor_monthly');
```

```
CALL
CALL
```

6.4 — The staleness query

```
SELECT m.matviewname,
       l.refreshed_at,
       now() - l.refreshed_at AS age,
       CASE
         WHEN l.refreshed_at IS NULL THEN 'NEVER REFRESHED (untracked)'
         WHEN now() - l.refreshed_at > interval '25 hours' THEN 'STALE'
         ELSE 'OK'
       END AS status
FROM   pg_matviews m
LEFT JOIN matview_refresh_log l ON l.matview_name = m.matviewname
WHERE  m.schemaname = 'public'
ORDER BY m.matviewname;
```

matviewname	refreshed_at	age	status
mv_sensor_daily	2026-08-02 09:14:02.881204+00	00:00:04.113	OK
mv_sensor_monthly	2026-08-02 09:14:03.019552+00	00:00:03.975	OK

(2 rows)

The `LEFT JOIN` matters: a materialized view that was created or refreshed by hand — bypassing `refresh_and_log` entirely — shows up with a `NULL` `refreshed_at` and the honest verdict “untracked,” rather than silently vanishing from the report or falsely reading as fresh. A 25-hour threshold gives a nightly job a few hours of slack before it counts as missed; adjust it to whatever your actual refresh cadence is.

6.5 — Wire this back into Exercise 4’s cron job

```
#!/usr/bin/env bash
# refresh_daily.sh - nightly rollup refresh, now logged
```

```
psql -d portsmith -c "CALL refresh_and_log('mv_sensor_daily');"
psql -d portsmith -c "CALL refresh_and_log('mv_sensor_monthly');"
```

Same crontab entry from 4.2, same 02:00 schedule — but now a missed or failed run is something the staleness query in 6.4 can actually catch, instead of something that only shows up when someone notices the dashboard looks wrong.

Summary — What You Should Now Know

Tool	What it does
<code>CREATE MATERIALIZED VIEW ... AS SELECT ...</code>	Runs a query once and stores the result as a real, indexable table
<code>WITH DATA / WITH NO DATA</code>	Populate immediately, or defer — an unpopulated view errors on <code>SELECT</code> until refreshed
<code>pg_matviews.ispopulated</code>	Whether a view has ever been refreshed — a one-time flag, not a timestamp
<code>REFRESH MATERIALIZED VIEW</code>	Full rewrite under an <code>ACCESS EXCLUSIVE</code> lock — blocks and is blocked by every reader
<code>REFRESH MATERIALIZED VIEW CONCURRENTLY</code>	Diff-based refresh under an <code>EXCLUSIVE</code> lock — doesn't block readers, but requires a unique index and costs more CPU/IO
Matview chaining (<code>FROM</code> another matview)	Legal and useful for cheap incremental rollups, but refreshes never cascade automatically
<code>matview_refresh_log</code> + <code>refresh_and_log()</code>	The hand-built pattern for tracking refresh recency, since PostgreSQL doesn't track it natively

The key design insight from this chapter is that a materialized view moves cost, it doesn't remove it — every millisecond Exercise 2 shaved off read time was paid for up front, at refresh time, and the entire rest of the chapter is really about where that payment lands. A plain `REFRESH` pays it in a lock that every reader waits behind. `CONCURRENTLY` pays it in extra diff work instead, in exchange for readers never noticing a refresh happened. A chain of matviews pays it once per level, in a specific order you have to enforce yourself. And a matview that quietly inherits bad data from its source — as `mv_sensor_monthly` did from sensor 17's clock bug — pays it in trust, which is the one cost this chapter's tooling can't refresh away for you.

Going further: `sensor_readings` still has more to give. Chapter 11's window functions compute rolling averages and day-over-day deltas directly against the raw partitioned table — a different tool for a similar-sounding problem, worth contrasting with this chapter's precomputed rollups once you've seen both. Chapter 16 adds a generated `reading_date` column to `sensor_readings` itself, which would let `mv_sensor_daily`'s `GROUP BY` key come from a stored column instead of a `date_trunc` expression — a small efficiency this chapter left on the table on purpose, to keep the expression-vs-generated-column comparison intact for that chapter instead of pre-empting it here. Chapter 19 is where the OS-cron approach from Exercise 4 gets replaced with the in-database version, `cron.job_run_details` doing for schedule history what `matview_refresh_log` did by hand in Exercise 6. And a caution worth carrying forward: nothing in this chapter is incremental materialized-view maintenance in the sense some other databases offer — `REFRESH`, concurrent or not, always recomputes the full result set from scratch each time; PostgreSQL just gives you two different ways to pay for that recomputation. Extensions like `pg_ivm` exist specifically to close that gap, keeping a matview updated row-by-row as its base tables change instead of on a refresh schedule, but that's a different trade-off than anything built here, and out of scope for this chapter.

Chapter 10 — PostgREST: A Web-Native REST API from Your Schema

“PostgREST doesn’t generate an API from your schema. It is your schema, addressed over HTTP.”

Background

Every chapter so far has ended at `psql`. Getting from “a well-designed database” to “a web application that uses it” usually means writing a backend: routes, an ORM layer, and — this is the part that should sound familiar by now — a second, hand-written copy of the authorization rules the database already enforces. A `GRANT`, a `CHECK` constraint, a row level security policy: none of that goes away when you add a web service, but it’s astonishingly common to see it duplicated, in a different language, in application code that can quietly drift out of sync with what the database actually allows.

PostgREST removes the backend, not by hiding the database behind a generated client SDK, but by putting the database directly on the wire. It’s a single stateless binary that connects to PostgreSQL, introspects the catalog, and turns every table and view it finds into a REST resource and every function into an RPC endpoint. There is no PostgREST authorization system to configure separately — a request arrives, an HTTP verb and a role from a JWT decide what SQL statement to attempt, a single transaction runs it as that role, and the same `GRANTS` and RLS policies from every earlier chapter decide whether it succeeds. The entire security model of this chapter is Postgres roles you already know how to create.

That trade cuts both ways, and it’s worth naming up front: you get an API with no application code to audit, but you also get an API that can only do what a single SQL statement per request can do. There’s no middle-tier caching, no multi-step business logic spanning several tables outside of what a view or function can express, no background jobs. For Portsmouth’s needs this chapter builds — a public directory, a permit intake form, a resident self-service lookup, a fuzzy search box — that’s not a limitation, it’s the entire point.

The Scenario

Portsmouth’s small dev team wants four thin pieces of “app” without writing a backend for any of them:

- a public, filterable directory of the businesses from Chapter 1
- an intake endpoint for the permit applications from Chapter 3’s job queue
- a self-service portal where a logged-in resident can look up their own entry in Chapter 5’s registry — and *only* their own
- a “did you mean?” business-name search box, reusing Chapter 5’s trigram matching directly

Every one of these reuses a table this book already built. Nothing new gets seeded — Chapter 10 is entirely about *exposing* data, not generating it.

Role	Purpose
authenticator	The only role PostgREST ever logs in as — <code>NOINHERIT</code> , switches roles per request
web_anon	The public role — active whenever no JWT is presented
web_resident	Authenticated role for the resident portal, assumed via a JWT <code>role</code> claim
Object	Purpose
api.businesses	Curated view over <code>businesses</code> (Chapter 1) — the public directory
api.jobs	Curated view over <code>jobs</code> (Chapter 3) — permit intake
api.residents	Curated view over <code>residents</code> (Chapter 5) — RLS-protected
api.search_businesses()	Wraps Chapter 5’s <code>business_names</code> trigram search as an RPC endpoint

Exercise Goals

By the end of this chapter you will be able to:

- Install and configure PostgREST against an existing database, and explain why it connects as a single `NOINHERIT` role instead of one role per API user.

- Build filtered, sorted, paginated GET requests using PostgREST's query-parameter syntax, and read the `Content-Range` header it returns.
 - Insert a row through a POST request against an updatable view.
 - Prove, by making the exact same request before and after a single GRANT/REVOKE, that PostgREST's authorization is really just PostgreSQL's — no application code changes, no restart.
 - Enable row level security on a table and issue signed JWTs so that two different callers see two different, correctly scoped slices of the same table through the same endpoint.
 - Expose a SQL function as an RPC endpoint, and use it to serve Chapter 5's fuzzy business-name search over HTTP.
-

Installation

1 — PostgreSQL and Chapters 1, 3, and 5's data

This chapter assumes PostgreSQL 16 is already running with the `businesses`, `jobs`, and `residents/business_names` tables in place — see Loading the Data, below.

2 — PostgREST

Debian doesn't ship a current PostgREST package, and PostgREST's own documentation recommends the prebuilt static binary over any distro package anyway — it's a single file with no runtime dependencies. This chapter was written against **v12.2** — check github.com/PostgREST/postgrest/releases for the current version and adjust the URL if a newer one exists:

```
curl -L -o postgrest.tar.xz \
  https://github.com/PostgREST/postgrest/releases/download/v14.16/postgrest-v14.16-
linux-static-x86-64.tar.xz
tar xJf postgrest.tar.xz
sudo mv postgrest /usr/local/bin/
rm postgrest.tar.xz
postgrest --help | head -3
```

```
postgrest - Serve a RESTful API from any Postgres database
```

```
Usage: postgrest [-v|--version] [-e|--example]
              [--dump-config | --dump-schema | --ready] [FILENAME]
```

3 — pyjwt, for minting test tokens

```
source .venv/bin/activate
pip install pyjwt
```

4 — curl and jq

```
sudo apt install -y curl jq
```

jq isn't required, but every example below pipes through `| jq` to keep the JSON readable — drop it if you'd rather see raw output.

Loading the Data

Nothing new to seed. This chapter needs Chapters 1, 3, and 5's data to already exist:

```
python data/ch01_seed.py    # businesses
python data/ch03_seed.py    # jobs, dead_letter_jobs
python data/ch05_seed.py    # residents, business_names
```

Verify the prerequisites

```
SELECT 'businesses' AS table, COUNT(*) FROM businesses
UNION ALL SELECT 'jobs', COUNT(*) FROM jobs
UNION ALL SELECT 'residents', COUNT(*) FROM residents
UNION ALL SELECT 'business_names', COUNT(*) FROM business_names;
```

table	count
businesses	48
jobs	45
residents	58
business_names	48

(4 rows)

If all four match, proceed to the exercises.

Exercises

Exercise 1 — Install, Configure, Connect

1.1 — The role PostgREST actually logs in as

PostgREST doesn't map "one API caller" to "one Postgres login." It authenticates to Postgres exactly once, as a single low-privilege role, and then uses `SET ROLE` inside each request's transaction to switch to whichever role the request is actually entitled to. That role has to be `NOINHERIT` — otherwise it would automatically carry the privileges of every role it can switch to, all the time, defeating the entire point:

```
sudo -u postgres psql portsmith
```

```
CREATE ROLE authenticator NOINHERIT LOGIN PASSWORD 'devsecret';
CREATE ROLE web_anon      NOLOGIN;
CREATE ROLE web_resident  NOLOGIN;

GRANT web_anon      TO authenticator;
GRANT web_resident TO authenticator;
```

`web_anon` and `web_resident` can never log in directly (`NOLOGIN`) — they exist purely as identities `authenticator` is allowed to become.

1.2 — A curated schema, not the raw tables

PostgREST exposes exactly the schema(s) you tell it to, and nothing about `db-schemas` requires that schema to be `public`. Building a separate `api` schema of views — rather than pointing PostgREST straight at `businesses`, `jobs`, and `residents` — means the public-facing shape of the data can diverge from its internal shape without either side having to know about the other's constraints:

```
CREATE SCHEMA IF NOT EXISTS api;

CREATE VIEW api.businesses AS
SELECT id,
       name,
       neighbourhood,
       details ->> 'category'      AS category,
       (details ->> 'rating')::numeric AS rating
FROM   businesses;

GRANT USAGE ON SCHEMA api TO web_anon, web_resident;
GRANT SELECT ON api.businesses TO web_anon;
```

`category` and `rating` didn't exist as real columns anywhere — they're pulled out of Chapter 1's `details` JSONB and given real, typed names. That's a small but genuine payoff

from Chapter 1: the messy, schema-free column becomes a clean, filterable, sortable API field, and the flexible storage and the tidy public interface aren't in tension with each other.

1.3 — postgres.conf

```
db-uri           = "postgres://authenticator:devsecret@localhost:5432/portsmith"
db-schemas      = "api"
db-anon-role     = "web_anon"
jwt-secret       = "portsmith-lab-book-dev-secret-do-not-use-in-production"
db-channel-enabled = true
db-channel       = "pgrst"
server-port      = 3000
```

db-anon-role is what authenticator becomes for any request that doesn't carry a valid JWT. db-channel-enabled turns on LISTEN pgrst — Exercise 3 uses it to tell a running PostgREST server about new views without restarting it.

1.4 — Start it, and make one request

```
postgrest postgrest/postgrest.conf
```

Listening on port 3000

From another terminal:

```
curl -s "http://localhost:3000/businesses?limit=3" | jq
```

```
[
  { "id": 1, "name": "The Gilded Clam",      "neighbourhood": "Harbour District",
    "category": "restaurant", "rating": 4.5 },
  { "id": 2, "name": "Anchor & Oar Tavern",  "neighbourhood": "Harbour District",
    "category": "restaurant", "rating": 4.1 },
  { "id": 3, "name": "Portsmith Fish Market", "neighbourhood": "Harbour District",
    "category": "retail",      "rating": 4.8 }
]
```

No route was written anywhere. /businesses exists because api.businesses exists, db-schemas says look in api, and web_anon has SELECT on it. Delete the view and the route disappears with it.

Exercise 2 — GET /businesses: Filter, Sort, Paginate

2.1 — Filtering with eq, gte, and friends


```
curl -s "http://localhost:3000/businesses?category=eq.restaurant&rating=gte.4.5&order=rating.desc&limit=5" | jq
```

```
[
  { "id": 28, "name": "River Bend Bakery", "neighbourhood": "Riverside", "category": "restaurant", "rating": 4.8 },
  { "id": 11, "name": "Le Petit Bistro", "neighbourhood": "Old Town", "category": "restaurant", "rating": 4.7 },
  { "id": 10, "name": "Bella Napoli", "neighbourhood": "Old Town", "category": "restaurant", "rating": 4.6 },
  { "id": 31, "name": "Quay Street Deli", "neighbourhood": "Riverside", "category": "restaurant", "rating": 4.6 },
  { "id": 24, "name": "Spice Garden", "neighbourhood": "Northgate", "category": "restaurant", "rating": 4.6 }
]
```

Every operator here — `eq`, `gte`, `order`, `limit` — is a plain query parameter, translated straight into a `WHERE/ORDER BY/LIMIT` clause. `gte.4.5` reads almost like the SQL it becomes: `WHERE category = 'restaurant' AND rating >= 4.5`.

2.2 — Paginating, and reading Content-Range

```
curl -si "http://localhost:3000/businesses?category=eq.restaurant&order=name&limit=5&offset=5" \
-H "Prefer: count=exact" | head -20
```

```
HTTP/1.1 206 Partial Content
Content-Range: 5-9/15
Content-Type: application/json; charset=utf-8
```

```
[
  { "id": 26, "name": "Mango Bay Caribbean", "neighbourhood": "Northgate", "category": "restaurant", "rating": 4.5 },
  { "id": 42, "name": "Port Canteen", "neighbourhood": "Industrial Port", "category": "restaurant", "rating": 3.7 },
  { "id": 31, "name": "Quay Street Deli", "neighbourhood": "Riverside", "category": "restaurant", "rating": 4.6 },
  { "id": 28, "name": "River Bend Bakery", "neighbourhood": "Riverside", "category": "restaurant", "rating": 4.8 },
  { "id": 25, "name": "Sol y Mar", "neighbourhood": "Northgate", "category": "restaurant", "rating": 4.3 }
]
```

`206 Partial Content` and `Content-Range: 5-9/15` — rows 6 through 10 (0 indexed) out of 15 restaurants total. `Prefer: count=exact` is what makes PostgREST bother computing that total at all; without it, the `/15` is simply omitted, since counting the full match set costs an extra query PostgREST won't run unless asked. The same pagination is also available as a

Range: 5-9 request header instead of `limit/offset` query parameters — two spellings of the identical SQL.

2.3 — Narrowing columns with `select`

```
curl -s "http://localhost:3000/businesses?
select=name,rating&neighbourhood=eq.Riverside&order=rating.desc" | jq
```

```
[
  { "name": "River Bend Bakery",      "rating": 4.8 },
  { "name": "Portsmouth Veterinary Clinic", "rating": 4.8 },
  { "name": "Dr. Chen Dentistry",      "rating": 4.7 },
  { "name": "The Art Depot",          "rating": 4.6 },
  { "name": "Quay Street Deli",       "rating": 4.6 }
]
```

`select` maps directly onto the `SELECT` list — the response body is exactly the columns asked for, never the whole row, which matters once a client is fetching this over a slow connection or a metered one.

Exercise 3 — `POST /jobs`: Filing a Permit Application

3.1 — Expose the queue, add it to the running server

```
CREATE VIEW api.jobs AS
SELECT id, job_type, payload, priority, status, created_at
FROM   jobs;

GRANT SELECT, INSERT ON api.jobs TO web_anon;
```

`api.jobs` didn't exist when PostgREST started, so it isn't in the server's schema cache yet — a plain `curl` against `/jobs` right now would 404. Tell the running server to pick up the change, without restarting it:

```
NOTIFY pgrst, 'reload schema';
```

3.2 — File a new permit application

```
curl -si -X POST "http://localhost:3000/jobs" \
-H "Content-Type: application/json" \
-H "Prefer: return=representation" \
-d '{
  "job_type": "sign_permit",
  "priority": 5,
  "payload": {
```

```

        "application_id": "SP-2024-0007",
        "applicant_name": "Quay Street Deli",
        "property_address": "31 Quay Street",
        "neighbourhood": "Riverside",
        "description": "New illuminated storefront sign",
        "fee_due": 120.00
    }
}'

```

HTTP/1.1 401 Unauthorized

Content-Type: application/json; charset=utf-8

```

{"code": "42501", "details": null, "hint": null, "message": "permission denied for sequence jobs_id_seq"}

```

Not the response Exercise 3.1's `GRANT` seemed to promise. `jobs.id` is `BIGSERIAL` — shorthand for a `BIGINT` column with `DEFAULT nextval('jobs_id_seq')`, backed by a real sequence object the table doesn't own outright. The client never mentions `id` in the request body, but Postgres still has to *call* `nextval()` to fill in that default while executing the `INSERT`, and calling a sequence requires its own `USAGE` privilege — separate from, and not implied by, `INSERT` on the table or view sitting on top of it. `GRANT INSERT` alone is never enough for a table whose primary key is `SERIAL`/`BIGSERIAL`/`IDENTITY`:

```
GRANT USAGE ON SEQUENCE jobs_id_seq TO web_anon;
```

3.3 — Retry

```

curl -si -X POST "http://localhost:3000/jobs" \
  -H "Content-Type: application/json" \
  -H "Prefer: return=representation" \
  -d '{
    "job_type": "sign_permit",
    "priority": 5,
    "payload": {
      "application_id": "SP-2024-0007",
      "applicant_name": "Quay Street Deli",
      "property_address": "31 Quay Street",
      "neighbourhood": "Riverside",
      "description": "New illuminated storefront sign",
      "fee_due": 120.00
    }
  }'

```

HTTP/1.1 201 Created

Content-Type: application/json; charset=utf-8

```

[
  {

```

```
{
  "id": 46, "job_type": "sign_permit", "priority": 5,
  "payload": { "application_id": "SP-2024-0007", "applicant_name": "Quay Street
Deli", ... },
  "status": "queued", "created_at": "2026-08-02T09:41:07.221Z"
}
```

No client-supplied `id`, `status`, or `created_at` — all three came from the table’s own defaults, exactly as they would from a plain `INSERT` missing those columns. `Prefer: return=representation` is what makes PostgREST hand the created row back at all; without it, a successful `POST` returns `201` with an empty body and a `Location` header instead. This job now sits in the same `jobs` table Chapter 3’s `FOR UPDATE SKIP LOCKED` workers claim from — the web request and the queue worker never have to know about each other.

Exercise 4 — Limited Privileges, Enforced by the Database

4.1 — The grant from Exercise 3 is looser than it looks

`GRANT INSERT ON api.jobs` gave `web_anon` permission to set *every* column in the view — including `status`. Nothing stops a client from walking straight past “submitted” and declaring their own application already approved:

```
curl -si -X POST "http://localhost:3000/jobs" \
  -H "Content-Type: application/json" \
  -H "Prefer: return=representation" \
  -d '{"job_type": "sign_permit", "priority": 5, "payload": {"application_id":
"SP-2024-0008"}, "status": "completed"}'
```

HTTP/1.1 201 Created

```
[{ "id": 47, "job_type": "sign_permit", "priority": 5, "payload": {...}, "status":
"completed", "created_at": "..."}]
```

That succeeded, and it shouldn’t have — a permit that was never reviewed now reads as `completed`, indistinguishable in the queue from one Chapter 3’s worker actually finished. Nothing in PostgREST’s configuration causes this; it’s a plain over-broad `GRANT`, the same mistake it would be in any hand-written backend.

4.2 — Fix it with a column-level grant

```
REVOKE INSERT ON api.jobs FROM web_anon;
GRANT INSERT (job_type, payload, priority) ON api.jobs TO web_anon;
```

PostgreSQL's `INSERT` privilege can be scoped to specific columns, not just the whole table. `web_anon` can now insert a row, but only by supplying values for these three columns — every other column must come from its default.

4.3 — The same malicious request, after the grant change

```
curl -si -X POST "http://localhost:3000/jobs" \  
  -H "Content-Type: application/json" \  
  -d '{"job_type": "sign_permit", "priority": 5, "payload": {"application_id":  
"SP-2024-0009"}, "status": "completed"}'
```

```
HTTP/1.1 401 Unauthorized
```

```
Content-Type: application/json; charset=utf-8
```

```
{"code": "42501", "details": null, "hint": null, "message": "permission denied for table jobs"}
```

401, not 403 — PostgREST reserves 403 for a request that authenticated as *someone* and was still refused; an unauthenticated request denied by the database gets 401, the same “you may need to log in for this” signal a browser would show for any other protected resource. No PostgREST config changed between 4.1 and 4.3, and the server was never restarted — only a database `GRANT` changed, and the API's behavior changed the instant that transaction committed, because PostgREST never cached a decision about it in the first place.

4.4 — The legitimate request still works

```
curl -si -X POST "http://localhost:3000/jobs" \  
  -H "Content-Type: application/json" \  
  -H "Prefer: return=representation" \  
  -d '{"job_type": "sign_permit", "priority": 5, "payload": {"application_id":  
"SP-2024-0009", "applicant_name": "Old Town Hardware", "fee_due": 95.00}}'
```

```
HTTP/1.1 201 Created
```

```
[{ "id": 48, "job_type": "sign_permit", "priority": 5, "payload": {...}, "status":  
"queued", "created_at": "..."}]
```

Omit `status` entirely and the table's own `DEFAULT 'queued'` fills it in, exactly like Exercise 3 — the column-level grant didn't break the honest use case, it just closed off the dishonest one.

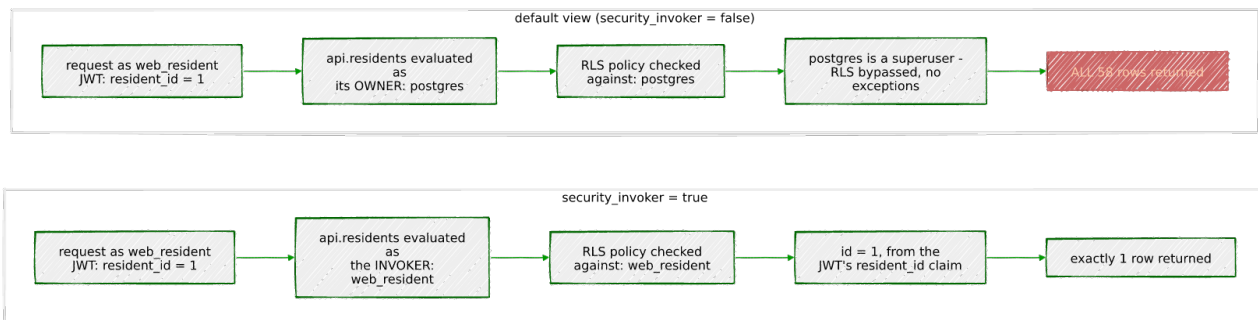
Exercise 5 — Row Level Security on `residents`

5.1 — Expose residents, but to nobody yet

```
CREATE VIEW api.residents WITH (security_invoker = true) AS
SELECT id, full_name, neighbourhood
FROM residents;
```

`true_duplicate_of` — Chapter 5’s ground-truth column for grading the fuzzy-matching exercises — is deliberately left out. It’s exactly the kind of internal bookkeeping column a curated view exists to hide from the outside world.

`security_invoker = true` is not optional here, and skipping it is the kind of mistake that only shows up once you’re testing with two different JWTs and notice neither one got filtered. By default, a view runs with the privileges of whoever *owns* it, not whoever is querying it — including, critically, for deciding which row security policies apply. That default is normally a feature: it’s exactly what lets `web_anon`, which has no direct grant on `businesses` at all, read anything through `api.businesses`. But every object in this chapter was created from the `sudo -u postgres psql` session back in Exercise 1.1, which makes `postgres` — a superuser — the owner of `api.residents`. Superusers bypass row level security unconditionally, no exceptions. Left at its default, this view would evaluate Exercise 5.2’s policy as `postgres` on every single request, regardless of which role or JWT actually called it, and a bypassed policy filters nothing — every resident would see every row. `security_invoker = true` makes the view check privileges and RLS as the real caller instead: `web_resident`, after PostgREST’s per-request `SET ROLE`, exactly as Exercise 5.2’s policy expects.



(If you’re changing an *existing* view rather than creating this one fresh, reach for `ALTER VIEW api.residents SET (security_invoker = true);` instead of dropping and recreating it — `DROP VIEW` followed by `CREATE VIEW` makes a brand-new object with none of the original’s grants, so Exercise 5.2’s `GRANT SELECT ON api.residents TO web_resident` would silently need to be re-run, and the endpoint would 403 with permission denied for view residents until it is.)

```
NOTIFY pgrst, 'reload schema';
```

No GRANT yet, on purpose — not even to `web_resident`. RLS goes on before any role can read a single row.

5.2 — Enable RLS and add a self-only policy

```
ALTER TABLE residents ENABLE ROW LEVEL SECURITY;

CREATE POLICY resident_self_only ON residents
    FOR SELECT
    USING (id = (current_setting('request.jwt.claims', true)::json ->>
'resident_id')::int);

GRANT SELECT ON api.residents TO web_resident;
GRANT SELECT (id, full_name, neighbourhood) ON residents TO web_resident;
```

`current_setting('request.jwt.claims', true)` reads the JSON claims object PostgREST sets, per request, from the caller's verified JWT — the `true` second argument means “return NULL instead of erroring if it's unset,” which matters because `web_anon` requests never carry one at all. The policy compares the row's own `id` against whatever `resident_id` claim a valid, signed JWT presents.

Both GRANTS are required, and it's easy to stop at the first one and not notice.

`security_invoker = true` from Exercise 5.1 means `api.residents` no longer borrows its owner's privileges to read the underlying table — the invoking role needs real privileges of its own on `residents`, not just on the view sitting in front of it. Skip the second GRANT and every request fails with `permission denied for table residents`, even though the role clearly has `SELECT` on the view; the view-level check passes, and the query still dies one level down, on the base table it was never given access to. The column list `(id, full_name, neighbourhood)` matters too, not just the table name: granting only these three columns — matching exactly what `api.residents` selects — keeps `true_duplicate_of` unreachable even if `web_resident` were ever queried against directly instead of through the view, which a bare `GRANT SELECT ON residents` (no column list) would not.

5.3 — Mint a JWT per resident

```
#!/usr/bin/env python3.12
# mint_jwt.py - issue a dev-only resident session token
import sys
import jwt

SECRET = "portsmith-lab-book-dev-secret-do-not-use-in-production"

resident_id = int(sys.argv[1])
token = jwt.encode(
```

```

    {"role": "web_resident", "resident_id": resident_id},
    SECRET,
    algorithm="HS256",
)
print(token)

```

The `role` claim is what PostgREST uses to decide which role to `SET ROLE` to for this request — it has to be a role `authenticator` was granted membership in back in Exercise 1.1, or PostgREST refuses the token outright.

5.4 — Two residents, two different answers from the same endpoint

```

TOKEN_1=$(python mint_jwt.py 1)    # Adrian Foscolo
TOKEN_2=$(python mint_jwt.py 2)    # Marisol Quintero

curl -s "http://localhost:3000/residents" -H "Authorization: Bearer $TOKEN_1" | jq
curl -s "http://localhost:3000/residents" -H "Authorization: Bearer $TOKEN_2" | jq

[ { "id": 1, "full_name": "Adrian Foscolo", "neighbourhood": "Old Town" } ]

[ { "id": 2, "full_name": "Marisol Quintero", "neighbourhood": "Riverside" } ]

```

Identical request, identical route, identical `GRANT` — the only difference is which JWT signed it, and RLS silently rewrote each query’s effective `WHERE` clause to match.

5.5 — The gotcha: RLS filters, it doesn’t refuse

```

curl -s "http://localhost:3000/residents?id=eq.2" -H "Authorization: Bearer $TOKEN_1" | jq

[]

```

Resident 1, asking directly for resident 2’s row by id, does **not** get a 403 — they get 200 OK and an empty array. This is a meaningfully different failure mode from Exercise 4’s column-privilege denial: a `GRANT` violation is an error, loud and explicit; an RLS mismatch is just a `WHERE` clause that happens to match nothing, indistinguishable at the HTTP layer from “that id doesn’t exist.” Compare with no credentials at all:

```
curl -si "http://localhost:3000/residents"
```

```
HTTP/1.1 401 Unauthorized
```

```
{"code":"42501","details":null,"hint":null,"message":"permission denied for table residents"}
```

This one errors, the same way Exercise 4.3 did — `web_anon` was never granted `SELECT` on `api.residents` at all, so it never gets far enough to run into the RLS policy in the first

place. Two different denials, two different HTTP shapes, and the difference between them is worth being able to explain: a missing `GRANT` is a wall; RLS is a filter.

Exercise 6 — An RPC Endpoint for Fuzzy Search

6.1 — Wrap Chapter 5’s trigram search as a function

```
CREATE OR REPLACE FUNCTION api.search_businesses(search_term text)
RETURNS TABLE (business_id integer, name text, similarity numeric)
LANGUAGE sql STABLE AS $$
    SELECT business_id, name, round(similarity(name, search_term)::numeric, 3) AS
similarity
    FROM    business_names
    ORDER BY name <-> search_term
    LIMIT   5;
$$;

GRANT EXECUTE ON FUNCTION api.search_businesses(text) TO web_anon;
GRANT SELECT ON business_names TO web_anon;

NOTIFY pgrst, 'reload schema';
```

The function body is Chapter 5, Exercise 5’s exact “did you mean?” query, verbatim — `name <-> search_term` ordering by trigram distance, backed by the `idx_business_names_trgm_gist` GiST index that chapter built. PostgREST doesn’t need to know anything about trigrams; it just sees a function in the `api` schema and exposes it at `/rpc/`.

Both `GRANTS` are required, for the same reason Exercise 5.2 needed two of them. SQL functions default to `SECURITY INVOKER` — this one was never told otherwise — so `api.search_businesses()` RUNS its `SELECT ... FROM business_names` as whichever role actually called it, `web_anon` here, not as the function’s owner. `EXECUTE` only grants permission to *call* the function; it says nothing about what the function is allowed to touch once it’s running. Skip the second `GRANT` and 6.2 fails with `permission denied for table business_names`, `EXECUTE` privilege notwithstanding — the same wall Exercise 5.2 hit with `residents`, just one function-call away instead of one view away.

6.2 — Call it

```
curl -s -X POST "http://localhost:3000/rpc/search_businesses" \
-H "Content-Type: application/json" \
-d '{"search_term": "Ironsyde Auto"}' | jq
```

```
[
  { "business_id": 45, "name": "Ironsides Auto", "similarity": 0.647 },
  { "business_id": 21, "name": "AutoFix Portsmouth", "similarity": 0.143 },
  { "business_id": 4, "name": "Harbour Inn", "similarity": 0.040 },
  { "business_id": 36, "name": "The Art Depot", "similarity": 0.037 },
  { "business_id": 34, "name": "Riverside Cinema", "similarity": 0.033 }
]
```

Same misspelling, same top match, same scores as Chapter 5's own `psql` session — this endpoint is that query, not a reimplement of it.

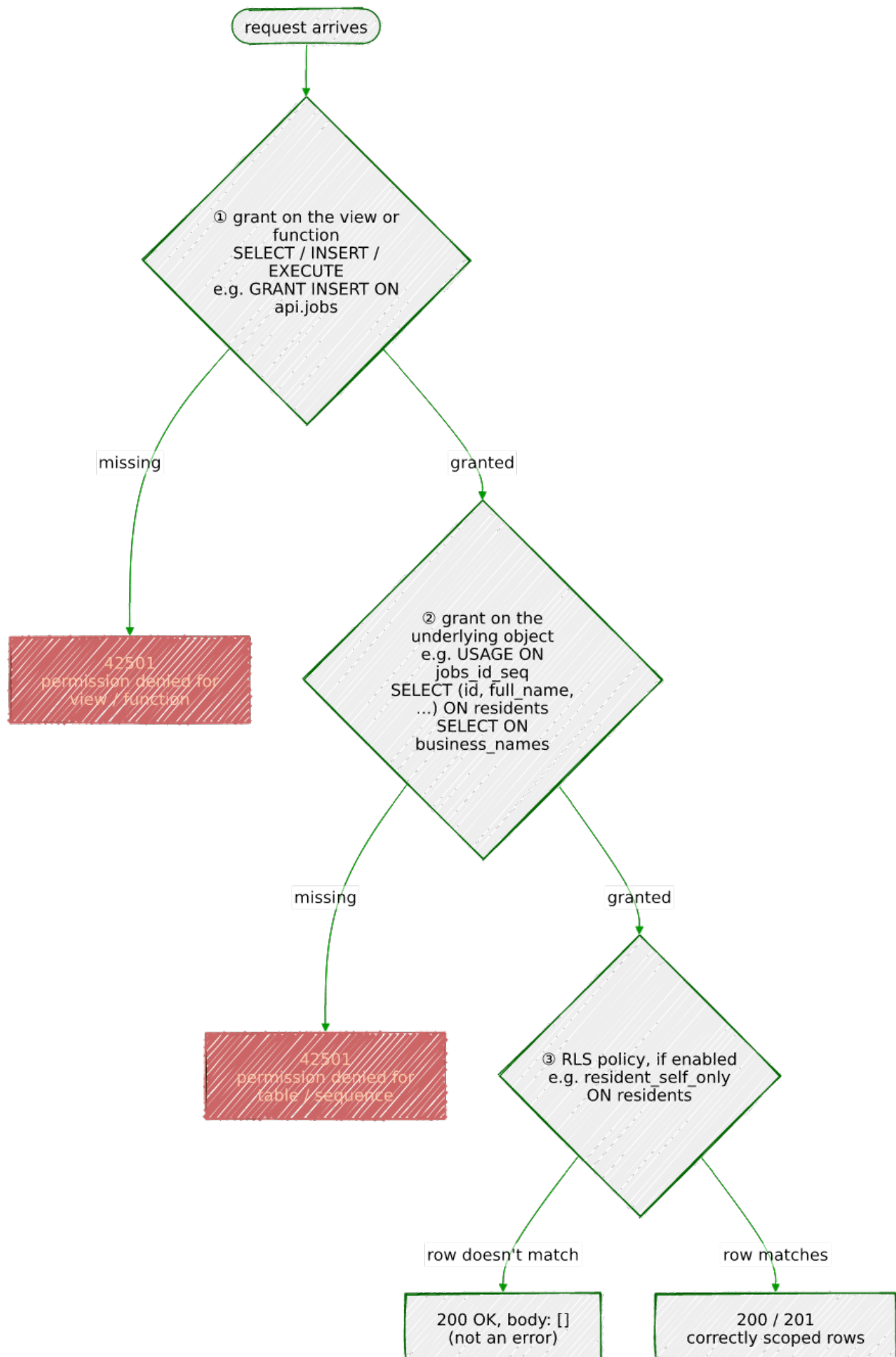
6.3 — GET works too, because the function is STABLE

```
curl -s "http://localhost:3000/rpc/search_businesses?search_term=Ironside+Auto" | jq -c '.[0]'
```

```
{"business_id":45,"name":"Ironside Auto","similarity":0.647}
```

PostgREST maps a function's SQL volatility onto which HTTP verbs it will accept: `VOLATILE` functions — anything that could plausibly write data — are reachable only by `POST`, on the theory that a `GET` should always be safe to retry, cache, or prefetch without side effects. `STABLE` and `IMMUTABLE` functions, like this one, are exposed under both `GET` and `POST`, because Postgres itself already promises they don't change anything. Declaring `LANGUAGE sql STABLE` back in 6.1 wasn't just documentation — it's the line that decided whether a browser could ever call this endpoint from a plain link.

Every real failure in this chapter — Exercise 3's `jobs_id_seq`, Exercise 5's `residents` columns, this exercise's `business_names` — turned out to be the same shape: one grant in place, a second, independent one missing, one layer further in than the error first suggests:



Summary — What You Should Now Know

Tool	What it does
authenticator role, NOINHERIT	The one role PostgREST logs in as; switches per-request via SET ROLE
db-anon-role	The role a request without a valid JWT runs as
api schema of views	Decouples the public API shape from the internal table shape
NOTIFY pgrst, 'reload schema'	Tells a running PostgREST server about new tables/views/ functions without a restart
?col=eq.x, ?order=, ? select=, ?limit=/? offset=	Query-parameter syntax mapping directly onto WHERE/ORDER BY/SELECT/LIMIT
Prefer: return=representation	Get the affected row(s) back in the response body
Prefer: count=exact / Content-Range	Ask for, and read, the total row count behind a paginated result
Column-level GRANT/ REVOKE	Restrict which fields a client can set on INSERT, enforced by Postgres itself
GRANT USAGE ON SEQUENCE ...	Needed alongside INSERT for any SERIAL/BIGSERIAL/IDENTITY column's default nextval() to succeed — table privileges don't imply sequence privileges
401 VS. 403	401 = denied and unauthenticated (or bad JWT); 403 = denied despite valid credentials
RLS policy vs. missing GRANT	RLS silently filters rows (still 200, possibly empty); a missing grant is a hard error
security_invoker = true on a view	Makes the view check privileges <i>and</i> RLS as the querying role, not the view's owner — required for RLS to mean anything through a view owned by a superuser or the table owner
/rpc/<function>	A SQL function exposed as an endpoint; GET-eligible only if STABLE/IMMUTABLE

The key design insight from this chapter is that every access-control decision PostgREST makes was already something PostgreSQL could do — this chapter never introduced a new authorization concept, only a new transport for reaching decisions the database was

always capable of making. `GRANT/REVOKE` from Exercise 4 and row level security from Exercise 5 aren't PostgREST features with database-flavored names; they are exactly the `GRANT` and `CREATE POLICY` you'd write for any other purpose, and PostgREST's only job was to authenticate a request, pick a role, and get out of the way. That's also this chapter's sharpest limitation, in the same sentence: anything that can't be expressed as one role running one SQL statement — multi-step workflows, calling another API mid-request, anything genuinely procedural — is out of scope by design, not by oversight.

Going further: Chapter 13's `LISTEN/NOTIFY` is the same primitive this chapter used for schema-reload notifications, applied to application data instead — a natural next step once `NOTIFY pgrst, 'reload schema'` feels familiar. Chapter 14's advisory locks are worth knowing about if an RPC function like `api.search_businesses()` ever needs to serialize access to a shared resource instead of just reading one. And the column-level `GRANT` from Exercise 4 generalizes: PostgreSQL's full privilege system — `SELECT`, `UPDATE`, and `REFERENCES` privileges can all be scoped per-column the same way `INSERT` was here — is worth a deliberate read through the `GRANT` documentation before exposing any table this way for real, well beyond what one chapter's exercises can cover.

Chapter 11 — Window Functions: Analytics Beyond `GROUP BY`

“`GROUP BY` answers a question by throwing away the rows that don’t fit in the answer. A window function answers the same question and keeps every row anyway.”

Background

If you’ve used `GROUP BY`, you already know the shape of the problem this chapter solves — and the shape of its one real limitation. `GROUP BY` answers “what’s the average rating per neighbourhood?” by collapsing every business in a neighbourhood down into a single output row: you get the average, but every individual business that went into computing it is gone from the result. Most of the time that’s exactly what you want. But plenty of real questions don’t fit that shape: “how does each business’s rating compare to its neighbourhood’s average, *while still showing me every business*?” “What’s this business’s running revenue total, quarter by quarter, without collapsing the quarters together?” `GROUP BY` cannot answer either one — the instant it groups, the individual rows that made up the group are gone for good.

A **window function** answers a `GROUP BY`-shaped question without paying that price. It looks at a set of rows related to the current row — its “window” — computes something over them, and attaches the result to the current row, which survives, completely unchanged, right alongside every other row. Nothing collapses. Start with 48 rows, end with 48 rows, every one of them now carrying an answer that depended on looking at some of its neighbours.

That’s the single idea this entire chapter builds on, and it’s worth sitting with before touching any syntax:

`GROUP BY` reduces row count. A window function never does.

Four pieces, defined before you see them used

Every example below is built from four ingredients. Knowing what each one means in plain language first should make the SQL read as sentences instead of unfamiliar syntax:

Piece	Plain-language meaning
<code>OVER (...)</code>	“Compute this using a window of related rows, not just this one row.” Attached after a function call — <code>AVG(x) OVER (...)</code> — it’s what turns an ordinary aggregate into a window function. An empty <code>OVER ()</code> means “the window is every row in the result.”
<code>PARTITION BY col</code>	“Only look at rows that share this row’s value of <code>col</code> .” The window equivalent of <code>GROUP BY</code> — it restricts <i>which rows count</i> , without collapsing any of them.
<code>ORDER BY col</code> (<i>written inside</i> <code>OVER (...)</code>)	“Put the rows in this order before computing.” This ordering lives entirely inside the window — it’s unrelated to the query’s own outer <code>ORDER BY</code> , and the two are free to differ or even conflict.
frame clause — e.g. <code>ROWS BETWEEN 6</code> <code>PRECEDING AND</code> <code>CURRENT ROW</code>	Out of the rows the partition and ordering make available, <i>exactly which ones</i> count toward this specific row’s calculation.

Exercise 1 walks through all four, one at a time, against a dataset small enough to check by hand — five numbers, no Portsmouth backstory required, so the mechanics stay in view before any real data shows up.

The Scenario

This chapter doesn’t tell one Portsmouth story — it reuses four different tables to show that window functions are a general-purpose tool, not a feature tied to any one kind of data:

Object	Source	Used for
<code>businesses</code>	Chapter 1	Ranking businesses within their own neighbourhood
<code>sensor_readings</code>	Chapter 8	7-day rolling averages and day-over-day change
<code>network_events</code>	Chapter 7	Detecting login “sessions” via gaps and islands
<code>business_revenue</code>		Running totals and percentage-of-category-total

Object	Source	Used for
	<i>(new, this chapter)</i>	

`business_revenue` is the one genuinely new thing here — a small, synthetic quarterly revenue figure for each of the 48 businesses from Chapter 1, built specifically so Exercise 6 has real running-total and percentage-of-partition data to work with.

Exercise Goals

By the end of this chapter you will be able to:

- Explain, without looking anything up, why `AVG(x) OVER ()` returns a value on every row while `AVG(x)` with `GROUP BY` returns one row — and predict, for any `OVER (...)` clause, exactly which rows are “in the window” for a given output row.
 - Rank rows within groups with `RANK()` and `DENSE_RANK()`, and state precisely how the two disagree the moment there’s a tie.
 - Write an explicit frame clause to compute a rolling average, and explain why adding `ORDER BY` to a window — with no frame clause at all — silently changes the default frame.
 - Use `LAG()`/`LEAD()` to compare a row to its neighbour without a self-join.
 - Recognize the “gaps and islands” pattern and use it to turn a raw event stream into sessions.
 - Combine two different `PARTITION BY` scopes — one for a running total, one for a percentage-of-total — in a single `SELECT`.
-

Installation

Nothing to install. Window functions have been part of core PostgreSQL since version 8.4 (2009) — no extension, no configuration.

Loading the Data

This chapter needs Chapters 1, 7, and 8’s data, plus one new small table:

```
python data/ch01_seed.py # businesses
python data/ch07_seed.py # network_events
python data/ch08_seed.py # sensors, then run Chapter 8's own exercises
                        # through Exercise 2 to get sensor_readings populated
python data/ch11_seed.py # business_revenue (new this chapter)
```

`business_revenue` only needs `businesses` to already exist — it doesn't depend on Chapters 7 or 8 at all.

Pin the session timezone

```
SET timezone = 'UTC';
```

Same reason as Chapter 8: Exercises 3 and 4 group `sensor_readings` by `date_trunc('day', recorded_at)`, and a day boundary computed in any timezone other than UTC quietly buckets a different set of 5-minute readings into “Feb 1” than the ones this chapter's numbers were computed from — not a rounding difference, a genuinely different average, since each bucket ends up averaging different rows entirely. Run this at the top of every session in this chapter, including the psql session that runs the prerequisite check below, or Exercises 3 and 4's numbers won't match what's printed here even though the query is identical.

Verify the prerequisites

```
SELECT 'businesses' AS table, COUNT(*) FROM businesses
UNION ALL SELECT 'network_events', COUNT(*) FROM network_events
UNION ALL SELECT 'sensor_readings', COUNT(*) FROM sensor_readings
UNION ALL SELECT 'business_revenue', COUNT(*) FROM business_revenue;
```

table	count
businesses	48
network_events	116
sensor_readings	9648000
business_revenue	192

(4 rows)

If all four match, proceed to the exercises.

Exercises

Exercise 1 — The Mental Model, Traced By Hand

Forget Portsmouth for a moment. Here are five temperature readings from one morning:

```
SELECT * FROM (VALUES
  ('06:00'::time, 52.1),
  ('07:00'::time, 53.4),
  ('08:00'::time, 55.8),
  ('09:00'::time, 58.2),
  ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);
```

reading_time	temp_f
06:00:00	52.1
07:00:00	53.4
08:00:00	55.8
09:00:00	58.2
10:00:00	60.5

(5 rows)

Five rows in, five rows to reason about. Keep this exact dataset in mind for everything below.

1.1 — GROUP BY collapses; OVER () doesn't

```
SELECT AVG(temp_f) FROM (VALUES
  ('06:00'::time, 52.1), ('07:00'::time, 53.4), ('08:00'::time, 55.8),
  ('09:00'::time, 58.2), ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);
```

avg
56.0000000000000000

(1 row)

Five rows go in, one number comes out. That's the `AVG(temp_f)` you already know. Now the window version — same aggregate, same result, one difference:

```
SELECT reading_time, temp_f,
  round(AVG(temp_f) OVER ()::numeric, 2) AS morning_avg
FROM (VALUES
  ('06:00'::time, 52.1), ('07:00'::time, 53.4), ('08:00'::time, 55.8),
  ('09:00'::time, 58.2), ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);
```

reading_time	temp_f	morning_avg
--------------	--------	-------------

06:00:00		52.1		56.00
07:00:00		53.4		56.00
08:00:00		55.8		56.00
09:00:00		58.2		56.00
10:00:00		60.5		56.00

(5 rows)

Five rows go in, five rows come out — every one of them now carrying 56.00, the exact number GROUP BY gave you, just not at the cost of the other four rows. An empty OVER () means “the window is every row here,” so every row gets the same whole-set average stamped onto it. This is the entire idea from the Background section, now sitting in front of you as actual output: nothing collapsed.

1.2 — The gotcha: adding ORDER BY silently changes the frame

Now add one thing — ORDER BY reading_time, *inside* the OVER (...):

```
SELECT reading_time, temp_f,
       round(AVG(temp_f) OVER (ORDER BY reading_time)::numeric, 2) AS running_avg
FROM (VALUES
      ('06:00'::time, 52.1), ('07:00'::time, 53.4), ('08:00'::time, 55.8),
      ('09:00'::time, 58.2), ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);
```

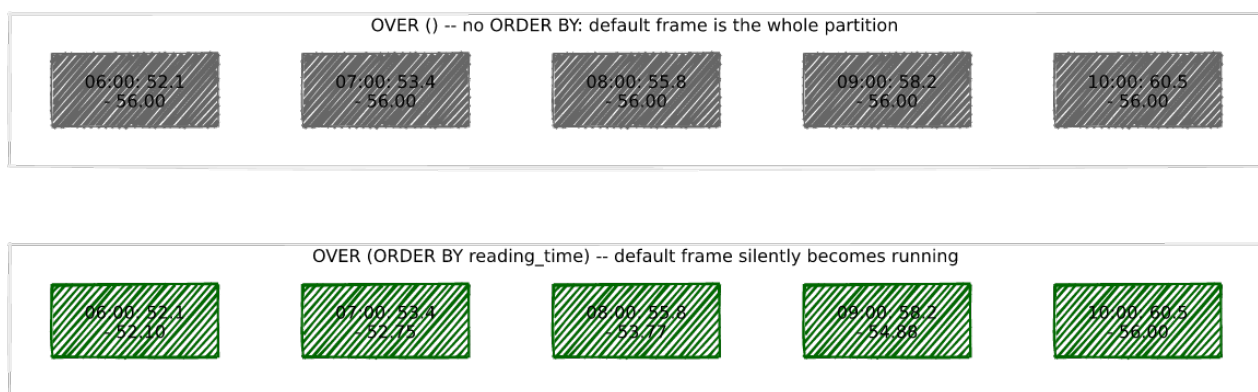
reading_time		temp_f		running_avg
-----+-----+-----				
06:00:00		52.1		52.10
07:00:00		53.4		52.75
08:00:00		55.8		53.77
09:00:00		58.2		54.88
10:00:00		60.5		56.00

(5 rows)

Same function, same data, completely different numbers — and only the *last* row still shows 56.00. This is the single most common surprise in all of window functions, so it’s worth tracing exactly what changed, row by row:

reading_time	temp_f	rows actually included in this row’s window	running_avg
06:00	52.1	[06:00]	52.10
07:00	53.4	[06:00, 07:00]	52.75
08:00	55.8	[06:00, 07:00, 08:00]	53.77
09:00	58.2	[06:00, 07:00, 08:00, 09:00]	54.88
10:00	60.5	[06:00, 07:00, 08:00, 09:00, 10:00]	56.00

Adding `ORDER BY` to a window did not just sort the rows — it silently changed *how many rows count* for each one, from “every row” down to “every row up to and including this one.” That’s PostgreSQL’s default frame the moment a window has an `ORDER BY` and no explicit frame clause: `RANGE UNBOUNDED PRECEDING AND CURRENT ROW`, i.e., a running calculation. Without `ORDER BY`, there’s nothing to run *up to*, so the default frame is the whole partition instead, which is exactly what 1.1’s flat `56.00` was. Nothing in the syntax announces this change — `ORDER BY` looks like it should only affect display order, and inside a window it affects something much bigger. Committing this one rule to memory now will save you from debugging a “wrong” running total later that was never actually wrong, just unexpectedly running.



1.3 — PARTITION BY: independent windows, still no collapsing

Add a second sensor to the tiny dataset:

```
SELECT sensor_label, reading_time, temp_f,
       round(AVG(temp_f) OVER (PARTITION BY sensor_label ORDER BY reading_time)::numeric,
2) AS running_avg
FROM (VALUES
      ('Temp-01', '06:00'::time, 52.1), ('Temp-01', '07:00'::time, 53.4), ('Temp-01',
'08:00'::time, 55.8),
      ('Temp-02', '06:00'::time, 48.9), ('Temp-02', '07:00'::time, 49.5), ('Temp-02',
'08:00'::time, 50.1)
) AS t(sensor_label, reading_time, temp_f)
ORDER BY sensor_label, reading_time;
```

sensor_label	reading_time	temp_f	running_avg
Temp-01	06:00:00	52.1	52.10
Temp-01	07:00:00	53.4	52.75
Temp-01	08:00:00	55.8	53.77
Temp-02	06:00:00	48.9	48.90
Temp-02	07:00:00	49.5	49.20
Temp-02	08:00:00	50.1	49.50

Temp-01’s running average never sees Temp-02’s numbers, and vice versa — `PARTITION BY` walled the two sensors off into completely independent windows, the same way `GROUP BY sensor_label` would have, except both sensors’ six rows are all still here. This is `PARTITION BY` doing to a window exactly what it would do to a `GROUP BY`: split the data into groups — just without ever throwing a row away.

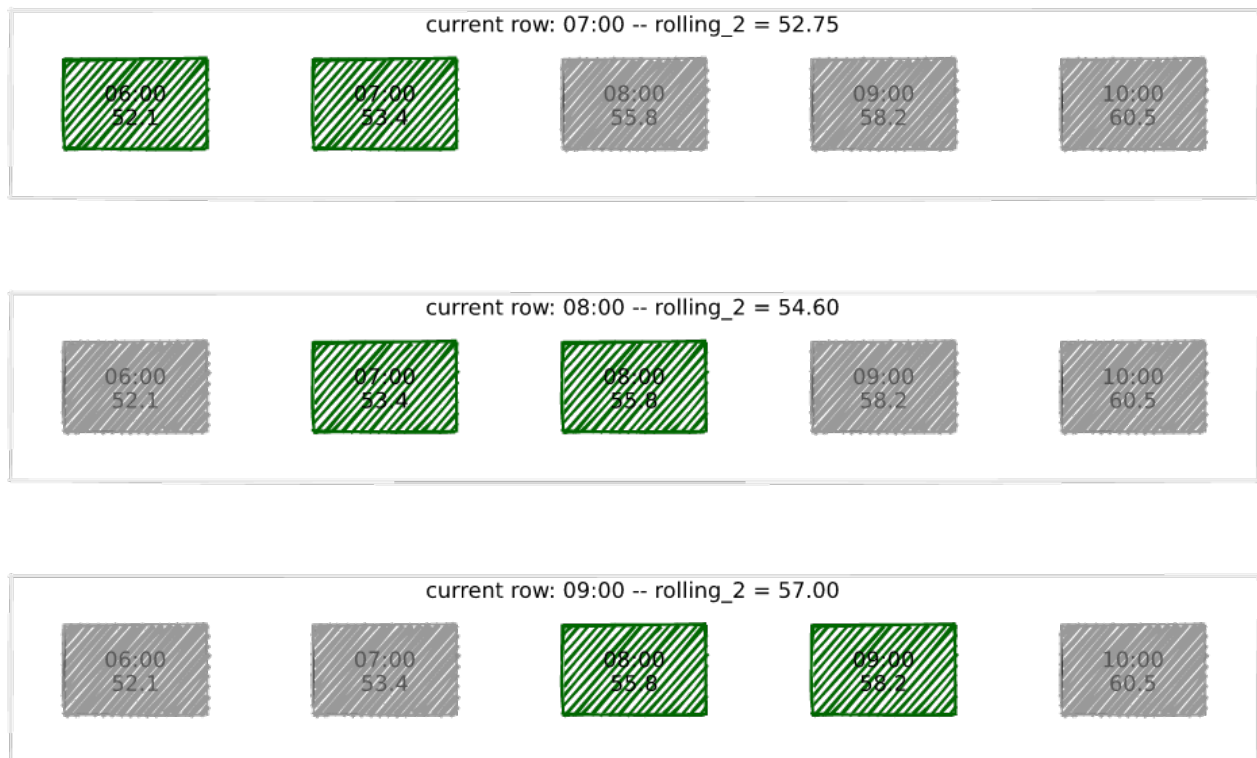
1.4 — An explicit frame clause: precise control

1.2’s running average grows to include more and more history as it goes — by the last row, it’s averaging all five readings. A **rolling** average instead asks “the last *N* readings only,” which needs an explicit frame clause instead of relying on the default:

```
SELECT reading_time, temp_f,
       round(AVG(temp_f) OVER (ORDER BY reading_time
                              ROWS BETWEEN 1 PRECEDING AND CURRENT ROW)::numeric, 2)
AS rolling_2
FROM (VALUES
      ('06:00'::time, 52.1), ('07:00'::time, 53.4), ('08:00'::time, 55.8),
      ('09:00'::time, 58.2), ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);
```

reading_time	temp_f	rolling_2
06:00:00	52.1	52.10
07:00:00	53.4	52.75
08:00:00	55.8	54.60
09:00:00	58.2	57.00
10:00:00	60.5	59.35

reading_time	rows in this row’s window (1 PRECEDING AND CURRENT ROW)	rolling_2
06:00	[06:00] — no prior row exists yet, so just itself	52.10
07:00	[06:00, 07:00]	52.75
08:00	[07:00, 08:00]	54.60
09:00	[08:00, 09:00]	57.00
10:00	[09:00, 10:00]	59.35



The window is now a fixed-size sliding pair, not an ever-growing history — 08:00's value depends on 07:00 and 08:00 only, never 06:00. ROWS BETWEEN 1 PRECEDING AND CURRENT ROW is deliberately the same shape of clause Exercise 3 uses next, just with a smaller number: this tiny example *is* a 2-reading rolling average, and Exercise 3 is nothing more than this same idea with 6 PRECEDING and real sensor data behind it.

Exercise 2 — Ranking Businesses Within Their Neighbourhood

2.1 — RANK() and DENSE_RANK(), side by side

```
SELECT name, neighbourhood, (details->>'rating')::numeric AS rating,
       RANK() OVER (PARTITION BY neighbourhood ORDER BY (details->>'rating')::numeric DESC) AS rank,
       DENSE_RANK() OVER (PARTITION BY neighbourhood ORDER BY (details->>'rating')::numeric DESC) AS dense_rank
FROM businesses
WHERE neighbourhood IN ('Harbour District', 'Riverside')
ORDER BY neighbourhood, rating DESC;
```

name	neighbourhood	rating	rank	dense_rank
Lighthouse Bookshop	Harbour District	5.0	1	1
Portsmith Fish Market	Harbour District	4.8	2	2
Saltbox Gallery	Harbour District	4.7	3	3

Mariners Rest B&B	Harbour District	4.7	3	3
Harbour View Theater	Harbour District	4.6	5	4
Tidal Wave Surf Shop	Harbour District	4.5	6	5
The Gilded Clam	Harbour District	4.5	6	5
Harbour Inn	Harbour District	4.3	8	6
Anchor & Oar Tavern	Harbour District	4.1	9	7
River Bend Bakery	Riverside	4.8	1	1
Portsmouth Veterinary Clinic	Riverside	4.8	1	1
Dr. Chen Dentistry	Riverside	4.7	3	2
Quay Street Deli	Riverside	4.6	4	3
The Art Depot	Riverside	4.6	4	3
Thai Orchid	Riverside	4.5	6	4
The Riverside Vegan	Riverside	4.5	6	4
Riverside Cinema	Riverside	4.4	8	5
Portsmouth Pharmacy	Riverside	4.3	9	6

(18 rows)

(Lighthouse Bookshop’s 5.0 is Chapter 1, Exercise 5’s `jsonb_set` update — if you’re seeing 4.9 instead, that exercise hasn’t run yet in this database, which is fine; the ranking logic below is identical either way.)

2.2 — Reading the tie exactly

Look at Harbour District’s two businesses tied at 4.7: both get `rank = 3`. The next business down, at 4.6, gets `rank = 5` under `RANK()` — 4 is simply never used, because two rows already claimed “3rd place” and `RANK()` counts every row ahead of you, ties included. `DENSE_RANK()` disagrees on principle: it counts *distinct rating values* seen so far, so 4.6 is the 4th distinct value in the list and gets `dense_rank = 4`, no gap. Neither is “more correct” — `RANK()` answers “how many businesses rate at or above me,” `DENSE_RANK()` answers “how many distinct rating tiers are at or above me” — but they give a different answer to “who’s in 4th place” the moment any tie exists, and Riverside’s 4.8 tie two rows later shows the same split happening again.

Exercise 3 — A 7-Day Rolling Average on `sensor_readings`

3.1 — Aggregate to daily first

`sensor_readings` reports every five minutes — a rolling average over raw readings would be a rolling average of noise. Roll up to one row per day first, the same shape of query Chapter 9 turned into a materialized view, just computed directly here instead:

```
WITH daily AS (
  SELECT date_trunc('day', recorded_at)::date AS reading_day,
```



```

        round(AVG(reading_value)::numeric, 2) AS daily_avg
    FROM    sensor_readings
    WHERE   sensor_id = 1
    AND     recorded_at >= '2024-02-01' AND recorded_at < '2024-02-15'
    GROUP BY 1
)
SELECT * FROM daily ORDER BY reading_day;

```

reading_day	daily_avg
2024-02-01	42.46
2024-02-02	42.37
2024-02-03	42.37
2024-02-04	42.40
2024-02-05	42.32
2024-02-06	42.43
2024-02-07	42.43
2024-02-08	42.38
2024-02-09	42.29
2024-02-10	42.40
2024-02-11	42.45
2024-02-12	42.48
2024-02-13	42.40
2024-02-14	42.41

(14 rows)

Sensor 1 over the first two weeks of February — the month right after Chapter 8’s dropped January partition, well clear of it. Check `sensors` and this is genuinely Temp-01, the same sensor Exercise 1.3’s toy example was named after — the tiny hand-crafted dataset was standing in for exactly this real one.

3.2 — Layer the rolling average on top

```

WITH daily AS (
    SELECT date_trunc('day', recorded_at)::date AS reading_day,
           round(AVG(reading_value)::numeric, 2) AS daily_avg
    FROM    sensor_readings
    WHERE   sensor_id = 1
    AND     recorded_at >= '2024-02-01' AND recorded_at < '2024-02-15'
    GROUP BY 1
)
SELECT reading_day, daily_avg,
       round(AVG(daily_avg) OVER (ORDER BY reading_day
                                   ROWS BETWEEN 6 PRECEDING AND CURRENT ROW)::numeric, 2)
AS rolling_7day_avg
FROM    daily
ORDER BY reading_day;

```

reading_day	daily_avg	rolling_7day_avg
2024-02-01	42.46	42.46
2024-02-02	42.37	42.42
2024-02-03	42.37	42.40
2024-02-04	42.40	42.40
2024-02-05	42.32	42.38
2024-02-06	42.43	42.39
2024-02-07	42.43	42.40
2024-02-08	42.38	42.39
2024-02-09	42.29	42.37
2024-02-10	42.40	42.38
2024-02-11	42.45	42.39
2024-02-12	42.48	42.41
2024-02-13	42.40	42.40
2024-02-14	42.41	42.40

(14 rows)

6 PRECEDING AND CURRENT ROW is 7 rows total — the exact same shape as Exercise 1.4’s 1 PRECEDING AND CURRENT ROW, just sized for a week instead of a pair. Watch the first week build up exactly like 1.4’s first row did: Feb 1 has no prior days, so its “7-day” average is really a 1-day average; Feb 2 averages 2 days; the window doesn’t reach a true 7 full days until Feb 7. ROWS BETWEEN ... PRECEDING never errors when fewer rows exist than requested — it just uses whatever’s actually available, which is worth knowing before trusting the first few values of any rolling window on real data.

Exercise 4 — LAG()/LEAD(): Comparing a Row to Its Neighbour

4.1 — Day-over-day change, without a self-join

```
WITH daily AS (
    SELECT date_trunc('day', recorded_at)::date AS reading_day,
           round(AVG(reading_value)::numeric, 2) AS daily_avg
    FROM   sensor_readings
    WHERE  sensor_id = 1
    AND    recorded_at >= '2024-02-01' AND recorded_at < '2024-02-15'
    GROUP BY 1
)
SELECT reading_day, daily_avg,
       LAG(daily_avg) OVER (ORDER BY reading_day) AS prev_day_avg,
       round((daily_avg - LAG(daily_avg) OVER (ORDER BY reading_day))::numeric, 2) AS
day_over_day_change,
       LEAD(daily_avg) OVER (ORDER BY reading_day) AS next_day_avg
FROM   daily
```

```
ORDER BY reading_day
LIMIT 6;
```

reading_day	daily_avg	prev_day_avg	day_over_day_change	next_day_avg
2024-02-01	42.46			42.37
2024-02-02	42.37	42.46	-0.09	42.37
2024-02-03	42.37	42.37	0.00	42.40
2024-02-04	42.40	42.37	0.03	42.32
2024-02-05	42.32	42.40	-0.08	42.43
2024-02-06	42.43	42.32	0.11	42.43

(6 rows)

`LAG(col)` reaches one row backward in the window's order; `LEAD(col)` reaches one row forward. Both are just `AVG(...)` `OVER (...)`'s siblings — ordinary window functions, not aggregates, so they don't need a frame clause at all. February 1st has no prior day, so `prev_day_avg` and `day_over_day_change` are both `NULL` for it — not zero, not an error, genuinely unknown, exactly the way a self-join against “yesterday” would also come up empty for the very first row. Before this chapter, the only way to compare a row to its neighbour was a self-join on `date - 1 = date`; `LAG()/LEAD()` is that same comparison with no join at all.

Exercise 5 — Gaps and Islands: Detecting Sessions in `network_events`

5.1 — The pattern, in words before SQL

“Gaps and islands” names a specific two-step trick: find the *gaps* (where consecutive rows, for the same actor, are far enough apart in time to count as separate events), then turn the space between gaps into *islands* (contiguous runs, numbered, that become your sessions). Step one is `LAG()` from Exercise 4. Step two is a cumulative `SUM()` — a running total of “did a new island start yet,” which is exactly Exercise 1.2's running-total behavior, repurposed as a counter instead of an average.

5.2 — Step one: flag where each session starts

```
SELECT source_ip, event_type, occurred_at,
       (occurred_at - LAG(occurred_at) OVER (PARTITION BY source_ip ORDER BY
occurred_at))
       > interval '5 minutes' AS is_new_session
FROM   network_events
WHERE  source_ip IN ('192.0.2.47', '192.0.2.151')
ORDER BY source_ip, occurred_at;
```

source_ip	event_type	occurred_at	is_new_session
192.0.2.47	api_call	2024-03-10 00:18:00-05	f
192.0.2.47	api_call	2024-03-10 00:20:00-05	f
192.0.2.47	api_call	2024-03-10 00:28:00-05	t
192.0.2.47	api_call	2024-03-10 00:36:00-05	t
192.0.2.47	api_call	2024-03-10 00:38:00-05	f
192.0.2.151	api_call	2024-03-09 23:53:00-05	f
192.0.2.151	api_call	2024-03-09 23:55:00-05	f
192.0.2.151	api_call	2024-03-09 23:56:00-05	f
192.0.2.151	api_error	2024-03-10 00:00:00-05	f
192.0.2.151	api_call	2024-03-10 00:07:00-05	t
192.0.2.151	api_call	2024-03-10 00:12:00-05	f

A 5-minute threshold: 192.0.2.47's second event lands 2 minutes after its first (f, still the same visit), but its third lands 8 minutes after that (t — long enough to count as a new visit). The first row for any IP has nothing before it, so LAG() returns NULL and the comparison is NULL, not true or false — handled explicitly in the next step.

5.3 — Step two: turn the flags into session numbers

```

WITH gapped AS (
    SELECT source_ip, event_type, occurred_at,
           (occurred_at - LAG(occurred_at) OVER (PARTITION BY source_ip ORDER BY
occurred_at))
           > interval '5 minutes' AS is_new_session
    FROM   network_events
),
islands AS (
    SELECT source_ip, event_type, occurred_at,
           SUM(CASE WHEN is_new_session IS NOT FALSE THEN 1 ELSE 0 END)
           OVER (PARTITION BY source_ip ORDER BY occurred_at) AS session_num
    FROM   gapped
)
SELECT source_ip, session_num, event_type, occurred_at
FROM   islands
WHERE  source_ip IN ('192.0.2.47', '192.0.2.151')
ORDER BY source_ip, occurred_at;

```

source_ip	session_num	event_type	occurred_at
192.0.2.47	1	api_call	2024-03-10 00:18:00-05
192.0.2.47	1	api_call	2024-03-10 00:20:00-05
192.0.2.47	2	api_call	2024-03-10 00:28:00-05
192.0.2.47	3	api_call	2024-03-10 00:36:00-05
192.0.2.47	3	api_call	2024-03-10 00:38:00-05
192.0.2.151	1	api_call	2024-03-09 23:53:00-05
192.0.2.151	1	api_call	2024-03-09 23:55:00-05
192.0.2.151	1	api_call	2024-03-09 23:56:00-05

192.0.2.151		1		api_error		2024-03-10 00:00:00-05
192.0.2.151		2		api_call		2024-03-10 00:07:00-05
192.0.2.151		2		api_call		2024-03-10 00:12:00-05

IS NOT FALSE — not = true — is what makes a row’s own first event (where `is_new_session` is NULL) correctly count as the start of session 1 instead of silently vanishing from every sum downstream; NULL = true and NULL AND anything are both NULL in SQL’s three-valued logic, never true, so a plain `WHEN is_new_session THEN 1` would skip every partition’s opening row. 192.0.2.47 splits into three short sessions; 192.0.2.151 splits into two. Neither IP did anything unusual — the same `api_calls`, just separated by an 8- and a 7-minute pause respectively, long enough to cross this query’s 5-minute line.

5.4 — Roll it up

```
WITH gapped AS (
    SELECT source_ip, occurred_at,
           (occurred_at - LAG(occurred_at) OVER (PARTITION BY source_ip ORDER BY
occurred_at))
           > interval '5 minutes' AS is_new_session
    FROM   network_events
),
islands AS (
    SELECT source_ip, occurred_at,
           SUM(CASE WHEN is_new_session IS NOT FALSE THEN 1 ELSE 0 END)
           OVER (PARTITION BY source_ip ORDER BY occurred_at) AS session_num
    FROM   gapped
)
SELECT COUNT(DISTINCT source_ip)           AS distinct_ips,
       COUNT(DISTINCT (source_ip, session_num)) AS total_sessions
FROM   islands;
```

distinct_ips	total_sessions
51	66

(1 row)

51 distinct IPs produced 66 sessions — 15 of them split into more than one visit under this threshold. Change `interval '5 minutes'` to `interval '10 minutes'` (the longest gap that exists anywhere in this dataset, per Chapter 7’s generator) and every one of those 66 collapses back down to exactly 51: the threshold you choose *is* the definition of “one visit,” and this query has no way of knowing which threshold is right for your actual users — that’s a judgment call the data alone can’t make for you.

Exercise 6 — Running Total and Percentage-of-Category-Total, Together

6.1 — Two different partitions, one query

```
WITH revenue_calc AS (  
    SELECT b.id, b.name, b.details->>'category' AS category, r.quarter, r.revenue,  
           round(SUM(r.revenue) OVER (PARTITION BY r.business_id ORDER BY r.quarter  
                                     ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT  
ROW))::numeric, 2)  
           AS running_total,  
           round(SUM(r.revenue) OVER (PARTITION BY b.id) * 100.0  
                 / SUM(r.revenue) OVER (PARTITION BY b.details->>'category'), 2)  
           AS pct_of_category_annual  
    FROM   business_revenue r  
    JOIN   businesses b ON b.id = r.business_id  
)  
SELECT name, quarter, revenue, running_total, pct_of_category_annual  
FROM   revenue_calc  
WHERE  name IN ('The Gilded Clam', 'Bella Napoli')  
ORDER BY name, quarter;
```

name	quarter	revenue	running_total	pct_of_category_annual
Bella Napoli	1	68827.50	68827.50	7.35
Bella Napoli	2	84983.85	153811.35	7.35
Bella Napoli	3	90452.38	244263.73	7.35
Bella Napoli	4	75346.83	319610.56	7.35
The Gilded Clam	1	75282.75	75282.75	8.04
The Gilded Clam	2	92954.38	168237.13	8.04
The Gilded Clam	3	98935.80	267172.93	8.04
The Gilded Clam	4	82413.52	349586.45	8.04

Three window functions, two different `PARTITION BY` scopes, in the exact same `SELECT`. `running_total` partitions by `business_id` and orders by `quarter` — Exercise 1.2’s running sum, applied to money instead of temperature. `pct_of_category_annual` partitions by `category` with **no `ORDER BY` at all** — Exercise 1.1’s flat, whole-partition total, computed twice with two different partitions (once for just this business, once for its whole category) and divided. Both restaurants’ percentages stay identical across all four of their own rows, exactly like 1.1’s 56.00 did, because neither has an `ORDER BY` to turn it into anything running.

6.2 — A gotcha worth hitting on purpose

Restrict the query further, to see what a narrower `WHERE` clause does to a percentage that’s supposed to mean “share of the whole category”:

```

SELECT b.name, r.quarter, r.revenue,
       round(SUM(r.revenue) OVER (PARTITION BY b.id) * 100.0
            / SUM(r.revenue) OVER (PARTITION BY b.details->'category'), 2) AS
pct_of_category
FROM   business_revenue r
JOIN   businesses b ON b.id = r.business_id
WHERE  b.name IN ('The Gilded Clam', 'Bella Napoli') -- filtered BEFORE the window runs
ORDER BY b.name, r.quarter;

```

name	quarter	revenue	pct_of_category
Bella Napoli	1	68827.50	47.76
Bella Napoli	2	84983.85	47.76
Bella Napoli	3	90452.38	47.76
Bella Napoli	4	75346.83	47.76
The Gilded Clam	1	75282.75	52.24
The Gilded Clam	2	92954.38	52.24
The Gilded Clam	3	98935.80	52.24
The Gilded Clam	4	82413.52	52.24

47.76 + 52.24 = 100.00 — these two restaurants apparently make up the *entire* restaurant category, when Portsmouth actually has 15. `WHERE` runs before `OVER (...)` ever sees a row: filtering down to two businesses filtered the `PARTITION BY category` window down to just those same two businesses, so “the category total” silently became “the total of the two rows I happened to ask for.” The fix is 6.1’s structure, not a different formula: compute every window function first, across the *entire* unfiltered table, inside a CTE — then filter the CTE’s output afterward, in an outer query the windows never see. A `WHERE` clause placed before a window and a `WHERE` clause placed after one are answering two different questions, and PostgreSQL will never warn you which one you actually wrote.

Summary — What You Should Now Know

Tool	What it does
<code>func(...) OVER (...)</code>	Computes across a window of related rows without collapsing the current row away
<code>PARTITION BY col</code>	Restricts the window to rows sharing this row’s value — <code>GROUP BY</code> without the collapsing
<code>ORDER BY col inside OVER (...)</code>	Orders the window — and, with no explicit frame, silently switches the default frame from “whole partition” to “running up to this row”

Tool	What it does
<code>ROWS BETWEEN x PRECEDING AND CURRENT ROW</code>	An explicit frame — precise, fixed-size control over which neighbouring rows count
<code>RANK()</code> VS. <code>DENSE_RANK()</code>	Agree with no ties; <code>RANK()</code> leaves gaps after a tie, <code>DENSE_RANK()</code> never does
<code>LAG()</code> / <code>LEAD()</code>	Reach one row backward/forward in window order — a neighbour comparison with no self-join
gaps and islands	<code>LAG()</code> flags where a new group starts; a cumulative <code>SUM()</code> turns those flags into group numbers
<code>IS NOT FALSE</code>	The three-valued-logic-safe way to treat a <code>NULL</code> flag (a partition's first row) as “start a new group”
Two <code>PARTITION BY</code> scopes, one query	Perfectly legal — e.g. a running total per entity alongside a percentage of a <i>different</i> , broader group
<code>WHERE</code> before <code>OVER (...)</code>	Filters the rows a window function ever sees — a narrow <code>WHERE</code> silently narrows what “the partition total” means

The key design insight from this chapter is the one from the very first paragraph, now proven six different ways: a window function is what you reach for the moment a `GROUP BY`-shaped question needs an answer *without* losing the rows that produced it. Ranking within a group, a rolling average, a neighbour comparison, a session boundary, a running total next to a category share — every one of these is the same underlying move, `OVER (...)` attached to a function that would otherwise collapse your data, with `PARTITION BY`, `ORDER BY`, and a frame clause as the three knobs that decide exactly which neighbouring rows a given row is allowed to see.

Going further: Chapter 9's materialized views precompute the exact kind of daily rollup Exercise 3 built on the fly — worth comparing directly now that you've seen both: a materialized view pays the aggregation cost once, at refresh time, while a window function pays it on every query but never goes stale. Chapter 20's `pg_stat_statements` work benefits from the running-total pattern in Exercise 6 when tracking cumulative query cost over time. And Chapter 12's recursive CTEs are this book's other tool for “a query that needs to see more than just the current row” — recursive CTEs walk relationships the data itself defines (a parent, a neighbour node), where window functions walk an ordering you impose yourself; knowing which kind of “related rows” a problem actually has is most of the work of picking the right one.

Chapter 12 — Recursive CTEs: Graphs and Hierarchies

“SQL doesn’t have loops. `WITH RECURSIVE` is a query that keeps calling itself until it runs out of new things to say.”

Background

Every table so far in this book has been “flat” — rows related to each other through a foreign key, sure, but never through a relationship of *unknown depth*. A resident belongs to one neighbourhood. A reading belongs to one sensor. But “who does this employee ultimately report to?” or “how do I get from this intersection to that one?” can’t be answered by following one foreign key — the answer might be one hop away, or ten, and a plain `JOIN` has to know in advance how many hops to write. A **recursive CTE**, written `WITH RECURSIVE`, is PostgreSQL’s answer to “I don’t know how many joins this needs — figure it out as you go.”

Structurally, a recursive CTE has two halves glued together with `UNION` or `UNION ALL`:

```
WITH RECURSIVE cte_name AS (  
    SELECT ...                -- ① the anchor: runs exactly once  
    UNION ALL  
    SELECT ...                -- ② the recursive term: references cte_name itself  
    FROM   some_table  
    JOIN   cte_name ON ...  
)  
SELECT * FROM cte_name;
```

The **anchor** runs once and seeds the result with a starting set of rows. The **recursive term** then runs *repeatedly* — each time, it sees only the rows the *previous* run just produced, joins them against the table again to find “the next hop,” and adds whatever new rows come out. This keeps going until one run of the recursive term produces zero new rows, at which point PostgreSQL stops and the final result is everything the anchor and every round of the recursive term ever added together. Nothing here is exotic under the hood — it’s an ordinary loop, just spelled out as a query instead of application code, and it stops for the most ordinary reason a loop ever stops: it ran out of new work.

Trees are graphs that promised to behave. A tree — an org chart, a category hierarchy — is a graph where every node has exactly one parent and there’s no way to walk back to somewhere you’ve already been. A road network makes no such promise: intersections connect to several other intersections, and it’s entirely possible to walk in a circle and end up back where you started. Recursive CTEs handle both shapes with the same syntax, but only the graph case needs you to actively guard against walking in circles forever — Exercises 1 through 3 build the tree intuition first, precisely so Exercise 4 can show you, concretely, what goes wrong without that guard, before Exercise 5 asks you to navigate a graph that has a real cycle built into it on purpose.

The Scenario

Object	Source	Shape
<code>city_org</code>	<i>(new)</i>	Tree — Portsmouth’s 30-person city government, 4 levels deep
<code>intersections,</code> <code>road_segments</code>	<i>(new, derived from Chapter 2’s</i> <i><code>city_infrastructure</code>)</i>	Graph — a real road network, cycles included
<code>categories</code>	<i>(new, derived from Chapter 1’s</i> <i>real category data)</i>	Tree — a 3-level faceted-search hierarchy

`intersections` and `road_segments` aren’t invented — every node is a real point where two of Chapter 2’s actual road `LINESTRINGS` cross, found with `ST_Intersects`, and every edge length is the real along-the-road distance for that stretch, not a straight-line guess. Ring Road bends around three sides of the city between some of its intersections, so a straight line between two of its crossing points would have understated the real distance by more than 5 kilometres in one case — Exercise 5 puts that exact gap to use.

Exercise Goals

By the end of this chapter you will be able to:

- Explain the anchor/recursive-term structure of `WITH RECURSIVE` well enough to predict how many times the recursive term will run for a given tree.

- Walk a hierarchy in both directions — up to the root, and down to every descendant — and know which direction each shape of question needs.
 - Compute each node’s depth and render an arbitrary-depth tree as indented text.
 - Recognize when a self-referencing table can contain a genuine cycle, and use `CYCLE ... SET ... USING` to detect one instead of hanging.
 - Write a breadth-first recursive CTE to find a shortest path through a graph, and explain why “fewest hops” and “shortest distance” are not always the same answer.
 - Flatten a category tree into an ancestor list for faceted search.
-

Installation

Nothing to install. `WITH RECURSIVE` has been part of core PostgreSQL since version 8.4 (2009) — the same release that introduced Chapter 11’s window functions.

Loading the Data

This chapter needs Chapters 1 and 2’s data, plus its own new tables:

```
python data/ch01_seed.py    # businesses (categories are derived from this)
python data/ch02_seed.py    # city_infrastructure (roads are derived from this)
python data/ch12_seed.py    # city_org, intersections, road_segments, categories
```

Verify the prerequisites

```
SELECT 'city_org' AS table, COUNT(*) FROM city_org
UNION ALL SELECT 'intersections', COUNT(*) FROM intersections
UNION ALL SELECT 'road_segments', COUNT(*) FROM road_segments
UNION ALL SELECT 'categories', COUNT(*) FROM categories;
```

table	count
city_org	30
intersections	15
road_segments	19
categories	48

(4 rows)

If all four match, proceed to the exercises.

Exercises

Exercise 1 — Walking to the Root

1.1 — The shape of city_org

```
\d city_org
```

Table "public.city_org"					
Column	Type	Collation	Nullable	Default	
id	integer		not null	nextval('city_org_id_seq'::regclass)	
name	text		not null		
title	text		not null		
manager_id	integer				

Foreign-key constraints:

"city_org_manager_id_fkey" FOREIGN KEY (manager_id) REFERENCES city_org(id)

A **self-referencing foreign key** — `manager_id` points to another row in the same table.

This one column is the entire tree: the Mayor's `manager_id` is NULL (the root has no manager), and everyone else's `manager_id` points to the row above them.

1.2 — From a patrol officer to the Mayor

```
WITH RECURSIVE chain AS (  
    SELECT id, name, title, manager_id, 0 AS depth  
    FROM city_org  
    WHERE name = 'Kwame Asante'  
  
    UNION ALL  
  
    SELECT o.id, o.name, o.title, o.manager_id, chain.depth + 1  
    FROM city_org o  
    JOIN chain ON o.id = chain.manager_id  
)  
SELECT depth, name, title FROM chain ORDER BY depth;
```

depth	name	title
0	Kwame Asante	Patrol Officer
1	Marcus Reilly	Patrol Captain
2	Diane Okonjo	Chief of Police
3	Coretta Vance	Mayor

(4 rows)

Trace exactly what happened, round by round, since this is the pattern every other exercise in this chapter builds on:

- **Anchor** (`depth = 0`): finds exactly one row, Kwame Asante.
- **Round 1** of the recursive term: joins `city_org` against the anchor's *one* row, looking for whoever has `id = Kwame's manager_id` — finds Marcus Reilly, tags him `depth = 1`.
- **Round 2**: joins against *only* Marcus Reilly's row (not Kwame's again) — finds Diane Okonjo, `depth = 2`.
- **Round 3**: joins against *only* Diane's row — finds Coretta Vance, `depth = 3`.
- **Round 4**: joins against *only* Coretta's row, looking for whoever has `id = Coretta's manager_id` — but `manager_id` is `NULL` for the Mayor, and nothing has `id = NULL`. Zero rows come back. The recursion stops here, on its own, having found the natural end of the chain.

Four rounds for four levels — that's not a coincidence. A recursive CTE walking one path up (or down) a tree always runs exactly as many times as the tree is deep along that path, no more.

Exercise 2 — Every Employee Under a Department Head

2.1 — Same idea, opposite direction

Exercise 1 walked *up*: “whose `id` equals my `manager_id`.” Finding everyone *under* a department head walks *down* instead: “whose `manager_id` equals my `id`.”

```
WITH RECURSIVE subtree AS (  
    SELECT id, name, title, manager_id, 0 AS depth  
    FROM city_org  
    WHERE name = 'Marcus Webb'  
  
    UNION ALL  
  
    SELECT o.id, o.name, o.title, o.manager_id, subtree.depth + 1  
    FROM city_org o  
    JOIN subtree ON o.manager_id = subtree.id  
)  
SELECT name, title, depth FROM subtree ORDER BY depth, name;
```

name	title	depth
Marcus Webb	Director of Public Works	0
Dana Ruiz	Streets & Sanitation Supervisor	1
Tom Delgado	Water & Sewer Supervisor	1

Ivy Chen		Water & Sewer Crew		2
Leo Park		Streets Crew		2
Noah Brandt		Water & Sewer Crew		2
Priya Nair		Streets Crew		2
Sam Okafor		Streets Crew		2

(8 rows)

Only the JOIN condition flipped — `o.manager_id = subtree.id` instead of `o.id = subtree.manager_id` — and the exact same query shape now answers a completely different question. This distinction matters more than it looks: get the join condition backward and PostgreSQL won't error, it will just silently return the anchor row and nothing else, since “everyone whose manager is Marcus Webb” and “whoever Marcus Webb's manager is” are both perfectly valid, perfectly different questions with perfectly valid, empty-looking answers if you meant to ask the other one.

Exercise 3 — Depth and an Indented Tree

3.1 — The whole org chart at once

Exercise 1's `depth` counter already generalizes to the entire tree — start from every root instead of one named employee, and every manager-report edge gets discovered in the same breadth-expanding way:

```
WITH RECURSIVE org_tree AS (
    SELECT id, name, title, manager_id, 0 AS depth, ARRAY[name] AS path
    FROM city_org
    WHERE manager_id IS NULL

    UNION ALL

    SELECT o.id, o.name, o.title, o.manager_id, org_tree.depth + 1, org_tree.path ||
o.name
    FROM city_org o
    JOIN org_tree ON o.manager_id = org_tree.id
)
SELECT repeat(' ', depth) || name || ' (' || title || ')' AS org_chart
FROM org_tree
ORDER BY path;
```

org_chart

```
-----
Coretta Vance (Mayor)
  Aisha Bonner (Director of Parks & Recreation)
    Felix Wren (Parks Maintenance Supervisor)
      Ezra Kowalski (Groundskeeper)
```

```

    Nora Villalobos (Groundskeeper)
Diane Okonjo (Chief of Police)
    Marcus Reilly (Patrol Captain)
    Bianca Ferro (Patrol Officer)
    Kwame Asante (Patrol Officer)
    Theo Lindqvist (Patrol Officer)
    Paula Mensah (Records Sergeant)
Helena Cross (Director of Permitting & Licensing)
    Grace Halloway (Senior Permit Reviewer)
    Mia Sorensen (Permit Clerk)
    Owen Fitch (Permit Clerk)
    Ray Castellano (Building Inspector)
Julian Ostrowski (Director of Finance)
    Colin Marsh (Budget Analyst)
    Renata Sikes (City Accountant)
Marcus Webb (Director of Public Works)
    Dana Ruiz (Streets & Sanitation Supervisor)
    Leo Park (Streets Crew)
    Priya Nair (Streets Crew)
    Sam Okafor (Streets Crew)
    Tom Delgado (Water & Sewer Supervisor)
    Ivy Chen (Water & Sewer Crew)
    Noah Brandt (Water & Sewer Crew)
Wendell Achebe (Director of IT)
    Hugo Petrakis (Database Administrator)
    Zara Lindholm (Systems Administrator)
(30 rows)

```

Two new pieces do all the work. `repeat(' ', depth)` turns the integer `depth` into visible indentation — two spaces per level, the plainest possible tree rendering. `path`, an array built up one name at a time as the recursion descends (`org_tree.path || o.name`), is what makes `ORDER BY path` produce a proper depth-first listing instead of grouping everyone by depth first — without it, `ORDER BY depth` alone would print all six directors before any of their reports, flattening the tree back into levels instead of branches. `path` is never displayed here; it only exists to sort correctly, which is a pattern worth remembering any time a recursive CTE's *output order* matters as much as its contents.

Exercise 4 — Cycle Detection

4.1 — A bad `UPDATE`, on purpose

`manager_id` is just a foreign key — PostgreSQL enforces that it points to a real row, but nothing stops it from pointing to a row that, somewhere further up the chain, points right back down to where you started. Simulate exactly that data-entry mistake:


```
UPDATE city_org
SET   manager_id = (SELECT id FROM city_org WHERE name = 'Leo Park')
WHERE name = 'Dana Ruiz';
```

Leo Park already reports to Dana Ruiz. This one `UPDATE` now also makes Dana report to Leo — a two-person cycle, and a perfectly valid row by every constraint the table has.

4.2 — Where the cycle actually bites

Exercise 3's root-to-leaves traversal doesn't even notice: Dana and Leo simply stop being reachable from the Mayor at all (their `manager_id` chain no longer leads back up to anyone the tree already found), and they silently vanish from that query's output — 26 rows instead of 30, no error, just two people and their two remaining reports quietly orphaned. It's Exercise 1's *upward* direction — the one this whole chapter has been building on — that walks straight into it. Before running this, set a safety net; you're about to run a query you already know doesn't terminate on its own:

```
SET statement_timeout = '3s';

WITH RECURSIVE chain AS (
  SELECT id, name, manager_id, 0 AS depth
  FROM   city_org
  WHERE  name = 'Priya Nair'

  UNION ALL

  SELECT o.id, o.name, o.manager_id, chain.depth + 1
  FROM   city_org o
  JOIN   chain ON o.id = chain.manager_id
)
SELECT COUNT(*) FROM chain;
```

```
ERROR: canceling statement due to statement timeout
```

Priya reports to Dana, Dana now reports to Leo, and Leo reports to Dana — the walk to the root never reaches a `NULL manager_id`, because there isn't one anywhere on this path anymore. Without the timeout, this query runs until it exhausts memory or disk, whichever comes first, generating an unbounded stream of alternating Dana/Leo rows forever. **Any time a recursive CTE walks a self-referencing column you don't personally control the integrity of, set a `statement_timeout` before the first run** — it's cheap insurance against exactly this.

```
RESET statement_timeout;
```

4.3 — `CYCLE ... SET ... USING`: detect it instead of hanging

```

WITH RECURSIVE chain AS (
    SELECT id, name, manager_id, 0 AS depth
    FROM   city_org
    WHERE  name = 'Priya Nair'

    UNION ALL

    SELECT o.id, o.name, o.manager_id, chain.depth + 1
    FROM   city_org o
    JOIN   chain ON o.id = chain.manager_id
)
CYCLE id SET is_cycle USING path
SELECT id, name, depth, is_cycle FROM chain;

```

id	name	depth	is_cycle
5	Priya Nair	0	f
3	Dana Ruiz	1	f
4	Leo Park	2	f
3	Dana Ruiz	3	t

(4 rows)

Four rows and it's done — no timeout needed. `CYCLE id` tells PostgreSQL to track every `id` this query has already visited, in a hidden array column named by `USING path`; the moment a round would revisit an `id` already in that array, it stops expanding *that* branch, flags the repeated row `is_cycle = t`, and moves on rather than looping. Dana Ruiz shows up twice — once as Priya's genuine manager, once again as proof the walk looped back to her — and that second appearance is exactly the signal that something in the data is wrong, ready to `WHERE is_cycle` for or alert on, instead of a hung connection and no explanation at all.



4.4 — Undo the damage

```

UPDATE city_org
SET    manager_id = (SELECT id FROM city_org WHERE name = 'Marcus Webb')
WHERE  name = 'Dana Ruiz';

```

Back to the real org chart before moving on.

Exercise 5 — Shortest Path Through a Graph

5.1 — Roads go both ways; road_segments only says one

```
SELECT road_name, from_intersection, to_intersection, length_m
FROM   road_segments
WHERE  road_name = 'Ring Road';
```

road_name	from_intersection	to_intersection	length_m
Ring Road	11	3	2002.4
Ring Road	3	5	10541.7
Ring Road	5	15	1888.9
Ring Road	15	8	2559.7
Ring Road	8	10	6111.4
Ring Road	10	13	443.9
Ring Road	13	11	3323.3

(7 rows)

Each row is stored once, in one direction, the same way Chapter 2's city_infrastructure stores each road as a single `LINESTRING` — but a car can drive either way down Ring Road. Build a bidirectional view of the graph before doing anything else with it:

```
CREATE VIEW road_graph AS
SELECT from_intersection AS a, to_intersection AS b, road_name, length_m FROM
road_segments
UNION ALL
SELECT to_intersection, from_intersection, road_name, length_m FROM road_segments;
```

5.2 — Breadth-first search: fewest hops

```
WITH RECURSIVE bfs AS (
  SELECT i.id AS node, ARRAY[i.id] AS path, 0 AS hops, 0.0 AS total_m
  FROM   intersections i
  WHERE  i.name = 'Fisherman's Row & Market Street'

  UNION ALL

  SELECT g.b, bfs.path || g.b, bfs.hops + 1, bfs.total_m + g.length_m
  FROM   road_graph g
  JOIN   bfs ON g.a = bfs.node
  WHERE  NOT g.b = ANY(bfs.path)
)
SELECT hops, total_m, path
FROM   bfs
JOIN   intersections dest ON dest.id = bfs.node
WHERE  dest.name = 'Bay Street & Ring Road (East)'
```

```
ORDER BY hops
LIMIT 1;
```

```
hops | total_m |      path
-----+-----+-----
    4 | 15379.5 | {9,12,11,3,5}
(1 row)
```

WHERE NOT g.b = ANY(bfs.path) is doing the same job Exercise 4’s CYCLE clause did — this graph genuinely contains a cycle (Ring Road loops back on itself), so without it, this query would walk in circles exactly like Exercise 4.2’s did. ORDER BY hops LIMIT 1 takes the first path that reaches the destination in the fewest steps: 4 hops, 15,379.5 metres.

5.3 — The gotcha: fewest hops isn’t shortest distance

Widen the search instead of stopping at the first match:

```
WITH RECURSIVE bfs AS (
  SELECT i.id AS node, ARRAY[i.id] AS path, 0 AS hops, 0.0 AS total_m
  FROM   intersections i
  WHERE  i.name = 'Fisherman's Row & Market Street'

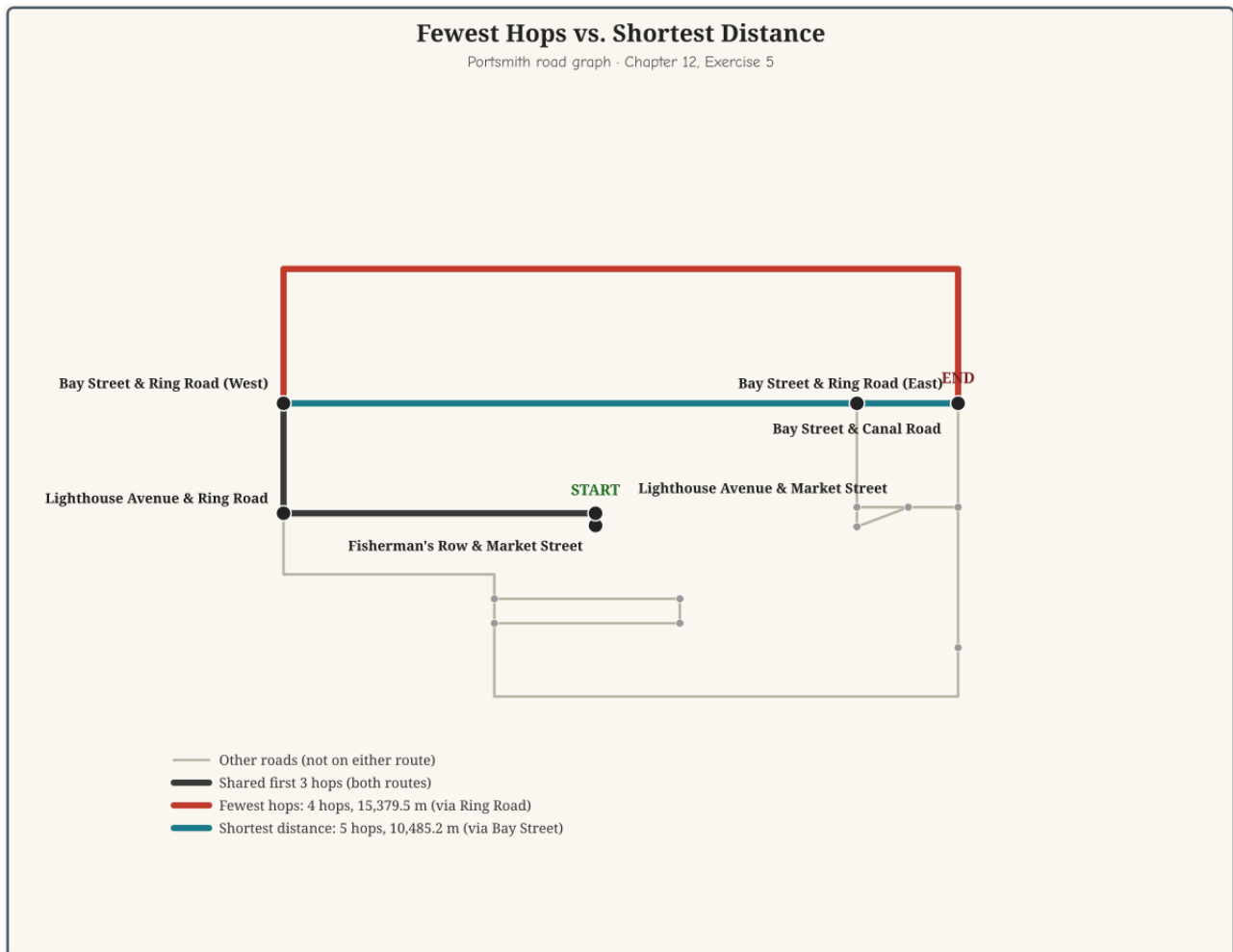
  UNION ALL

  SELECT g.b, bfs.path || g.b, bfs.hops + 1, bfs.total_m + g.length_m
  FROM   road_graph g
  JOIN   bfs ON g.a = bfs.node
  WHERE  NOT g.b = ANY(bfs.path) AND bfs.hops < 6
)
SELECT hops, total_m, path
FROM   bfs
JOIN   intersections dest ON dest.id = bfs.node
WHERE  dest.name = 'Bay Street & Ring Road (East)'
ORDER BY total_m;
```

```
hops | total_m |      path
-----+-----+-----
    5 | 10485.2 | {9,12,11,3,4,5}
    4 | 15379.5 | {9,12,11,3,5}
(2 rows)
```

The 4-hop path 5.2 found is *not* the shortest one — a 5-hop route covers nearly 5 kilometres less. The 4-hop route takes the single giant Ring Road segment straight from Bay Street & Ring Road (West) to Bay Street & Ring Road (East) — the 10,541.7 m stretch Chapter 2’s ST_LineSubstring measured back in this chapter’s data setup, the one a straight-line guess would have badly understated. The 5-hop route detours one extra

intersection down Bay Street itself instead, trading one more turn for two much shorter segments (4,800.3 m + 847.1 m instead of 10,541.7 m).



`ORDER BY hops LIMIT 1` — plain breadth-first search — optimizes for *number of turns*, not *distance travelled*, and those are only the same question when every edge in the graph costs roughly the same to traverse. They don't, here, and a recursive CTE has no built-in concept of "cost" unless a query explicitly asks it to minimize one, the way 5.3's `ORDER BY total_m` does instead of 5.2's `ORDER BY hops`. Real routing (turn-by-turn navigation, Dijkstra's algorithm, `pgRouting`) is built entirely around taking that distinction seriously; this exercise is the smallest possible version of the same lesson.

Exercise 6 — Ancestors of a Leaf, for Faceted Search

6.1 — A category tree built from real Chapter 1 data

```
SELECT c.name AS category, p.name AS parent
FROM   categories c
```

```
JOIN categories p ON p.id = c.parent_id
WHERE c.name IN ('bakery', 'restaurant');
```

category	parent
restaurant	All Categories
bakery	restaurant

categories isn't invented data — its two levels under “All Categories” are exactly Chapter 1's real category and subcategory/cuisine values, pulled straight out of `businesses.details` and organized into a tree. “Bakery” is a real cuisine value on a real business (River Bend Bakery, Chapter 1) three levels deep in this hierarchy.

6.2 — Walk up, then flatten into a breadcrumb

A faceted search UI wants “all ancestors of this leaf” to build a breadcrumb trail — the exact same upward walk as Exercise 1, applied to a different tree, finished off with `string_agg`:

```
WITH RECURSIVE ancestors AS (
  SELECT id, name, parent_id, 0 AS depth
  FROM categories
  WHERE name = 'bakery'

  UNION ALL

  SELECT c.id, c.name, c.parent_id, ancestors.depth + 1
  FROM categories c
  JOIN ancestors ON c.id = ancestors.parent_id
)
SELECT string_agg(name, ' > ' ORDER BY depth DESC) AS breadcrumb
FROM ancestors;
```

breadcrumb
All Categories > restaurant > bakery

`ORDER BY depth DESC` inside `string_agg` puts the root first and the leaf last — the walk itself discovers ancestors leaf-to-root (depth increasing outward, exactly like Exercise 1's officer-to-Mayor chain), so displaying them root-to-leaf means reversing that order at the very end, not changing how the recursion runs. Nothing about this query knows or cares that the tree is only three levels deep — the identical query against a ten-level category tree would produce a ten-element breadcrumb without a single line changing.

Summary — What You Should Now Know

Tool	What it does
<code>WITH RECURSIVE name AS</code> <code>(anchor UNION ALL</code> <code>recursive_term)</code>	Anchor runs once; the recursive term reruns against only the previous round's new rows until a round adds nothing
<code>o.id = cte.manager_id</code>	Walk <i>up</i> a hierarchy — toward the root
<code>o.manager_id = cte.id</code>	Walk <i>down</i> a hierarchy — toward the leaves
depth counter	Increment it once per recursive round to know how deep any given row is
<code>path array + ORDER BY</code> <code>path</code>	Make a recursive CTE's output print in proper depth-first tree order
<code>CYCLE col SET flag USING</code> <code>path</code>	Detect a repeated value instead of looping forever; flags the row where it happened
<code>WHERE NOT x = ANY(path)</code>	The hand-rolled version of cycle prevention, for graphs that need per-branch path tracking <code>CYCLE</code> alone doesn't give you
<code>SET statement_timeout</code>	Cheap insurance before running any recursive query over data you don't control the integrity of
<code>BFS (ORDER BY hops)</code> vs. shortest-distance <code>(ORDER BY total_m)</code>	Fewest steps and least total cost are different questions unless every edge costs the same
<code>string_agg(name, ' > '</code> <code>ORDER BY depth DESC)</code>	Turn an ancestor walk into a breadcrumb, root first

The key design insight from this chapter is that a recursive CTE is just an ordinary query, run in a loop, over data that happens to describe its own structure — the anchor decides where to start, the join direction decides which way to walk, and everything past that is the same `SELECT` you already know how to write. Trees are the friendly case, where that loop is guaranteed to end because nothing points back at itself. The moment real data — a bad `manager_id`, a road network that loops — stops guaranteeing that, the loop needs its own explicit exit condition, either `CYCLE`'s built-in bookkeeping or a hand-rolled `path` array doing the same job. Every exercise in this chapter is one of exactly two ideas: which direction to walk, and how to know when to stop.

Going further: Chapter 21's placeholder chapter on PostgreSQL 19's SQL/PGQ property graphs picks up directly from Exercise 5 — the same `road_segments` graph, queried with

*syntax purpose-built for path traversal instead of a hand-rolled `path` array and a `WHERE NOT ... ANY(...)` guard. Reading that chapter (once PostgreSQL 19 is out of beta and this book can say something definite about it) right after this one, rather than waiting for its place in the numbering, is a reasonable way to see the same problem solved twice, eight chapters apart in the table of contents but adjacent in what they're actually about. Chapter 17's foreign data wrappers occasionally combine with recursive CTEs when a hierarchy spans more than one database — the recursion itself doesn't care where a row physically lives, only that the self-referencing column resolves. And `pgRouting`, mentioned briefly in Exercise 5, is the production answer to "I need real shortest-path routing, not the smallest example that demonstrates the idea" — turn restrictions, one-way streets, and genuine Dijkstra/A implementations, all built on top of the same PostGIS geometry this chapter's road graph came from.**

